GUIDED REVISION

JEE(Main + Advanced) 2025
ENTHUSIAST+LEADER COURSE
SCORE ADVANCED

ANSWER KEY

(VECTOR-3D) GUIDED REVISION # 1

SOLUTIONS

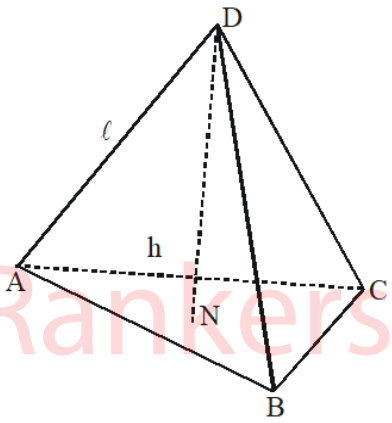
Maths

SECTION-I (i)

1. Ans (A)

Ans. (A)

Let D is origin and position vectors of A, B, C are $\vec{a}$, $\vec{b}$, $\vec{c}$ respectively,



then $|\vec{a}| = |\vec{b}| = |\vec{c}| = \ell$ and

$|\vec{a}\,.\,\vec{b}| = |\vec{b}\,.\,\vec{c}| = |\vec{c}\,.\,\vec{a}| = \ell^2\cos 60° = \dfrac{\ell^2}{2}$

Now $[\vec{a}\,\vec{b}\,\vec{c}]^2 = \begin{vmatrix} \vec{a}.\vec{a} & \vec{a}.\vec{b} & \vec{c}.\vec{c} \\ \vec{b}.\vec{a} & \vec{b}.\vec{b} & \vec{b}.\vec{c} \\ \vec{c}.\vec{a} & \vec{c}.\vec{b} & \vec{c}.\vec{c} \end{vmatrix} = \dfrac{\ell^6}{2}$

now volume of tetrahedron $= \dfrac{1}{3}\times$ Area of base $\times$ height

$\Rightarrow \dfrac{1}{6}[\vec{a}\,\vec{b}\,\vec{c}] = \dfrac{1}{3}\times\left(\dfrac{\sqrt{3}}{4}\ell^2\right)\times$ height $\Rightarrow$

hight $= \dfrac{\frac{\ell^3}{\sqrt{2}}\times\frac{1}{2}}{\frac{\sqrt{3}}{4}\ell^2} = \sqrt{\dfrac{2}{3}}\ell$

2. Ans (D)

Ans. (D)

Lines L_1 & L_2 are coplanar if

$\begin{vmatrix} 3-2 & 2-3 & 1-2 \\ 2 & 3 & \lambda \\ 3 & 2 & 3 \end{vmatrix} = 0 \Rightarrow \lambda = 4$

Let $L_1 : \dfrac{x-3}{2} = \dfrac{y-2}{3} = \dfrac{z-1}{4} = t$

$\therefore$ Any point on it is P(2t + 3, 3t +2, 4t +1)

It lies on the plane x + y + z = 15 $\Rightarrow$ 9t + 6 = 15 or t = 1

$\therefore$ Distance of P(5, 5, 5) from (3, 4, 3) = 3

3. Ans (D)

Ans. (D)

$\vec{a}\times\vec{b} = |\vec{a}|\,|\vec{b}|\sin 30°\,\hat{n} = 3\hat{n}$

Since $\vec{d}$ is perpendicular to both $\vec{a}$ & $\vec{b}$ $\Rightarrow$ $\hat{n} = \pm\vec{d}$

$\therefore\ \vec{a}\times\vec{b} = \pm 3\vec{d}$

Now $(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d}) = \pm 3\vec{d}\times(\vec{c}\times\vec{d})$ $\Rightarrow$

$\pm 3\{(\vec{d}.\vec{d})\vec{c} - (\vec{d}.\vec{c})\vec{d}\} = \dfrac{\hat{i}}{6} - \dfrac{\hat{j}}{3} + \dfrac{\hat{k}}{3}$ or

$\pm 3\vec{c} = \dfrac{\hat{i} - 2\hat{j} + 2\hat{k}}{6}$

$\therefore\ |\vec{c}.\hat{i}| = \dfrac{1}{18},\ |\vec{c}.\hat{j}| = \dfrac{1}{9},\ |\vec{c}.\hat{k}| = \dfrac{1}{9}$

4. Ans (C)

Ans. (c)

Diagonals of rhombus are perpendicular

$\Rightarrow \vec{d}_1.\vec{d}_2 = 0 \Rightarrow 2|\vec{a}|^2 - 2|\vec{b}|^2 + 3\vec{a}\,.\,\vec{b} = 0$ ....(i)

& $\vec{d}_3.\vec{d}_4 = 0 \Rightarrow 2|\vec{a}|^2 - 2|\vec{b}|^2 - 3\vec{a}\,.\,\vec{b} = 0$ ....(ii)

$\Rightarrow \vec{a}\,.\,\vec{b} = 0 \Rightarrow \vec{a}$ & $\vec{b}$ are perpendicular

$\Rightarrow [\vec{a}\,\vec{b}\,\vec{a}\times 2\vec{b}] = (\vec{a}\times\vec{b}).(\vec{a}\times 2\vec{b}) = 2|\vec{a}\times\vec{b}|^2 = 2|\vec{a}|^2\,|\vec{b}|^2$

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5. Ans (D)

Ans. (D)

$\vec{c} = \alpha\left(\hat{i}+\hat{j}\right) + \beta\left(\hat{j}+\hat{k}\right)$

$= \alpha\hat{i} + (\alpha+\beta)\hat{j} + \beta\hat{k}$

$\because \vec{c}, \hat{i}-2\hat{j}+\hat{k}, \ 3\hat{i}+2\hat{j}-\hat{k}$ are coplanar

$$\therefore \begin{vmatrix} \alpha & \alpha+\beta & \beta \\ 1 & -2 & 1 \\ 3 & 2 & -1 \end{vmatrix} = 0$$

$\alpha(2-2) - (\alpha+\beta)(-1-3) + \beta(2+6) = 0$

$4\alpha + 4\beta + 8\beta = 0$

$\frac{\alpha}{\beta} = -3$

Maths

SECTION-I (ii)

6. Ans (B,C)

Ans. B,C

Let equation of plane be $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$

it cuts coordinates axes at A(a,0,0), B(0,b,0), C(0,0,c)

its centroid is $G\left(\frac{a}{3},\frac{b}{3},\frac{c}{3}\right) \equiv (1, r, r^2)$

$\Rightarrow a = 3, b = 3r, c = 3r^2$

$\Rightarrow$ equation of plane is $x + \frac{y}{r} + \frac{z}{r^2} = 3$

$\because$ it posses through $(4,-8,-15)$

$\therefore 4 - \frac{8}{r} + \frac{15}{r^2} = 3 \Rightarrow r^2 - 8r + 15 = 0$

$\Rightarrow r = 5, r = 3$

7. Ans (B,D)

Ans. B,D

Let normal vector n1 perpendicular to plane determining $\hat{i}, \hat{j}+\hat{k}$ is

$n_1 = \hat{i} \times (\hat{i}+\hat{j}) = \hat{k}$

similarly $n_2 = (\hat{i}-\hat{j}) \times (\hat{i}-\hat{k}) = \hat{i}+\hat{j}+\hat{k}$

Now vector parallel to intersection of plane $= \vec{n}_2 \times \vec{n}_1$

$= \vec{k} \times (\hat{i}+\hat{j}+\hat{k}) = -(\hat{j}-\hat{i}) \Rightarrow$

$\frac{x-0}{1-0} = \frac{y-0}{1-0} = \frac{z-0}{1-0}$

Angle between $\lambda(-\hat{j}+\hat{i})$ and $(\hat{i}-2\hat{j}+2\hat{k})$

$\cos\theta = \frac{\lambda(-\hat{j}+\hat{i}).(\hat{i}-2\hat{j}+2\hat{k})}{\lambda\sqrt{2}\times 3} = \frac{1}{\sqrt{2}} \Rightarrow$

$\theta = \frac{\pi}{4}$ or $\theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$

8. Ans (A,B,C)

Ans. (A,B,C)

We have $\hat{i} \cdot \hat{j} = 0$

given $\vec{p} = 3\hat{i} + 4\hat{j}, \vec{q} = 5\hat{i}$

$4\vec{r} = \vec{p} + \vec{q}, 2\vec{s} = \vec{p} - \vec{q}$

$\Rightarrow \vec{r} = 2\vec{i} + \vec{j}, \vec{s} = -\vec{i} + 2\vec{j}$

for option A :

$|\vec{r} + k\vec{s}| = \left|(2-k)\hat{i} + (1+2k)\hat{j}\right| = \sqrt{(2-k)^2 + (1+2k)^2}$

$= \sqrt{5+5k^2}$ ..... (i)

& $|\vec{r} - k\vec{s}| = \left|(2+k)\hat{i} + (1-2k)\hat{j}\right| = \sqrt{(2+k)^2 + (1-2k)^2}$

$= \sqrt{(5+5k^2)}$ ..... (ii)

from (i) and (ii)

$|\vec{r} + k\vec{s}| = |\vec{r} - k\vec{s}|$

option B :

$\because \vec{r} \cdot \vec{s} = 0$

$\vec{r} \perp \vec{s}$

option C :

$\because \vec{r} + \vec{s} = \hat{i} + 3\hat{j}$ and $\vec{r} - \vec{s} = 3\hat{i} - \hat{j}$

$(\vec{r} + \vec{s}) \cdot (\vec{r} - \vec{s}) = 0$

Hence $(\vec{r} + \vec{s})$ is perpendicular to $(\vec{r} - \vec{s})$

option D :

$|\vec{r}| = \sqrt{5}, |\vec{s}| = \sqrt{5}, |\vec{p}| = 5, |\vec{q}| = 5$

$\therefore |\vec{r}| = |\vec{s}| \neq |\vec{p}| = |\vec{q}|$

9. Ans (A,C)

Ans. (A,C)

The required vector, $\vec{r} = \lambda\left(\vec{a} + \vec{b}\right)$, λ is a scalar

or

$\vec{r} = \lambda\left(\frac{1}{9}\left(7\hat{i} - 4\hat{j} - 4\hat{k}\right) + \frac{1}{3}\left(-2\hat{i} - \hat{j} + 2\hat{k}\right)\right)$

$= \frac{\lambda}{9}\left(\hat{i} - 7\hat{j} + 2\hat{k}\right)$

$\because |\vec{r}|^2 = 54 \Rightarrow \frac{\lambda^2}{81}(1+49+4) = 54 \Rightarrow 1 = \pm 9$

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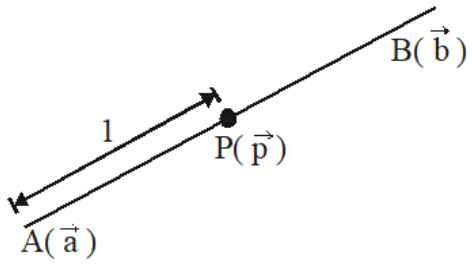
10. **Ans (A,D)**

Ans. (a,D)

$\vec{a} = \hat{i} + \hat{j} - 2\hat{k}$, $\vec{b} = \hat{i} - 3\hat{j} + \hat{k}$

$\vec{p} - \vec{a} = \lambda(\vec{b} - \vec{a}) = \lambda(-4\hat{j} + 3\hat{k})$

$|\vec{p} - \vec{a}| = \lambda\sqrt{16+9}$



$25\lambda^2 = 1 \Rightarrow \lambda = \pm\frac{1}{5}$

$\therefore \vec{p} = \vec{a} + \lambda(\vec{b} - \vec{a}) = \begin{bmatrix} \frac{1}{5}(5\hat{i} + \hat{j} - 7\hat{k}) \\ \frac{1}{5}(5\hat{i} + 9\hat{j} - 13\hat{k}) \end{bmatrix}$

Maths

SECTION-I (iii)

11. **Ans (A)**

Plane containing the line

$L_1 : 2x - y + 3z - 5 = 0 = x - 2y + 2z + 1$

is in the from $P_1 + \lambda P_2 = 0$

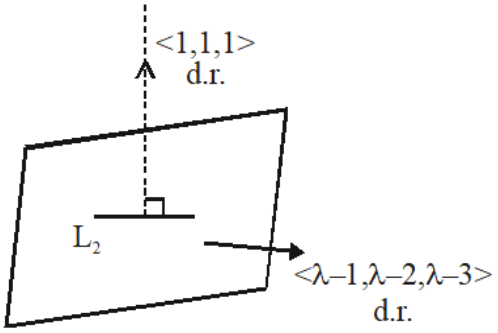
where $P_1 : 2x - y + 3z - 5 = 0$, $P_2 : x - 2y + 2z + 1 = 0$

$(2x - y + 3z - 5) + k(x - 2y + 2z + 1) = 0$

it passes through (1,2,3) [$\because$ plane contains lie L_2]

$(2 - 2 + 9 - 5) + k(1 - 4 + 6 + 1) = 0$

$4 + k(4) = 0 \quad \Rightarrow \quad k = -1$



Ans. (A)

Equation of plane $x + y + z = 6$

12. **Ans (C)**

Plane containing the line

$L_1 : 2x - y + 3z - 5 = 0 = x - 2y + 2z + 1$

is in the from $P_1 + \lambda P_2 = 0$

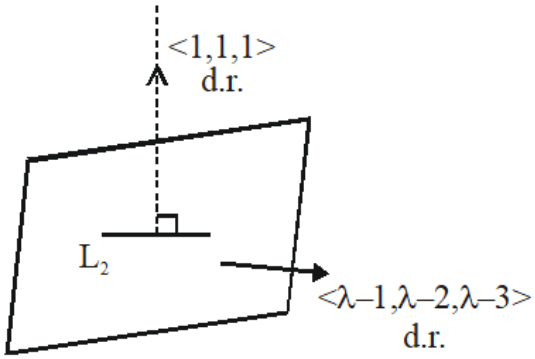
where $P_1 : 2x - y + 3z - 5 = 0$, $P_2 : x - 2y + 2z + 1 = 0$

$(2x - y + 3z - 5) + k(x - 2y + 2z + 1) = 0$

it passes through (1,2,3) [$\because$ plane contains lie L_2]

$(2 - 2 + 9 - 5) + k(1 - 4 + 6 + 1) = 0$

$4 + k(4) = 0 \quad \Rightarrow \quad k = -1$



Ans. (C)

L_2 also lies on the plane

$(\lambda - 1) \cdot 1 + (\lambda - 2) \cdot 1 + (\lambda - 3) \cdot 1 = 0$

$\Rightarrow \lambda = 2$

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13. **Ans (C)**

Plane containing the line

$L1 : 2x - y + 3z - 5 = 0 = x - 2y + 2z + 1$

is in the from $P_1 + \lambda P_2 = 0$

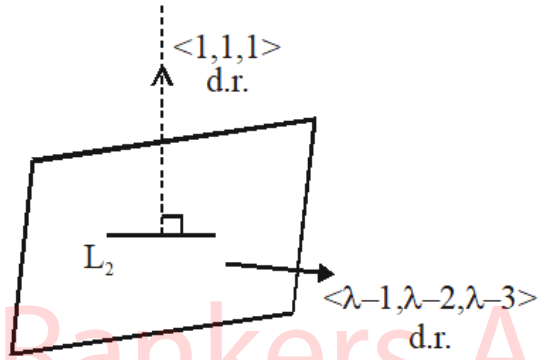
where $P_1 : 2x - y + 3z - 5 = 0$, $P_2 : x - 2y + 2z + 1 = 0$

$(2x - y + 3z - 5) + k(x - 2y + 2z + 1) = 0$

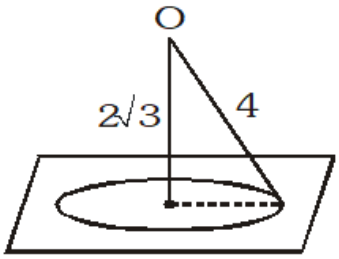
it passes through (1,2,3) [$\because$ plane contains lie L_2]

$(2 - 2 + 9 - 5) + k(1 - 4 + 6 + 1) = 0$

$4 + k(4) = 0 \quad \Rightarrow \quad k = -1$



Ans. (C)



Perpendicular distance of

(0, 0, 0) from the plane

$\Rightarrow \quad \frac{6}{\sqrt{3}} = 2\sqrt{3}$

Hence radius of circle be 2.

$\Rightarrow$ required area is 4π

Maths

SECTION-II (ii)

1. **Ans (3.96)**

Ans. 3.96

$\because \quad \left(\vec{a} \times (\vec{a} \times \vec{b})\right) \times \vec{c} = \left((\vec{a}.\vec{b})\vec{a} - (\vec{a}.\vec{a})\vec{b}\right) \times \vec{c}$

$= -|\vec{a}|^2 \vec{b} \times \vec{c} = -6\left(\vec{b} \times \vec{c}\right)$

Similarly $\left(\vec{b} \times (\vec{b} \times \vec{c})\right) \times \vec{a} = -|\vec{b}|^2 \vec{c} \times \vec{a} = -2(\vec{c} \times \vec{a})$

and $(\vec{c} \times (\vec{c} \times \vec{a})) \times \vec{b} = -|\vec{c}|^2 \vec{a} \times \vec{b} = -3(\vec{a} \times \vec{b})$

Now, $\vec{v} = -\left[3(\vec{a} \times \vec{b}) + 6(\vec{b} \times \vec{c}) + 2(\vec{c} \times \vec{a})\right]$

$\because \vec{a}$, $\vec{b}$ and $\vec{c}$ are mutually perpendicular

$\therefore \quad \vec{a} \times \vec{b}$,$\vec{b} \times \vec{c}$ and $\vec{c} \times \vec{a}$ are also mutually perpendicular

$\Rightarrow |\vec{v}| = \sqrt{9|\vec{a}|^2 |\vec{b}|^2 + 36|\vec{b}|^2|\vec{c}|^2 + 4|\vec{c}|^2|\vec{a}|^2} = 6\sqrt{11}$

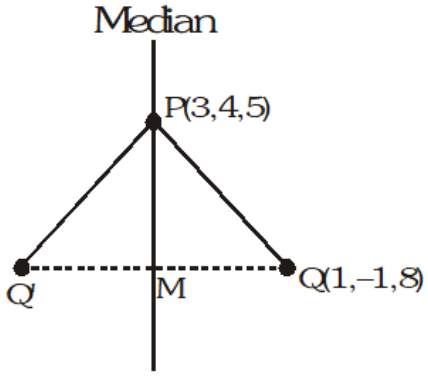
$M = 6\sqrt{11}$

$M_2 = 36 \times 11$

$\frac{M^2}{100} = 3.96$

2. **Ans (3.00)**

Ans. 3.00



$\Delta PQQ'$ is an isosceles triangle & $\overrightarrow{PM} = 2\hat{i} - 2\hat{j} + \hat{k}$

Now, $PM = \left|\overrightarrow{PQ}\right| \cos\theta$, $\overrightarrow{PQ} = 2\hat{i} + 5\hat{j} - 3\hat{k}$

$PM = \left|\frac{\overrightarrow{PQ}.\overrightarrow{PM}}{\left|\overrightarrow{PM}\right|}\right| = \left|\frac{4 - 10 - 3}{3}\right| = 3$

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3. Ans (0.5)

Ans. 0.5

lines are $\frac{x-\frac{1}{b}}{c}=\frac{y}{-b}=\frac{z-\frac{1}{b}}{0}$ &

$\frac{x-\frac{2}{b}}{a}=\frac{y}{b}=\frac{z-\frac{3}{b}}{c}$ are perpendicular

$\Rightarrow \quad ac-b^2=0 \Rightarrow b^2=ac$

Now $\cos B=\frac{a^2+c^2-b^2}{2ac}$

$=\frac{1}{2}\left(\frac{a}{c}+\frac{c}{a}\right)-\frac{1}{2}\geqslant\frac{1}{2}(2)-\frac{1}{2}$

$\left(\because x+\frac{1}{x}\geqslant 2, x>0\right)$

$\Rightarrow$ minimum value of $\cos B=\frac{1}{2}$ exist when $a=c$

4. Ans (1.00)

Ans. 1.00

$\hat{a}+2\hat{b}+\hat{c}=\hat{a}\times\hat{c}$ ....(i)

Taking dot with $\hat{a}$ we get

$1+2\hat{a}.\hat{b}+\hat{a}.\hat{c}=0$

Let $\hat{a}.\hat{c}=x$

$\Rightarrow 2\hat{a}.\hat{b}=-1-x$

From (i), $\hat{a}+2\hat{b}=-\hat{c}+\hat{a}\times\hat{c}$

Squaring $1+4+4\hat{a}.\hat{b}=1+1-(\hat{a}.\hat{c})^2$

$\Rightarrow 5-2-2x=2-x^2 \quad \Rightarrow x=1$

$\Rightarrow \hat{a}.\hat{c}=1 \Rightarrow \hat{a}=\hat{c} \Rightarrow$ From (i)

$\hat{a}=-\hat{b}$

$\Rightarrow \left|\hat{a}\times\hat{b}+\hat{b}\times\hat{c}+\hat{a}+\hat{b}+\hat{c}\right|$

$|(\hat{a}\times-\hat{a})+(-\hat{a}\times\hat{a})+\hat{a}-\hat{a}+\hat{c}|=|\hat{c}|=1$

5. Ans (3.12 or 3.13)

Ans. 3.12 or 3.13

$\vec{b}.(\vec{b}\times\vec{\alpha}+\vec{a})=\vec{b}.\vec{\alpha} \quad \Rightarrow \vec{b}.\vec{a}=\vec{b}.\vec{\alpha}$ ...(1)

$\because$ proejction of $\vec{\alpha}$ on $\vec{b}=\vec{\alpha}.\hat{b}$

$\Rightarrow \vec{\alpha}.\hat{b}=3 \quad \Rightarrow \quad \vec{\alpha}.\vec{b}=3 \left(\because |\vec{b}|=1\right)$

$\therefore \vec{b}.\vec{a}=3$ (from (1))

Now $\vec{b}\times(\vec{b}\times\vec{\alpha}+\vec{a})=\vec{b}\times\vec{\alpha}$

$\Rightarrow (\vec{b}.\vec{\alpha})\vec{b}-b^2\vec{\alpha}+\vec{b}\times\vec{a}=\vec{b}\times\vec{\alpha}$

$\Rightarrow 3\vec{b}-\vec{\alpha}+\vec{b}\times\vec{a}=\vec{b}\times\vec{\alpha}$

$\Rightarrow 3\vec{b}-\vec{\alpha}+\vec{b}\times\vec{a}=\vec{\alpha}-\vec{a}$

$\Rightarrow \vec{\alpha}=\frac{1}{2}\left(3\vec{b}+\vec{a}+\vec{b}\times\vec{a}\right)$

Projection of $\vec{\alpha}$ along $\vec{a}$

$=\vec{\alpha}.\hat{a}=\vec{\alpha}.\frac{\vec{a}}{4}=\frac{1}{8}\left[3\vec{b}+\vec{a}+\vec{b}\times\vec{a}\right].\vec{a}$

$=\frac{1}{8}\left[3\vec{b}.\vec{a}+|\vec{a}|^2\right]=\frac{1}{8}[3\times3+16]=\frac{25}{8}=3.125$

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Maths

SECTION–IV

1. **Ans (A->QS,B->T,C->P,D->R)**

Ans.(A) $\longrightarrow$ (Q,S); (B) $\longrightarrow$ (T); (C) $\longrightarrow$ (P); (D) $\longrightarrow$ (R)

(A) Let $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$

$c_1^2 + c_2^2 + c_3^2 = 2,$

$\frac{\vec{a}.\vec{b}}{|\vec{a}||\vec{b}|} = \frac{1}{2} = \frac{\vec{a}.\vec{c}}{|\vec{a}||\vec{c}|} = \frac{c_1 + c_2}{2} \Rightarrow c_1 + c_2 = 1$

$c_2 = 1 - c_1$

simillarly $c_3 = 1 - c_1$

$\Rightarrow c_1^2 + (1 - c_1)^2 + (1 - c_1)^2 = 2 \Rightarrow c_1 = 0, \frac{4}{3}$

(B) Let $\vec{r} = r_1\hat{i} + r_2\hat{j} + r_3\hat{j}$

$\vec{r} \times (2\hat{i} + \hat{j} + \hat{k}) = \hat{j} - \hat{k}$

$\Rightarrow (r_2 - r_3)\hat{i} - \hat{j}(r_1 - 2r_3) + \hat{k}(r_1 - 2r_2) = \hat{j} - \hat{k}$

comparing corresponding components

$r_2 = r_3, r_1 - 2r_3 = -1, r_1 - 2r_2 = -1$

$\Rightarrow r_1 - 2r_2 = -1$ There are infinite solutions.

Let $r_2 = \lambda = r_3$ $\because r_1 = 2r_2 - 1 \setminus r_1 = 2\lambda - 1$

$\vec{r} = -\hat{i} + \lambda(2\hat{i} + \hat{j} + \hat{k})$

$\Rightarrow \vec{r} = 3\hat{i} + 2\hat{j} + 2\hat{k}$ for $\lambda = 2$

(C) Let equation of plane be $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

it pass through (3,4,5) $\Rightarrow \frac{3}{a} + \frac{4}{b} + \frac{5}{c} = 1$ ...(1)

and A(a,0,0), B(0,b,0), C(0,0,c)

equation of plane throught A and parallel to yz plane is

$x = a$

equation of plane throught B and parallel to zx plane is

$y = b$

equation of plane throught C and parallel to xy plane is

$z = c$

$\Rightarrow$ point of intersection of three planes is (a,b,c)

$a = 6, b = 16, c = 20$ satisfy ...(1)

(D) $(\vec{r} - \vec{a}) \times \vec{b} = 0$

$\Rightarrow \vec{r} = \vec{a} + t\vec{b}$

$= \left(\hat{i} + \hat{j}\right) + t\left(2\hat{i} - \hat{k}\right)$

$= (1 + 2t)\hat{i} + \hat{j} - t\hat{k}$

simillarly for second line

$\vec{r} = \vec{b} + s\vec{a}$

$\vec{r} = (2 + s)\hat{i} + s\hat{j} - \hat{k}$

if two lines intersect

$1 + 2t = 2 + s$

$1 = s$

$t = 1$

$\Rightarrow$ Point of intersection is $(3,1,-1)$

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ANSWER KEY

(VECTOR-3D) GUIDED REVISION # 2

TIME : 60 Min.

M.M. : 102

SOLUTIONS

MATHS

Section–I(i) Single Correct Type

1. Ans (C)

Ans. (C)

$\vec{x}$, $\vec{y}$ & $\vec{x} \times \vec{y}$ are linearly independent

$a\alpha^2 + b\alpha + c = 0$

$a\beta^2 + b\beta + c = 0$

$a\gamma^2 + b\gamma + c = 0$

$\Rightarrow$ quadratic $ax^2 + bx + c = 0$ has three roots α, β and γ

$\Rightarrow ax^2 + bx + c = 0$ is an identity

$\Rightarrow a = b = c = 0$

2. Ans (C)

Ans. (C)

$\hat{a} \times \hat{b} + \hat{a} = \vec{c}$ ...(i)

$\hat{a} \times \hat{b} = \vec{c} - \hat{a}$

Dot with $\vec{c} \Rightarrow v = \left[\hat{a}\ \hat{b}\ \vec{c}\right] = |\vec{c}|^2 - \hat{a}.\hat{c}$ ...(ii)

for (i) : dot with

$\Rightarrow 0 + \hat{a}.\hat{b} = \hat{b}.\vec{c} = 0 \Rightarrow \hat{a}.\hat{b} = 0$

for (i) dot with $\hat{a} \Rightarrow 0 + 1 = \hat{a}.\vec{c}$

(ii) $\Rightarrow \left[\hat{a}\ \hat{b}\ \vec{c}\right] = |\vec{c}|^2 - 1$

now $\hat{a} = \vec{c} - \hat{a} \times \hat{b}$

$|\hat{a}|^2 = \left|\vec{c} - \left(\vec{a} \times \hat{b}\right)\right|^2$

$|\hat{a}|^2 = \left|\vec{c} - \left(\vec{a} \times \hat{b}\right)\right|^2$

$\Rightarrow 1 = |c|^2 + \left|\hat{a} \times \hat{b}\right|^2 - 2\left[\hat{a}\ \hat{b}\ \vec{c}\right]$

$\Rightarrow 1 = \left[\hat{a}\ \hat{b}\ \vec{c}\right] + 1 + 1 - 2\left[\hat{a}\ \hat{b}\ \vec{c}\right]$

$\Rightarrow \left[\hat{a}\ \hat{b}\ \vec{c}\right] = 1$

3. Ans (B)

Ans. (B)

$\vec{d}.\vec{c} = \lambda\left[\vec{a}\ \vec{b}\ \vec{c}\right] + \mu.0 + v.0$

$\vec{d}.\vec{a} = \mu\left[\vec{a}\ \vec{b}\ \vec{c}\right]$

$\vec{d}.\vec{b} = v\left[\vec{a}\ \vec{b}\ \vec{c}\right]$

$\vec{d}.\left(\vec{a}\ \vec{b}\ \vec{c}\right) = \left[\vec{a}\ \vec{b}\ \vec{c}\right](\lambda + \mu + v)$

$\lambda + \mu + v = 64$

4. Ans (A)

Ans. (A)

Let $|\vec{U}| = 2\cos\alpha\,\hat{i} + 2\sin\alpha\,\hat{j}$

$$\left([\vec{U}\ \vec{V}\ \vec{W}]\right)^2 = \begin{vmatrix} 2\cos\alpha & 2\sin\alpha & 0 \\ 2 & 1 & -1 \\ 1 & 0 & 3 \end{vmatrix}^2$$

$= |6\cos\alpha - 14\sin\alpha|^2$

Maximum value = 36 + 196 = 232

5. Ans (B)

Ans. (B)

Let A $(\vec{a})$, B $(\vec{b})$ and C $(\vec{c})$

$\Rightarrow G_1\left(\frac{\vec{a}+\vec{b}+\vec{c}}{3}\right)$ and $G_2\left(\frac{\vec{a}+\vec{c}}{3}\right)$

$\overrightarrow{OG_1} \cdot \overrightarrow{BG_2} = 0 \Rightarrow \frac{\vec{a}+\vec{b}+\vec{c}}{3} \cdot \frac{a+\vec{c}-3\vec{b}}{3} = 0$

$\Rightarrow a^2 + c^2 - 3b^2 + 2\vec{a}\cdot\vec{c} - 2\vec{b}\cdot\vec{c} - 2\vec{a}\cdot\vec{b} = 0$

Now, $|\vec{c} - \vec{a}|^2 = b^2, |\vec{c} - \vec{b}|^2 = a^2$ and $|\vec{a} - \vec{b}|^2 = c^2$

$\{\because AC = b, BC = a$ and $AB = c\}$

$\Rightarrow 2\vec{a}\cdot\vec{c} = a^2 + c^2 - b^2, 2\vec{b}\cdot\vec{c}$

$= b^2 + c^2 - a^2, 2\vec{a}\cdot\vec{b} = a^2 + b^2 - c^2$

Putting in the above result, we get

$2a^2 + 2c^2 - 6b^2 = 0 \Rightarrow \frac{a^2 + c^2}{b^2} = 3$

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6. Ans (B)

Ans. (B)

Required plane

$(1 + 2\lambda)x + (2 - 3\lambda)y + (3 + 7\lambda)z + 5 + \lambda = 0$

it is parallel to given line

$\Rightarrow$

$(1 + 2\lambda)\hat{i} + (2 - 3\lambda)\hat{j} + (3 + 7\lambda)\hat{k}.\left(8\hat{i} - 7\hat{j} - 4\hat{k}\right) = 0$

$\Rightarrow 8(1+ 2\lambda) - 7(2 - 3\lambda) - 4(3 + 7\lambda) = 0$

$\Rightarrow \lambda = 2$

x intercept $= \dfrac{-5 - \lambda}{1 + 2\lambda} = -\dfrac{7}{5}$

7. Ans (B)

Ans. (B)

Let the plane be $a(x - 1) + b(y - 0) + c(z - 0) = 0$

$\Rightarrow ax + by + cz - a = 0$

Pass through $(0,1,0) \Rightarrow a.0 + b.1 + c.0 - a = 0$

$\Rightarrow a = b$ ...(i)

Normal to plane : $a\hat{i} + b\hat{j} + c\hat{k}$

Makes angle $\dfrac{\pi}{4}$ with $\hat{i} + \hat{j}$

$\therefore \dfrac{a + b}{\sqrt{a^2 + b^2 + c^2}.\sqrt{2}} = \dfrac{1}{\sqrt{2}}$

$\Rightarrow (a + b)^2 = a^2 + b^2 + c^2$ ...(ii)

By (i) and (ii) : $4a^2 = 2a^2 + c^2 \Rightarrow 2a^2 = c^2$

$\Rightarrow \sqrt{2}a = c; -\sqrt{2}a = c$

$\therefore a : b : c = 1 : 1 : \sqrt{2}$

8. Ans (B)

Ans. (B)

$\dfrac{x - 1}{3} = \dfrac{y - 2}{1} = \dfrac{z - 3}{2} = t$

$\dfrac{x - 3}{1} = \dfrac{y - 1}{2} = \dfrac{z - 2}{3} = \mu$

for point of intersection

$3t + 1 = \mu + 3$ ...(1)

$t + 2 = 2\mu + 1$ ...(2)

$2t + 3 = 3\mu + 2$

from (1) and (2)

$5 = 5\mu$

$\Rightarrow \mu = 1$

$\therefore t = 1$

$\therefore$ Point of intersection P(4,3,5)

Now, Plane passing through P and at greatest distance from (0,0,0) will have its normal vector along $\overrightarrow{OP}$.

$\therefore$ Required equaiton : $4 (x - 4) + 3(y - 3) + 5(z - 5) = 0$

$4x + 3y + 5z = 50$

option (B)

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9. Ans (A)

Ans. (A)

Since the line $\dfrac{x - 1}{2} = \dfrac{y}{-1} = \dfrac{z + 2}{1}$ is parallel to the plane $2x + y - 3z = 4$

($\because 2 \times 2 - 1 - 3 = 4 - 4 = 0$).

so the direction cosines of the projected line remains the same as the given line

so dc's $= \dfrac{2}{\sqrt{6}}, -\dfrac{1}{\sqrt{6}}, \dfrac{1}{\sqrt{6}}$

10. Ans (A)

Ans. (A)

Let $\vec{a} = x\hat{i} + y\hat{j} + z\hat{k}$

$\therefore x^2 + y^2 + z^2 = 1$ ...(1)

also $\vec{a} \times \left(\hat{i} + \hat{j} + \hat{k}\right) = \hat{i} - \hat{k}$

$\Rightarrow \hat{i}(y - z) - \hat{j}(x - z) + \hat{k}(x - y) = \hat{i} - \hat{k}$

comparing, we get

$z = x;\ y = 1 + x$

$\therefore$ By (1) : $x = -\frac{2}{3};\ y = \frac{1}{3};\ z = -\frac{2}{3}$ or $x = z = 0,\ y = 1$

$\therefore \hat{a} = -\frac{2}{3}\hat{i} + \frac{1}{3}\hat{j} - \frac{2}{3}\hat{k}$

$\therefore \hat{a}.\hat{b} = -\frac{2}{3} + \frac{1}{3} - \frac{2}{3} = -\frac{3}{3} = -1$

MATHS

Section–I(ii) Multiple Correct Type

11. Ans (A,C)

Ans. (A,C)

$$\begin{vmatrix} 1 & 6 & 3 \\ 3 & 2 & 1 \\ \alpha + 1 & \beta - 1 & 1 \end{vmatrix} = 0 \Rightarrow \beta = 3$$

$|\vec{c}| = \sqrt{6} \Rightarrow a = 0$ or -2

12. Ans (A,C,D)

Ans. (A,C,D)

$a_1 + a_2\cos 2x + a_3 \sin^2 x = 0\ \forall\ x \in R$

$\Rightarrow (a_1 + a_2) + (a_3 - 2a_2)\sin^2 x = 0\ \forall\ x \in R$

$\therefore\ a_1 = -a_2 = -\frac{a_3}{2} = \lambda$

$\therefore\ \vec{a} = \lambda\hat{i} - \lambda\hat{j} - 2\lambda\hat{k}$

if $|\vec{a}| = \sqrt{6} \Rightarrow \lambda = \pm 1$

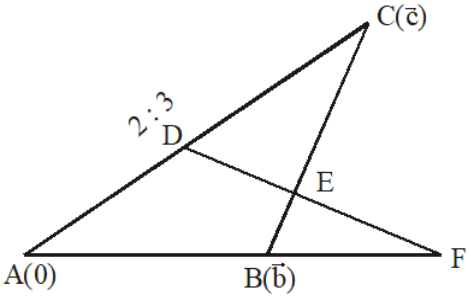
$\Rightarrow$ ordered triplet $(1, -1, -2)$ or $(-1, 1, 2)$

if $2a_1 + 3a_2 + 6a_3 = 26\ \ \lambda = -2$

$\Rightarrow |\vec{a}| = 2\sqrt{6}$

13. Ans (A,C)

Ans. (A,C)



$D\left(\frac{2\vec{c}}{5}\right)$ & $E\left(\frac{\vec{c} + 2\vec{b}}{3}\right)$

Let DE : EF = λ : 1 & AB : BF = μ : 1

$$\frac{\vec{c} + 2\vec{b}}{3} = \frac{\lambda\vec{f} + \frac{2\vec{c}}{5}}{\lambda + 1} \quad \text{.....(i)}$$

$$\& \ \vec{b} = \frac{\mu\vec{f}}{\mu + 1} \Rightarrow \vec{f} = \frac{\mu + 1}{\mu}\vec{b}$$

$$\frac{\vec{c} + 2\vec{b}}{3} = \frac{\frac{\lambda(\mu+1)\vec{b}}{\mu} + \frac{2\vec{c}}{5}}{\lambda + 1}$$

$$\frac{1}{3} = \frac{2}{5(\lambda + 1)} \quad \& \ 5\lambda + 5 = 6$$

$\lambda = \frac{1}{5}$ & $\mu = \frac{1}{3}$

14. Ans (A,B,C,D)

Ans. (A,B,C,D)

$$\cos\alpha = \frac{\vec{u}.\vec{v}}{|\vec{u}|.|\vec{v}|} = \frac{2x^2 - 3x + 1}{\sqrt{x^2 + 9 + 1}.\sqrt{4x^2 + x^2 + 1}} > 0$$

$\Rightarrow (2x - 1)(x - 1) > 0$

$\Rightarrow x \in \left(-\infty, \frac{1}{2}\right) \cup (1, \infty)$ .....(1)

$$\cos\beta = \frac{\vec{v}.\hat{j}}{|\vec{v}|} = \frac{x}{\sqrt{4x^2 + x^2 + 1}} < 0$$

$\Rightarrow x < 0$ .....(2)

from (1) & (2), $x < 0 \Rightarrow x^2 \in R^+$

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15. Ans (A,B,C)

Ans. (A,B,C)

$\bar{a}\times\bar{b}+\bar{b}=\bar{c}\times\bar{a}+\bar{b}\times\bar{c}$ .....(i)

taking dot product with $\bar{a}$ and $\bar{b}$ respectively

$\bar{a}.\bar{b}=[\bar{a}\,\bar{b}\,\bar{c}]$

& $|\bar{b}|^2=[\bar{a}\,\bar{b}\,\bar{c}]$

Let $|\bar{a}|=|\bar{b}|=k$

$\Rightarrow [\bar{a}\,\bar{b}\,\bar{c}]=k^2\cos\theta$

& $[\bar{a}\,\bar{b}\,\bar{c}]=k^2$

$\Rightarrow \cos\theta=1$ where $\theta=\bar{a}\wedge\bar{b}$

$\Rightarrow \bar{a}=\bar{b}$

putting in (i)

$\bar{a}=\bar{c}\times\bar{a}+\bar{a}\times\bar{c}$

$\Rightarrow \bar{a}=\bar{0}$

$\Rightarrow \bar{b}=\bar{0}$

and $\bar{c}$ can be any general vector

16. Ans (A,B,C)

Ans. (A,B,C)

Equation of P_1

$2(x-1)-2(y+1)+2(z-1)=0$

$\Rightarrow x-y+z-3=0$ 1

Equation of P_2 is

$(-2)(x-1)+3(y-2)+2(z-1)=0$

$\Rightarrow 2x-3y-2z+6=0$ 2

Vector along line of intersection of plane P_1 and P_2 is

$$\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & -1 & 1\\ 2 & -3 & -2\end{vmatrix}=5\hat{i}+4\hat{j}-\hat{k}$$

dr of line is 5, 4, −1

in (1) and (2) put z = 0, we get

x = 15, y = 12

Equation of line of intersection of P_1 and P_2 is

$\frac{x-15}{5}=\frac{y-12}{4}=\frac{z-0}{-1}$

$P_3: P_1+\lambda P_2=0$

$(x-y+z-3)+\lambda(2x-3y-2z+6)=0$

Passes through (1, 0, −1)

$\Rightarrow$ 16 x − 19y + 4z = 12

$\therefore$ length of $\perp$ from (0, 0, 0) to P_3 is $=\frac{12}{\sqrt{633}}$

MATHS

Section–I(iii) Comprehension Type

17. Ans (C)

Ans. (C)

Equation of line is $(\vec{r}-\vec{a})\times\vec{b}=\vec{0}\Rightarrow\vec{r}=\vec{a}+t\vec{b}$

$\Rightarrow \vec{r}=\hat{i}+2\hat{j}-3\hat{k}+t\left(2\hat{i}-\hat{j}+\hat{k}\right)$

Let M be the foot of $\perp$ from P(0,1,2) to the line

$\therefore \overline{PM}=(2t+1)\hat{i}+(1-t)\hat{j}+(t-5)\hat{k}$

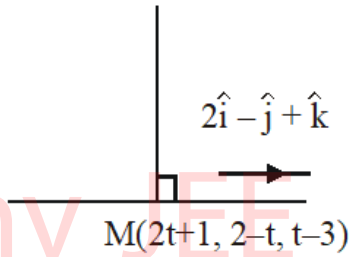
$\because \overline{PM}.\left(2\hat{i}-\hat{j}+\hat{k}\right)=0$

$2(2t+1)-1.(1-t)+1.(t-5)=0$

$4t+2-1+t+t-5=0$

$\Rightarrow 6t=4\Rightarrow t=\frac{2}{3}$

$\therefore$



$M=\left(\frac{7}{3},\frac{4}{3},-\frac{7}{3}\right)$ $\therefore$ Shortest distance = PM

$=\sqrt{\frac{49}{9}+\frac{1}{9}+\frac{169}{9}}=\frac{\sqrt{219}}{3}$

18. Ans (B)

Ans. (B)

Let $\vec{R}=x\hat{i}+y\hat{j}$ then curve 'C' is circle

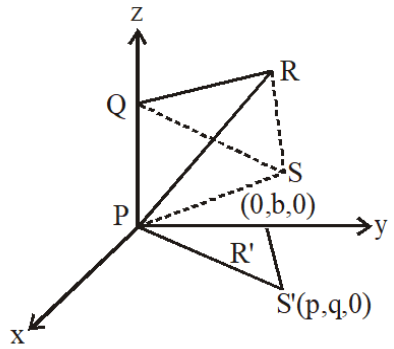
$x^2+y^2-2x+2y+1=0$ whose centre = (1,−1)

Now proceeding as above problem we can find minimum distance of line 'L' from the point (1,−1,0). Min. distance = $2\sqrt{3}$.

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19. Ans (B)

Ans. (B)



P(0, 0, 0); Q(0, 0, 4)

R(0, b, c); S(p, q, r)

P is same as P'

$\text{Ar}\left(\Delta P'R'S'\right) = \frac{1}{2}bp = 15$

$bp = 30$

$\text{Vol}(PQRS) = \frac{1}{3} \times (\text{Ar}PQR) \times \text{height}$

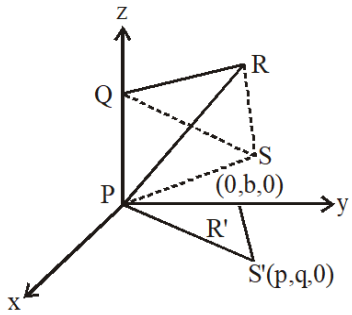
$= \frac{1}{3} \times \left(\frac{1}{2} \times 4 \times b\right) \times p$

$= \frac{1}{3} \times \frac{1}{2} \times 4 \times 30 = 20$ cu. units.

if $P = 5, b = 6$

20. Ans (B)

Ans. (B)



P(0, 0, 0); Q(0, 0, 4)

R(0, b, c); S(p, q, r)

P is same as P'

$\text{Ar}\left(\Delta P'R'S'\right) = \frac{1}{2}bp = 15$

$bp = 30$

$\text{Vol}(PQRS) = \frac{1}{3} \times (\text{Ar}PQR) \times \text{height}$

$= \frac{1}{3} \times \left(\frac{1}{2} \times 4 \times b\right) \times p$

$= \frac{1}{3} \times \frac{1}{2} \times 4 \times 30 = 20$ cu. units.

if $P = 5, b = 6$

21. Ans (C)

Ans. (C)

Let $P(\cos\theta, \sin\theta, 0)$ be a point on the circle then its distance d from the

$$\text{plane} = d = \left|\frac{3 - \cos\theta - \sin\theta}{\sqrt{3}}\right| = \left|\frac{3 - \sqrt{2}\sin\left(\theta + \frac{\pi}{4}\right)}{\sqrt{3}}\right|$$

for maxima, $\theta = \frac{-3\pi}{4}$

and for minima, $\theta = \frac{\pi}{4}$

$\Rightarrow$ A $\left(-\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0\right)$ and B $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$

$\Rightarrow |x_1 + y_1 - x_2 - y_2| = 2\sqrt{2}$

22. Ans (D)

Ans. (D)

Plane P is $x + y + z = 3$

get the image of $P(\cos\theta, \sin\theta, 0)$ in $x + y + z = 3$ which comes out as

$$P'\left(2 + \frac{\cos\theta - 2\sin\theta}{3}, 2 + \frac{\sin\theta - 2\cos\theta}{3}, 2 - \frac{2}{3}(\sin\theta + \cos\theta)\right)$$

i.e. $x = 2 + \frac{\cos\theta - 2\sin\theta}{3}$

$y = 2 + \frac{\sin\theta - 2\cos\theta}{3}$

and $z = 2 - \frac{2}{3}(\sin\theta + \cos\theta)$

$\Rightarrow$ eliminating θ, we get $2x + 2y - z = 6$

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MATHS

Section–I(iv) Matrix Match Type

23. Ans (A)

Ans. (A)

(P) $\vec{x} \times \vec{b} = \vec{c} \times \vec{b}$

$\Rightarrow \vec{x} - \vec{c} = t\vec{b}$, t is scalar

$\vec{x} = \vec{c} + t\vec{b}$

$\vec{x} = \vec{c} + t\vec{b}$

$\vec{x}.\vec{a} = 0 \Rightarrow t = -\left(\frac{\vec{c}.\vec{a}}{\vec{b}.\vec{a}}\right) = -\left(\frac{8+7}{2+1}\right) = 5$

$\vec{x} = \vec{c} - 5\vec{b}$

$\vec{x} - \vec{c} = -5\vec{b} = -5\left(\hat{i} + \hat{j} + \hat{k}\right)$

(Q) $\vec{x} \times \vec{a} + \vec{x} \times \vec{b} = 0$

$\vec{x} \times (\vec{a} + \vec{b}) = 0$

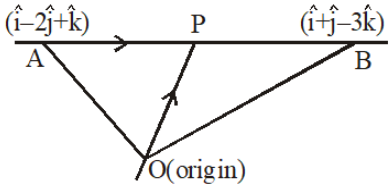
$\vec{x} = \mu(\vec{a} + \vec{b}) = \mu(3\hat{i} + 2\hat{j} - \hat{k})$

$|\vec{x}| = \mu\sqrt{14}$

$\left(\frac{\vec{x}}{|\vec{x}|}.\sqrt{14}\right) = \frac{3\hat{i} + 2\hat{j} - \hat{k}}{\sqrt{14}}$

(R) Let AP = 1

$\Rightarrow \overrightarrow{OP} = \overrightarrow{OA} \pm \overrightarrow{AP} = \overrightarrow{OA} \pm 1.\frac{\overrightarrow{AB}}{|\overrightarrow{AB}|}$



$= \hat{i} - 2\hat{j} + \hat{k} \pm \frac{3\hat{j} - 4\hat{k}}{5}$

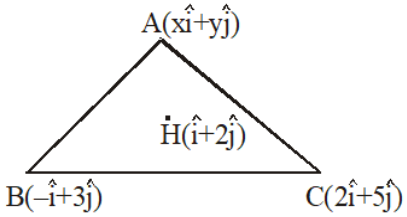
taking +ve sign $\frac{5\hat{i} - 7\hat{j} + \hat{k}}{5}$

(S) AH ⊥ BC $\Rightarrow$ $\overrightarrow{AH}.\overrightarrow{BC} = 0$

$3x + 2y = 7$ .....(1)

BH ⊥ CA

$\Rightarrow 2x - y = -1$ .....(2)



$\Rightarrow$

$x = \frac{5}{7}, \; y = \frac{17}{7}$

$\Rightarrow \; A = \frac{1}{7}\left(5\hat{i} + 17\hat{j}\right)$

24. Ans (A)

(P) Let D ($\vec{O}$), A ($\vec{a}$), B ($\vec{b}$), C ($\vec{c}$)

given volume of tetrahedron is $\left|\frac{1}{6}[\bar{a}\,\bar{b}\,\bar{c}]\right| = \frac{81}{2}$

$|[\bar{a}\,\bar{b}\,\bar{c}]| = 3^5$

Centroids by faces are $G_1\left(\frac{\vec{a}+\vec{b}+\vec{c}}{3}\right)$, $G_2\left(\frac{\vec{a}+\vec{b}}{3}\right)$, $G_3\left(\frac{\vec{b}+\vec{c}}{3}\right)$, $G_4\left(\frac{\vec{c}+\vec{a}}{3}\right)$

Volume of parallelopiped = $\left|[\overrightarrow{G_1G_2}\;\overrightarrow{G_1G_3}\;\overrightarrow{G_1G_4}]\right|$

$= \left|\left[-\frac{\vec{c}}{3} - \frac{\vec{a}}{3} - \frac{\vec{b}}{3}\right]\right| = \left|\frac{[\vec{a}\,\vec{b}\,\vec{c}]}{3^3}\right| = 9$

(Q) Let the image of the point (1, 0, 1) in the plane $x - y - z = 1$ is A′(a,b,c)

D.R. of AA′ are (a − 1, b, c − 1)

D. R. of normal are (1, −1, −1)

$\frac{a-1}{1} = \frac{b}{-1} = \frac{c-1}{-1} = \lambda \Rightarrow a = \lambda + 1, b = -\lambda, c = 1 - \lambda$

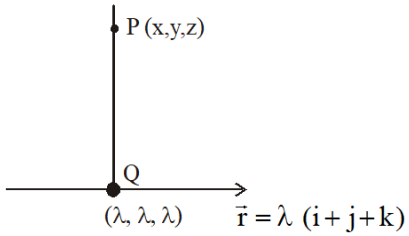
mid point of AA′ lie on plane $x - y - z = 1$

$\Rightarrow \lambda = \frac{2}{3}$

$a = \frac{5}{3}, \; b = \frac{-2}{3}, \; c = \frac{1}{3} \Rightarrow 3(a - b + c) = 3\left(\frac{8}{3}\right) = 8$

(R) $\vec{QP}.(i + j + k) = 0 \Rightarrow (x - \lambda) + (y - \lambda) + (z - \lambda) = 0$

$\lambda = \frac{x+y+z}{3}$



$(PQ)^2 = 9 \Rightarrow (x - \lambda)^2 + (y - \lambda)^2 + (z - \lambda)^2 = 9$

$x^2 + y^2 + z^2 + 3\lambda^2 - 2\lambda(x + y + z) = 9$

$x^2 + y^2 + z^2 + 3\lambda^2 - 2\lambda.\; 3\lambda = 9$

$\Rightarrow x^2 + y^2 + z^2 - 3\left(\frac{x+y+z}{3}\right)^2 = 9$

$\Rightarrow x^2 + y^2 + z^2 - xy - yz - zx = \frac{27}{2}$

$k = 27, \quad \frac{k}{9} = 3$

(S) $\vec{C} = \lambda(\bar{a} \times \bar{b})$ also $\left|\frac{1}{6}[\bar{a}\,\bar{b}\,\bar{c}]\right| = 1 \Rightarrow |[\bar{a}\,\bar{b}\,\bar{c}]| = 6$

$\Rightarrow |(\vec{a} \times \bar{b}).\vec{c}| = 6 \Rightarrow \frac{|c|^2}{|\lambda|} = 6 \Rightarrow 6 = 6|\lambda| \Rightarrow |\lambda| = 1$

$\therefore |c|^2 = \lambda^2|a|^2|b|^2\sin^2\theta \Rightarrow 6 = 12\sin^2\theta = \theta = \frac{3\pi}{4}(a.\,b < 0)$

$p + q = 7$

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MATHS

Section–II (iii) Numerical Type

1. Ans (1.25)

Ans. 1.25

$\vec{r}.\left(\vec{a}+\vec{b}+\vec{c}\right)=0$

$\Rightarrow (\sin x+\cos y+2)\left[\vec{a}\ \vec{b}\ \vec{c}\right]=0$

$\Rightarrow \sin x+\cos y+2=0$

$\Rightarrow \sin x=\cos y=-1$

$\Rightarrow$ we take $x=-\frac{\pi}{2}$, $y=p$ so that x^2+y^2 obtain its minimum value.

$\therefore$ Minimum value of $\frac{(x^2+y^2)}{\pi^2}=\frac{\frac{\pi^2}{4}+\pi^2}{\pi^2}=\frac{5}{4}=1.25$

2. Ans (1.00)

Ans. 1.00

Let $\vec{a}=x\hat{i}+y\hat{j}+z\hat{k}$

$\therefore x^2+y^2+z^2=1$ ...(1)

also $\vec{a}\times\left(\hat{i}+\hat{j}+\hat{k}\right)=\hat{i}-\hat{k}$

$\Rightarrow \hat{i}(y-z)-\hat{j}(x-z)+\hat{k}(x-y)=\hat{i}-\hat{k}$

comparing, we get

$z=x;\ y=1+x$

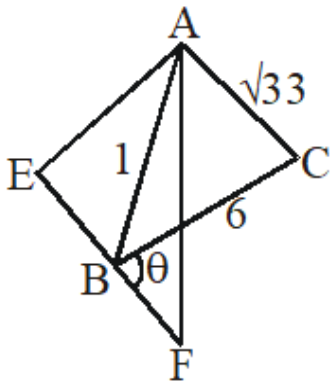
$\therefore$By (1) : $x=-\frac{2}{3};\ y=\frac{1}{3};\ z=-\frac{2}{3}$ or $x=z=0,\ y=1$

$\therefore \hat{a}=-\frac{2}{3}\hat{i}+\frac{1}{3}\hat{j}-\frac{2}{3}\hat{k}$

$\therefore \hat{a}.\hat{b}=-\frac{2}{3}+\frac{1}{3}-\frac{2}{3}=-\frac{3}{3}=-1$

3. Ans (2.00)

Ans. 2.00



$\overrightarrow{AB}.\overrightarrow{AE}+\overrightarrow{AC}.\overrightarrow{AF}=2$

$\overrightarrow{AB}.\left(\overrightarrow{AB}+\overrightarrow{BE}\right)+\overrightarrow{AC}.\left(\overrightarrow{AB}+\overrightarrow{BF}\right)=2$

$\left|\overrightarrow{AB}\right|^2+\overrightarrow{AB}.\overrightarrow{BE}+\overrightarrow{AC}.\overrightarrow{AB}+\overrightarrow{AC}.\overrightarrow{BF}=2$ ...(1)

$\overrightarrow{AB}+\overrightarrow{CA}=-\overrightarrow{BC}$

$\left|\overrightarrow{AB}+\overrightarrow{CA}\right|=\left|-\overrightarrow{BC}\right|$

$\Rightarrow 1+33-2\overrightarrow{AB}.\overrightarrow{AC}=36$

$\overrightarrow{AB}.\overrightarrow{AC}=-1$

(1) $\Rightarrow \overrightarrow{AB}.\overrightarrow{BE}+\overrightarrow{AC}.\overrightarrow{BF}=2$

$\left(-\overrightarrow{AB}+\overrightarrow{AC}\right).\overrightarrow{BF}=2\quad\left[\because \overrightarrow{BE}=-\overrightarrow{BF}\right]$

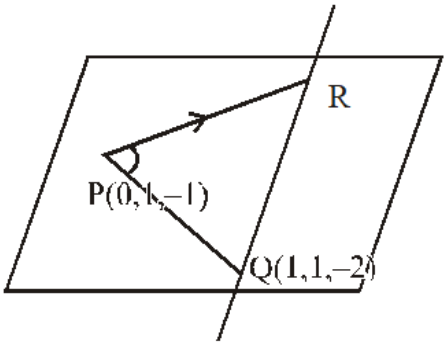
$\left(\overrightarrow{BA}+\overrightarrow{AC}\right).\overrightarrow{BF}=2$

$\overrightarrow{BC}.\overrightarrow{BF}=2$

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4. Ans (10.80)

Ans. 10.80



$\frac{x-1}{2} = \frac{y-1}{3} = \frac{z+2}{1} = \lambda$

$\therefore$ R$(2\lambda + 1, 3\lambda + 1, \lambda - 2)$

$\overrightarrow{PR}.\overrightarrow{PQ} = 0$

$((2\lambda+1)\hat{i} + 3\lambda\hat{j} + (\lambda-1)\hat{k}).(\hat{i} - \hat{k}) = 0$

$2\lambda + 1 - \lambda + 1 = 0 \quad \Rightarrow \quad \lambda = -2$

$\therefore$ R is $(-3, -5, -4)$

$PR = \sqrt{9+36+9} = \sqrt{54}$

5. Ans (0.85 or 0.86)

Ans. 0.85 or 0.86

Let M being origin then one possible equation of face LNO will be $\Rightarrow x + \frac{y}{2} + \frac{z}{3} = 1$

$\therefore$ required altitude = distance of LNO from M = $\frac{6}{7}$

6. Ans (4.50)

Ans. 4.50

equation of P_3 is $P_1 + \lambda P_2 = 0$

$\Rightarrow (x + y - 2z - 1) + \lambda(\alpha x - y + z - 2) = 0$

$\because$ P and Q lie on P_3

$\Rightarrow (\alpha - 6) + \lambda(\alpha^2) = 0$

and $\left(\frac{5}{3} - 2\alpha\right) + \lambda\left(\frac{5}{3}\alpha - 3 + \alpha\right) = 0$

$\Rightarrow 2\alpha^3 + \alpha^2 - 19\alpha + 18 = 0$

$\Rightarrow$ a = 2 is only rational value.

$\Rightarrow$ equation of planes are

$P_1 : x + y - 2z = 1$ and $P_2 : 2x - y + z = 2$

vector along L is

$$\vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -2 \\ 2 & -1 & 1 \end{vmatrix} = -\hat{i} - 5\hat{j} - 3\hat{k}$$

$\Rightarrow$ its direction cosines are

$$\pm\left(\frac{1}{\sqrt{35}}, \frac{5}{\sqrt{35}}, \frac{3}{\sqrt{35}}\right)$$

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ANSWER KEY

(COMPLEX NUMBER) GUIDED REVISION # 3

TIME : 60 Min.

M.M. : 60

SOLUTIONS

MATHS

Section–I(i)

1. Ans (D)

Ans. (D)

$(1+z)^6 + z^6 = 0$

$(1+z)^6 = -z^6$

$\left(\frac{z+1}{z}\right)^6 = -1$

$\left|\frac{z+1}{z}\right|^6 = 1$

$|z+1| = |z|$

Thus : z lies on perpendicular bisector of line segment joining (−1,0) and (0,0)

Hence : all roots lie on line and arc collinear.

2. Ans (B)

Ans. (B)

$P = \cos\frac{2\pi}{7} + i\sin\frac{2\pi}{7}$

$P = e^{\frac{2\pi}{7}i} \Rightarrow P^7 = \left(e^{\frac{2\pi}{7}i}\right) = e^{2\pi i}$

$a + b = P + P^2 + P^4 + P^3 + P^5 + P^6$

$a + b + 1 = 1 + P + P^2 + ... + P^6$

$= \frac{P^7 - 1}{P - 1}\{P^7 = 1\}$

$a + b + 1 = 0$

$\therefore a + b = -1$

$ab = (P + P^2 + P^4)(P^3 + P^5 + P^6)$

$= P^4 + P^6 + P^7 + P^5 + P^7 + P^8 + P^7 + P^9 + P^{10}$

$= P^4 + P^6 + P^5 + P^8 + P^9 + P^{10} + 3$

$P^4 + P^5 + P^6 + P^7(P + P^2 + P^3) + 3$

$P + P^2 + ... + P^6 + 3$

$-1 + 3 = 2$

$\therefore |(a+b) + abi| = |-1 + 2i| = \sqrt{(-1)^2 + (2)^2} = \sqrt{5}$

3. Ans (D)

Ans. (D)

since $Im(z_1) + Im(z_2) + Im(z_3) = Im(z_1 + z_2 + z_3)$

$= Im(\text{sum of roots})$

$= Im(1 + 3i) = 3$

Alternate :

$(z^3 - z^2 - 2z + 2) + 3i(z - z^2) = 0$

$(z-1)(z^2 - 2) + 3iz(1 - z) = 0$

$(z-1)(z^2 - 2 - 3iz) = 0$

$z = 1,\ z = 2i, z = i$

$Im(z_1) + Im(z_2) + Im(z_3) = 3.$

4. Ans (B)

Ans. (B)

$|z - 2i| = 2$ is a circle with centre 2i

$z(1 - i) - \bar{z}(1 + i) = 4i$ is a line

This line passes through center of circle.

$\therefore$ Total 2 points of intersection.

5. Ans (D)

Ans. (D)

$\sum_{k=1}^{100} i^{k!} + \sum_{k=1}^{100} \omega^{k!}$

$\sum_{k=1}^{100} i^{k!} = i^{1!} + i^{2!} + i^{3!} + i^{4!} + ... + i^{100!}$

$= i - 1 + i^6 + 1 + 1 + 1 + ... + 1 \quad = i - 2 + 97$

$= i + 95.$

$\sum_{k=1}^{100} \omega^{k!} = \omega^1 + \omega^{2!} + \omega^{3!} + \omega^{4!} + ... + \omega^{100!}$

$= \omega + \omega^2 + 1 + 1 + 1 + ... + 1 = 97$

sum = i + 95 + 97 = i + 192

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6. Ans (D)

Ans. (D)

$$\arg\left(\frac{z_1}{z_4}\right)+\arg\left(\frac{z_2}{z_3}\right)=\arg\left(\frac{z_1z_2}{z_3z_4}\right)$$

$$\arg\left(\frac{z_1\bar{z}_1}{z_3\bar{z}_3}\right)=\arg\left(\frac{|z_1|^2}{|z_3|^2}\right)=0$$

7. Ans (D)

Ans. (D)

$z = x + iy$

$|(x-3)+(y-2)i| = |x+(y+2)i|$

$(x-3)2+(y-2)^2 = x^2+(y+2)^2$

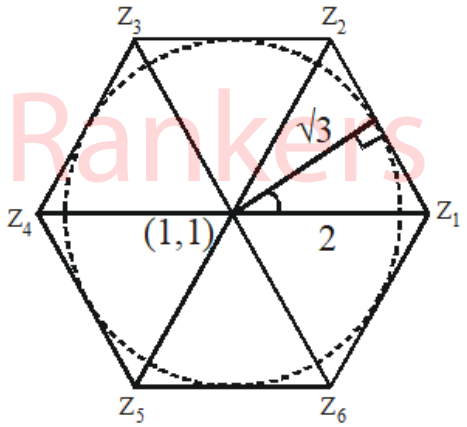
$6x + 8y = 9$

$|z|$ means distance from origin

$$\text{So } |z|_{\min.} = \left|\frac{0+0-9}{\sqrt{6^2+8^2}}\right| = \frac{9}{10}$$

8. Ans (C)

Ans. (C)



$\left(2+\left(1+\sqrt{3}\right)i\right)$

$z_2-(1+i) = \left(2+\left(1+\sqrt{3}\right)i-(1+i)\right)e^{i\pi/3}$

$\Rightarrow \quad z_2-(1+i) = \left(1+i\sqrt{3}\right)e^{i\pi/3}$

$$= \left(1+i\sqrt{3}\right)\left(\frac{1+\sqrt{3}i}{2}\right) = -1+\sqrt{3}i$$

$\Rightarrow z_2 = \left(1+\sqrt{3}\right)i$

similarly using concept of rotation find z_6 and z_5

$\Rightarrow z_6 = 3+i$

$z_5 = 2+\left(1-\sqrt{3}\right)i$

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MATHS

Section–I(ii)

9. Ans (A,B,C,D)

Ans.(A,B,C,D)

$|2z_1-3z_2-4z_3| = |2z_1-3(z_2-2)-4(z_3-5)-26|$

$\le 2|z_1|+3|z_2-2|+4|z_3-5|+|26| \le 2+9+24+26 \le 61$

10. Ans (B,D)

Ans.(B,D)

$zw + w^2 = \bar{z}$

$1 + zw = w^2\bar{z}$

$\Rightarrow \bar{z}-w^2 = \bar{z}w^2-1 \quad \Rightarrow (\bar{z}+1)\left(1-w^2\right)=0$

$\Rightarrow \bar{z} = -1 \quad \text{or } w^2 = 1$

Case I : $\bar{z} = -1 \Rightarrow \arg z = \pi$

$$\Rightarrow \quad -1+w = -\frac{1}{w} \Rightarrow w^2-w+1=0$$

$$\Rightarrow \quad \arg(w) = -\frac{\pi}{3} \text{ or } \frac{\pi}{3}$$

Case II : $w = -1$

$$\Rightarrow \quad z-1 = -\bar{z} \quad \Rightarrow \quad \text{Re}(z) = \frac{1}{2}$$

11. Ans (B,C)

Ans. (B,C)

$$\because |z_3| = \frac{|z_2|.|z_1-z_4|}{|\bar{z}_1z_4-1|} = \frac{|z_1-z_4|}{|\bar{z}_1z_4-1|} \leqslant 1$$

$\Rightarrow |z_1-z_4|^2 \le |\bar{z}_1z_4-1|^2$

$\Rightarrow (z_1-z_4)(\bar{z}_1-\bar{z}_4) \leqslant (\bar{z}_1z_4-1).(z_1\bar{z}_4-1)$

$\Rightarrow |z_1|^2+|z_4|^2-|z_1|^2|z_4|^2-1 \le 0 \Rightarrow (|z_1|^2-1)(|z_4|^2-1) \ge 0$

$\because |z_1| < 1 \Rightarrow |z_4| \le 1$

MATHS

Section–I(iii)

12. Ans (C)

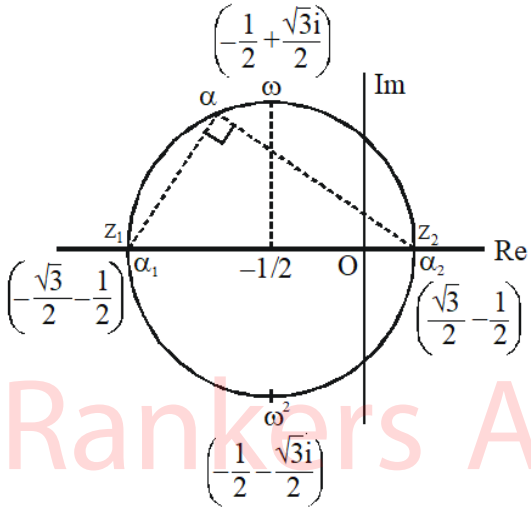
$(z-\omega)^2+(z-\omega^2)^2=0$

$\Rightarrow 2z^2-2z(\omega+\omega^2)+\omega^4+\omega^2=0$

$\Rightarrow z_1=-\frac{\sqrt{3}}{2}-\frac{1}{2};\ z_2=\frac{\sqrt{3}}{2}-\frac{1}{2}$

α lies on circle with centre $-\frac{1}{2}$ and radius $\frac{\sqrt{3}}{2}$

$\alpha_1=-\frac{\sqrt{3}}{2}-\frac{1}{2}$; $\alpha_2=\frac{\sqrt{3}}{2}-\frac{1}{2}$



Ans. (C)

z_1, z_2 and ω, ω^2 are diametrical ends of given circle

$\therefore |z_1-\alpha|^2+|z_2-\alpha|^2+|\omega-\alpha|^2+|\omega^2-\alpha|^2$

$=(\sqrt{3})^2+(\sqrt{3})^2=6$

13. Ans (B)

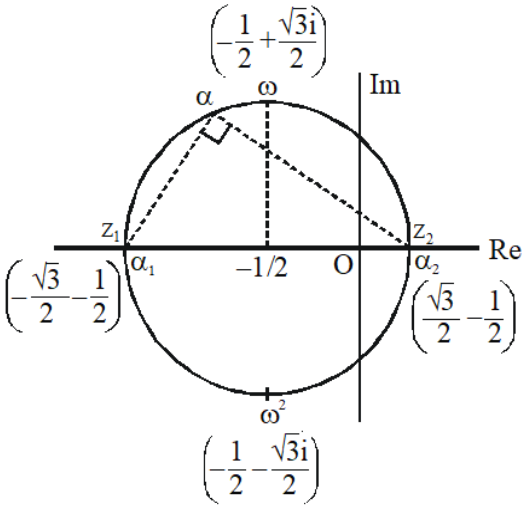
$(z-\omega)^2+(z-\omega^2)^2=0$

$\Rightarrow 2z^2-2z(\omega+\omega^2)+\omega^4+\omega^2=0$

$\Rightarrow z_1=-\frac{\sqrt{3}}{2}-\frac{1}{2};\ z_2=\frac{\sqrt{3}}{2}-\frac{1}{2}$

α lies on circle with centre $-\frac{1}{2}$ and radius $\frac{\sqrt{3}}{2}$

$\alpha_1=-\frac{\sqrt{3}}{2}-\frac{1}{2}$; $\alpha_2=\frac{\sqrt{3}}{2}-\frac{1}{2}$



Ans. (B)

$\alpha_1=z_1$ and $\alpha_2=z_2$

$\therefore$ (B) is correct.

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14. Ans (C)

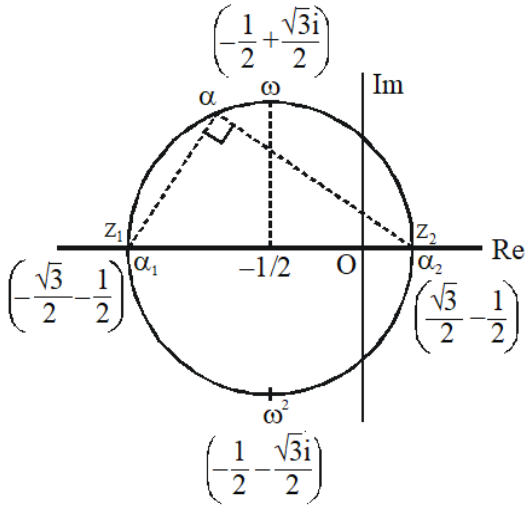
$(z-\omega)^2+(z-\omega^2)^2=0$

$\Rightarrow 2z^2-2z(\omega+\omega^2)+\omega^4+\omega^2=0$

$\Rightarrow z_1=-\frac{\sqrt{3}}{2}-\frac{1}{2};\ z_2=\frac{\sqrt{3}}{2}-\frac{1}{2}$

α lies on circle with centre $-\frac{1}{2}$ and radius $\frac{\sqrt{3}}{2}$

$\alpha_1=-\frac{\sqrt{3}}{2}-\frac{1}{2}\ ;\ \alpha_2=\frac{\sqrt{3}}{2}-\frac{1}{2}$



Ans. (C)

$|\alpha-\alpha_1|+|\alpha-\alpha_2|=\sqrt{6}$

Here α moves on an ellipse having its focii as α_1 and α_2 and length of minor axis as $\sqrt{3}$ units.

$\therefore$ 2 points for $|\alpha-\alpha_1|+|\alpha-\alpha_2|=\sqrt{6}$

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MATHS

Section–II (iii)

1. Ans (4.50)

Ans. 4.50

Let $z=x+iy$

$\Rightarrow |x-2+iy|=3|x+iy+2|$

$\Rightarrow (x-2)^2+y^2=9(x+2)^2+9y^2$

$\Rightarrow 8x^2+8y^2+40x+32=0$

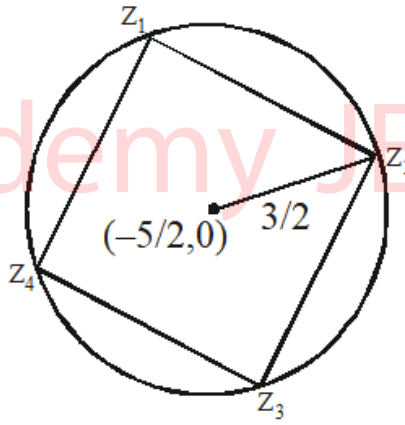
$\Rightarrow x^2+y^2+5x+4=0$

Clearly locus is a circle with centre $\left(-\frac{5}{2},0\right)$ and radius

$=\sqrt{\frac{24}{4}-4}=\frac{3}{2}$

since $z_1,z_2,z_3,z_4 \in A$

$\Rightarrow$ all these points must lie on this circle and forms a parallelogram, since $z_1+z_3=z_2+z_4$



Now to have maximum area of parallelogram, it must be a square whose area will be

$\frac{d^2}{2}=\frac{9}{2}$

2. Ans (1.00)

$\alpha=\frac{1\pm\sqrt{3}\,i}{2}\ \Rightarrow\ \alpha=-\omega\ \text{ or }\ -\omega^2$

$\therefore\ (-\omega)^{333}+(-\omega)^{666}+(-\omega)^{999}=-1+1-1=-1$

3. Ans (5.00)

Ans. 5.00

$z_1 = 4 + 3i$

After moving $\sqrt{2}$ units on the direction of $-i + \hat{j}$

$z_2 = (4 - 1) + (3 + 1)\,\hat{i}$

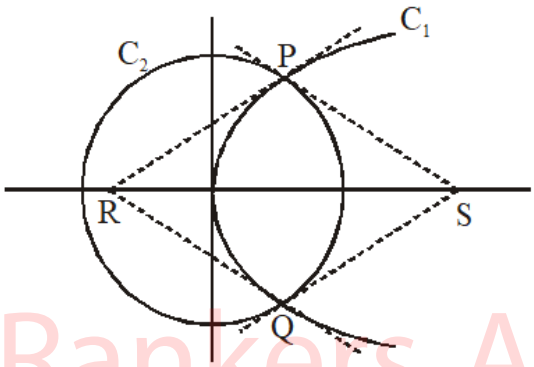
$= 3 + 4i$

z_2 is rotated through an angle $\frac{\pi}{2}$ in clockwise direction

so $p + iq = (3 + 4i)\, e^{-i\frac{\pi}{2}} = (3 + 4i)\,(-i) = 4 - 3i$

4. Ans (100.00)

Ans. 100.00



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Put $z = x + iy$

$C_1 : y^2 = 8x$

$C_2 : x^2 + y^2 = 9$

On solving C_1 & C_2

$P(1,\ 2\sqrt{2})$

Tangent to C1 at P : $y(2\sqrt{2}) = 8\frac{(x + 1)}{2}$

at $y = 0 : x = -1 \Rightarrow R(-1, 0)$

Tangent to C_2 at P : $x(1) + y(2\sqrt{2}) = 9$

at $y = 0 : x = 9 \Rightarrow S(9, 0)$.

$\therefore$ Area $\Delta PRS : \frac{1}{2} \times (9 - (-1)) \times 2\sqrt{2} = 10\sqrt{2}$

$\therefore \lambda = 10 \Rightarrow \lambda_2 = 100$

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ANSWER KEY | **(P & C, BT) GUIDED REVISION # 4**

TIME : 60 Min. | **M.M. : 60**

SOLUTIONS

MATHS

Section–I (i)

1. Ans (B)

Ans. (B)

Clearly $9 - x \geq 0,\ x - 1 \geq 0$ & $9 - x - (x - 1) \geq 0$

$\Rightarrow 1 \leq x \leq 5,\ x \in I$

$\therefore m = 5$

Range of $f(x)$ $\Rightarrow$ Domain = {1,2,3,4,5}

{1, 7, 15, 10}

$\therefore \quad n = 4$

2. Ans (B)

Ans. (B)

Case-I :

a, b, c having exactly one 5 with :

(i) 2 even digits (different or same) : ${}^4C_2.3! + {}^4C_1.\frac{3!}{2!} = 48$

(ii) one digit divisible by 4 and one odd : ${}^2C_1.{}^4C_1.\ 3! = 48$

Case-II :

a, b, c having exactly two 5 with :

(i) one digit divisible by 4 : $2\ .\ \frac{3!}{2!} = 6$

Hence number of ways = 102

3. Ans (C)

Ans. (C)

A — A — A

Separate A's by gap method

$$\frac{\lfloor 4}{\lfloor 2\ \lfloor 2} \times {}^5C_3 = 60 \qquad ...(1)$$

A — A — A N N

$$\frac{\lfloor 3}{\lfloor 2} \times {}^4C_3 = 12 \qquad ...(2)$$

(1) – (2) $\Rightarrow$ A – A –A and N – N = 60 – 12 = 48

T T A – A –A

$$\frac{\lfloor 3}{\lfloor 2} \times {}^4C_3 = 12 \qquad ...(3)$$

T T A – A –A N N

$$\lfloor 2 \times {}^3C_2 = 2 \qquad ...(4)$$

(3) – (4) $\Rightarrow$ T T A – A –A N N

12 – 2 = 10

$\Rightarrow$ 48 – 10 = 38

4. Ans (C)

Ans. C

$(e^x - 1)^2 = e^{2x} - 2e^x + 1 \qquad e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!}$

$e^{2x} = 1 + \frac{2x}{1!} + \frac{(2x)^2}{2!} + \frac{(2x)^3}{3!} + \frac{(2x)^4}{4!}$

co-efficient of $x^4 = \frac{2^4}{4!} - \frac{2}{4!} = \frac{7}{12}$

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5. Ans (B)

Ans. B

k^{th} term $\longrightarrow$ $T_k = (k^2 + 1)\, k\,!$

$(k^2 + 1) = k^2 + k - (k - 1) = k(k + 1) - (k - 1)$

$T_k = (k^2 + 1)\, k\,! = k(k + 1)\,! - (k - 1)\, k\,!$

$T_1 = 1(1 + 1)\,! - 0 \text{ ´ } 1\,!$

$T_2 = 2(2 + 1)\,! - 1 \text{ ´ } 2\,!$

$T_3 = 3(3 + 1)\,! - 2 \text{ ´ } 3\,!$

⋮

$T_n = n(n + 1)\,! - (n - 1)\, n\,!$

Add, we get $\Sigma\ T_k = n(n + 1)\,!$

6. Ans (C)

Ans. (C)

Let required sum = S = coeff. of x^n in

$(1+x)^{3n} - {}^nC_1(1+x)^{3n-3} + {}^nC_2(1+x)^{3n-6} - {}^nC_3(1+x)^{3n-9} + \ldots\ldots$

= Coeff. of x^n in $\{(1+x)^3 - 1\}^n$

= Coeff. of x^n in $x^n (x^2 + 3x + 3)n$

= 3n

7. Ans (A)

Ans. (A)

coefficient of x^4

${}^{15}C_4 + {}^{16}C_4 + {}^{17}C_4 + \ldots. + {}^{15}C_4$

${}^{15}C_5 + {}^{15}C_4 + {}^{16}C_4 + \ldots. + {}^{15}C_4 - {}^{15}C_5$

${}^{16}C_5 + {}^{16}C_4 + \ldots\ldots + {}^{15}C_4 - {}^{15}C_5$

${}^{17}C_5 + \ldots\ldots\ldots + {}^{50}C_4 - {}^{15}C_5$

$\Rightarrow {}^{51}C_5 - {}^{15}C_5$

MATHS

Section–I (ii)

8. Ans (B,C)

Ans. (B,C)

$$\sum_{r=0}^{n} \frac{r}{{}^nC_r} = \sum_{r=0}^{n} \frac{n-r}{{}^nC_{n-r}} = \frac{n}{2}\sum_{r=0}^{n} \frac{1}{{}^nC_r} \quad \Rightarrow$$

$$\frac{n}{2} = \frac{n^2 - 4n + 6}{2} \quad \Rightarrow \quad n = n^2 - 4n + 6$$

$\Rightarrow n^2 - 5n + 6 = 0 \quad \Rightarrow \quad n = 2, 3$

9. Ans (A,C)

Ans. (A,C)

$\frac{m-1}{2}$ = integer $\Rightarrow$ m = 2(integer) + 1

$\therefore$ m must be odd

Number of ways of formation of subsets B and C is

$= {}^mC_1 \cdot {}^{m-1}C_{\frac{m-1}{2}} = m.\ {}^{(m-1)}C_{\frac{m-1}{2}}$

(${}^mC_1 \longrightarrow$ Number of selection of common element,

${}^{m-1}C_{\frac{m-1}{2}} \longrightarrow$ Distribution of (m − 1) elements into two equal parts to B & C

10. Ans (A,B,C,D)

Ans. (A,B,C,D)

Number of terms in the expansion of

$(x_1 + x_2 + \ldots\ldots x_r)^n$ is ${}^{n+r-1}C_{r-1}$

(A) Number of terms in $(2x - 3y + z)^{10}$ is ${}^{10+3-1}C_{3-1}$=66

(B) Put $x = \frac{1}{2}$, $y = -\frac{1}{3}$ & z=1 we get

$$\sum_{\substack{0\leq\alpha,\beta,\gamma\leq 10\\ \alpha+\beta+\gamma=10}} \frac{10!}{\alpha!\beta!\gamma!} = 3^{10}$$

(C) Cofficient of $x^2y^3z^5 = \frac{10!}{2!3!5!}\, 2^2.(-3)^3.(1)^5$

$= -\frac{10!(9)}{5!}$

(D) Put x=1, y=1 & z=1

$$\sum_{\substack{0\leq\alpha,\beta,\gamma\leq 10\\ \alpha+\beta+\gamma=10}} \frac{10!}{\alpha!\beta!\gamma!} 2^{\alpha}(-3)^{\beta} = 0$$

11. Ans (A,B,D)

Ans. (A,B,D)

* at least one fruit is chosen.

(4 + 1) (3 + 1) (2 + 1) (2 + 1) − 1 = 179

* at least 2 fruits are chosen.

(4 + 1) (3 + 1) (2 + 1) (2 + 1) − 1 − (4) = 175

* at most 2 fruits can be chosen .

selection of No fruit ⇒ 1 method

selection of 1 fruit ⇒ 4 method.

selection of 2 fruits ⇒ $4 + {}^4C_2 = 10$

= 1 + 4 + 10 = 15

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MATHS

Section–I (iii)

12. Ans (D)

Ans. (D)

A(1, 2) indicates k = 2

The man has to climb 6 steps

(i) 1, 1, 1, 1, 1, 1 $= {}^6C_0$ ways = 1

(ii) 1, 1, 1, 1, 2 $= {}^5C_1$ ways = 5

(iii) 1, 1, 2, 2 $= {}^4C_2$ ways = 6

(iv) 2, 2, 2 $= {}^3C_3$ ways = 1

$\therefore$ Total ways = 13

13. Ans (C)

Ans. (C)

A(1, k) denotes the number of ways in which the man has to climb either 1 step or k steps at a time

He has to climb 3k steps

(i) $\underbrace{1, 1, 1, \ldots\ldots, 1}_{\text{3k times}}$ = 1 way

(ii) $\underbrace{k, 1, 1, \ldots\ldots, 1}_{\text{2k times}} = {}^{2k+1}C_1$ ways

(iii) $\underbrace{k, k, 1, 1, 1, \ldots\ldots, 1}_{\text{k times}} = {}^{k+2}C_2$ ways

(iv) k, k, k = 1 way

$\therefore$ Total number of ways

$\Rightarrow 1 + {}^{2k+1}C_1 + {}^{k+2}C_2 + 1 = 26$

$\therefore$ k = 4

14. Ans (A)

Ans. (A)

$\because \lambda_k = A(1, k+1) - A(1, k)$

$= 1 + {}^{2k+3}C_1 + {}^{k+3}C_2 + 1 - \left(1 + {}^{2k+1}C_1 + {}^{k+2}C_2 + 1\right)$

$= 2k + 3 + \frac{(k+3)(k+2)}{2} - (2k + 1) - \frac{(k+2)(k+1)}{2}$

$= 2 + \frac{k+2}{2}(k + 3 - k - 1)$

$\lambda_k = k + 4$

$\lambda_{k+1} - \lambda_k = k + 5 - (k + 4) = 1$

MATHS

Section–II

1. Ans (5.00)

Ans. 5.00

Let $\sum_{r=0}^{10} {}^{10}C_r . (r - 5)^3 = S$

replace r by 10 – r

$\Rightarrow \sum_{r=10}^{0} {}^{10}C_{10-r} . (10 - r - 5)^3 = S \quad \Rightarrow \quad \sum_{r=0}^{10} {}^{10}C_r (5 - r)^3 = S$

$\Rightarrow \quad 2S = 0 \quad \Rightarrow \quad S = 0$

$\Rightarrow \sum_{r=0}^{10} {}^{10}C_r (r - 5)^3 = 0$

$\Rightarrow {}^{10}C_0(-5)^3 + \sum_{r=1}^{10} {}^{10}C_r (r - 5)^3 = 0$

$\Rightarrow \sum_{r=1}^{10} {}^{10}C_r (r - 5)^3 = 125$

2. Ans (0.00)

Ans. 0.00

$(57)^{n+3} = (5 \times 11 + 2)^{n+3}$ = a multiple of $11 + 2^{n+3}$

$(79)^{n+1} = (7 \times 11 + 2)^{n+1}$ = a multiple of $11 + 2^{n+1}$

$(46)^n = (4 \times 11 + 2)^n$ = a multiple of $11 + 2^n$

$(57)^{n+3} + (79)^{n+1} + (46)^n = 11k + 2^n(2^3 + 2^1 + 1)$ for some $k \in N$

$= 11k + 11.2^n = 11(k + 2^n)$

Therefore $(57)^{n+3} + (79)^{n+1} + (46)^n$ is divisible by 11.

3. Ans (196.00)

Ans. 196.00

Number of ways = No condition – (Two friends always together)

$= {}^{10}C_5 - {}^8C_3 = 252 - 56 = 196$

4. Ans (8.00)

Ans. 8.00

$\because$ $10! = 2^8 . 3^4 . 5^2 . 7$ and

$\because$ the divisors are odd and not a multiple of 3

$\therefore$ the number of such divisors = (2 + 1) . (1 + 1) = 6

and they are 1, 5, 7, 25, 35, 175

out of them 5, 35 are of type 3n – 1 and 1, 7, 25, 175 are of type 3n + 1

$\Rightarrow S_1 = 208, S_2 = 40$

$\Rightarrow$ remainder obtained by dividing S_1 by S_2 is 8.

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GUIDED REVISION

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ANSWER KEY

(Probability) GUIDED REVISION # 5

TIME : 60 Min.

M.M. : 72

SOLUTIONS

MATHS

Section–I (i)

1. Ans (C)

Ans. (C)

E : none of the drawn balls are green

E_1 : 2R balls are drawn

E_2 : 2W balls are drawn

E_3 : 1R + 1W balls are drawn

$P(E_1/E)$

$$= \frac{P(E_1).P(E/E_1)}{P(E_1)P(E/E_1)+P(E_2).P(E/E_2)+P(E_3).P(E/E_3)}$$

$$= \frac{\frac{{}^3C_2}{{}^7C_2}\times 1}{\frac{{}^3C_2}{{}^7C_2}\times 1+\frac{{}^2C_2}{{}^7C_2}\times 1+\frac{{}^3C_1\times{}^2C_1}{{}^7C_2}} = \frac{3}{10}$$

2. Ans (C)

Ans. (C)

$P(A) - P(A \cap B) = \frac{1}{3}$ &

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{11}{15}$

$\Rightarrow P(B) = \frac{6}{15} = \frac{2}{5}$

$P(A) - P(A)\,P(B) = \frac{1}{3} \Rightarrow P(A) = \frac{5}{9}$

$\Rightarrow P(A \cap B) = P(A)\,P(B) = \frac{2}{5}\times\frac{5}{9} = \frac{2}{9}$

3. Ans (B)

Ans. (B)

$\because \quad \{P\} = P \Rightarrow P \in [0, 1)$

$\Rightarrow \quad P \neq 1$

$P(E) \neq 1 \Rightarrow$ E is not a sure event.

4. Ans (A)

Ans. (A)

T : took a taxi

R : Took a rickshaw

W : Walked by foot

$P(T) = \frac{1}{2}, P(R) = \frac{1}{3}, P(W) = \frac{1}{6}$

C : Catching the train

$P(C/R) = \frac{1}{2}\,P(C/T)$; $P(C/W) = \frac{1}{4}\,P(C/R)$

let $P(C/R) = x$

$\therefore P(C/T) = 2x$; $P(C/W) = \frac{x}{4}$

$$P(W/C) = \frac{\frac{1}{6}\times\frac{x}{4}}{\frac{1}{2}\times 2x+\frac{1}{3}\times x+\frac{1}{6}\times\frac{x}{4}}$$

$$= \frac{\frac{1}{24}}{1+\frac{1}{3}+\frac{1}{24}} = \frac{1}{24+8+1} = \frac{1}{33}$$

5. Ans (D)

Ans. (D)

All throws are independent

$\therefore P = \frac{1}{6}$

6. Ans (C)

Ans. (C)

$P\left(A \cap \bar{C}\right) = P\left(A \cap \bar{C} \cap B\right) + P\left(A \cap \bar{C} \cap \bar{B}\right)$

$= \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$

$P(A) = P(A \cap C) + P\left(A \cap \bar{C}\right)$

$\Rightarrow P(A \cap C) = \frac{3}{4} - \frac{2}{3} = \frac{1}{12}$

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7. Ans (A)

Ans. (A)

$P(i) = ki$

$\because \sum_{i=1}^{6} P(i) = 1 \Rightarrow k(1+2+3+4+5+6) = 1$

$\Rightarrow k = \frac{1}{21}$

$\therefore P(\text{even}) \times P(\text{odd})$

$= \left(\frac{2}{21} + \frac{4}{21} + \frac{6}{21}\right) \times \left(\frac{1}{21} + \frac{3}{21} + \frac{5}{21}\right) = \frac{12}{49}$

8. Ans (B)

Ans. (B)

$P(WBWB) + P(BWBW)$

$= \frac{5}{8} \times \frac{3}{8} \times \frac{5}{8} \times \frac{3}{8} + \frac{3}{8} \times \frac{5}{8} \times \frac{3}{8} \times \frac{5}{8}$

$= 2\left(\frac{15}{64}\right)^2 = \frac{15^2}{2^{11}}$

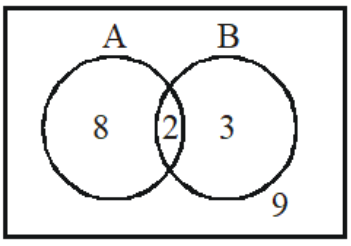
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MATHS

Section–I (ii)

9. Ans (B,C,D)

Ans. (B,C,D)



$P(\bar{A}) = \frac{6}{11}; P(A) = \frac{5}{11}$

$P(B) = \frac{5}{22}, P(A \cap B) = \frac{1}{11}$

$P(A \cup B) = \frac{13}{22}, P(\bar{A} \cap \bar{B}) = \frac{9}{22}$

$P(\bar{A} \cup \bar{B}) = \frac{10}{11}, P(\bar{B} \cap A) = \frac{8}{22} = \frac{4}{11}$

10. Ans (A,B,C,D)

Ans.(A,B,C,D)

(A) $P(\bar{A} \cup \bar{B}) = P(\overline{A \cap B}) = 1 - P(A \cap B)$

$1 - P(A)\,P(B) = \frac{5}{6}$

(B) $= P(B) - P(B \cap C) = P(B) - P(B)P(C)$

$= \frac{5}{18}$

(C) $= P(\overline{A \cup B}) = 1 - P(A \cup B)$

$= 1 - P(A) - P(B) + P(A \cap B)$

$= 1 - P(A) - P(B) + P(A)P(B) =$

(D) $= P(B) - P(A \cap B) + P(C) - P(B \cap C) + P(A) - P(A \cap C)$

$= P(B) - P(A)\,P(B) + P(C) - P(B)\,P(C) + P(A) - P(C)\,P(A) = \frac{25}{36}$

11. Ans (A,B,C,D)

Ans.(A,B,C,D)

(A) $P(A \cup B \cup C) = 1 - P(\bar{A} \cap \bar{B} \cap \bar{C})$

$1 - \frac{2}{3} \cdot \frac{1}{2} \cdot \frac{3}{4} = \frac{18}{24} = \frac{3}{4}$

(B) $\Sigma P(AB) - 3P(ABC) = \frac{1}{6} + \frac{1}{8} + \frac{1}{12} - \frac{3}{24}$

$= \frac{4+3+2-3}{24} = \frac{1}{4}$

(C)

$\Sigma P(AB) - 2P(ABC) = \frac{1}{6} + \frac{1}{8} + \frac{1}{12} - \frac{2}{24} = \frac{7}{24}$

(D) $1 - P(A \cap B \cap C) = 1 - \frac{1}{3} \times \frac{1}{2} \times \frac{1}{4} = \frac{23}{24}$

MATHS

Section–I (iii)

12. Ans (C)

Ans. (C)

$P(B) = \frac{5}{6}p_1 + \frac{5}{6}(1-p_1)(1-p_2) \cdot \frac{5}{6}p_1 + \left(\frac{5}{6}\right)^3 (1-p_1)^2(1-p_2)^2 p_1 + \ldots\ldots$

$P(B) = \frac{\frac{5}{6}p_1}{1 - \frac{5}{6}(1-p_1)(1-p_2)}, P(C) = \frac{\frac{5}{6}(1-p_1)p_2}{1 - \frac{5}{6}(1-p_1)(1-p_2)}$

$\frac{P(B)}{P(C)} = \frac{p_1}{p_2(1-p_1)}$

13. Ans (C)

Ans.(C)

If game is equally likely

$\Rightarrow$ Probability becomes

$\frac{1}{3}$ for each player $\Rightarrow \frac{P(B)}{P(C)} = 1$

$\Rightarrow p_1 = p_2(1 - p_1)$ ...(i)

$$\& \ P(A) = \frac{\frac{1}{6}}{1 - \frac{5}{6}(1-p_1)(1-p_2)} = \frac{1}{3}$$

$\Rightarrow (1 - p_1)(1 - p_2) = \frac{3}{5}$ ...(ii)

solve (i) & (ii) $\Rightarrow p_2 = \frac{1}{4}, p_1 = \frac{1}{5}$

$\Rightarrow p_1 : p_2 = 4:5$

14. Ans (A)

Ans. (A)

P(student will pass)

$$= \frac{1}{7}.\frac{1}{2} + \frac{1}{3}.\frac{1}{3} + \frac{11}{21}.\frac{1}{6} = \frac{17}{63}$$

15. Ans (C)

Ans. (C)

$$P(\text{y/ student is fail}) = \frac{\frac{1}{3}.\frac{2}{3}}{\frac{1}{7}.\frac{1}{2} + \frac{1}{3}.\frac{2}{3} + \frac{11}{21}.\frac{5}{6}} = \frac{7}{23}$$

MATHS

Section–II

1. Ans (2.00)

Ans. 2.00

Prime numbers

{2, 3, 5, 7, 11, 13, 17, 19, 23, 29}

P(atleast one) = 1 − P(no prime)

$$= 1 - \frac{{}^{20}C_2}{{}^{30}C_2} = 1 - \frac{20.19}{30.29} = \frac{49}{87}$$

$$\therefore \ \frac{b-a}{19} = \frac{87-49}{19} = 2$$

2. Ans (106.00)

Ans. 106.00

Two balls transferred from I to II can be 1W and 1B or both W or both B.

Then the required probability

$$= \left(\frac{{}^4C_1\,{}^3C_1}{{}^7C_2}\right).\frac{6}{9} + \left(\frac{{}^4C_2}{{}^7C_2}\right)\frac{7}{9} + \left(\frac{{}^3C_2}{{}^7C_2}\right).\frac{5}{9}$$

$$= \frac{43}{63} = \frac{p}{q} \ \therefore p + q = 106$$

3. Ans (4.00)

Ans. 4.00

Possible points that Humpty Not scores are

$$= \begin{cases} 9 & \text{i.e.} \quad 333 \\ 8 & \text{i.e.} \quad 233 \\ 7 & \text{i.e} \quad 223 \end{cases}$$

Required probability = 1−P(he score 7,8 or 9)

$$= 1 - \{{}^3C_2(0.3)^2.(0.5) + {}^3C_2.(0.5)^2.(0.3) + {}^3C_3.(0.5)^3\}$$

$$= 1 - \left\{\frac{3.9.5 + 3.25.3 + 1.125}{1000}\right\} = \frac{103}{200}$$

4. Ans (0.49)

Ans. 0.49

E_r : hunters shoots the animal at r distance

$P(E) = P(\text{escapes}) = 1 - P(\bar{E}_{2a}\bar{E}_{3a}.....\bar{E}_{na})$

$$= 1 - \left(1 - \frac{a^2}{4a^2}\right)\left(1 - \frac{a^2}{9a^2}\right)....\left(1 - \frac{a^2}{n^2a^2}\right)$$

$$= 1 - \left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{3^2}\right)....\left(1 - \frac{1}{n^2}\right)$$

$$= 1 - \left(\frac{1.3}{2.2}\right)\left(\frac{2}{3}.\frac{4}{3}\right)\left(\frac{3}{4}.\frac{5}{4}\right)....\left(\frac{n-1}{n}\frac{n+1}{n}\right)$$

$$= 1 - \frac{n+1}{2n} = \frac{n-1}{2n} = \frac{49}{100} = 0.49$$

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MATHS

Section–IV

1. **Ans (A->P,B->QT,C->P,D->ST)**

Ans. (A) ⟶ (P); (B) ⟶ (Q,T); (C) ⟶ (P); (D) ⟶ (S,T)

(A) Let number of black balls = x

$$\Rightarrow \frac{{}^{2}C_1 \cdot {}^{2}C_1 + {}^{2}C_1 \cdot {}^{x}C_1 + {}^{2}C_1 \cdot {}^{x}C_1}{{}^{x+4}C_2} = \frac{5}{7}$$

$$\Rightarrow 7(4+4x) = \frac{5(x+4)(x+3)}{2}$$

$$\Rightarrow 56 + 56x = 5x^2 + 35x + 60$$

$$\Rightarrow 5x^2 - 21x + 4 = 0$$

$$\Rightarrow (5x-1)(x-4) = 0 \quad \Rightarrow \quad x = 4$$

(B) ∵ A,B are independent

$\Rightarrow$ A, $\bar{B}$ are also independent

$$\Rightarrow P\left(A/\bar{B}\right) = P(A) = \frac{1}{3}$$

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(C) 1 – P(1 on none of the throw) > 0.5

$$\Rightarrow 1 - \left(\frac{5}{6}\right)^n > 0.5 \Rightarrow \left(\frac{6}{5}\right)^n > 2 \Rightarrow n_{min} = 4$$

(D) $P(3H \ \& \ 5T) = {}^{8}C_3 \times \left(\frac{1}{2}\right)^3 \times \left(\frac{1}{2}\right)^5$

$$= 56 \times \frac{1}{8 \times 32}$$

$$\Rightarrow \frac{7}{32}$$

$\Rightarrow$ odds in favour = 7 to 25

GUIDED REVISION

JEE(Main + Advanced) 2025
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SCORE ADVANCED

ANSWER KEY — **(Straight Line and Circle) GUIDED REVISION # 6**

TIME : 60 Min. — **M.M. : 76**

SOLUTIONS

MATHS

Section–I(i) Single Correct Type

1. Ans (C)

Ans. (c)

As the point $(3\sqrt{3},\ 2)$ lies on the line $y = \sqrt{3}x - 7$ so locus of P will be a line perpendicular to given line and passing through $(3\sqrt{3},\ 2)$

$$y - 2 = -\frac{1}{\sqrt{3}}(x - 3\sqrt{3})$$

$$y - 2 = \frac{-x}{\sqrt{3}} + 3$$

$$\frac{x}{\sqrt{3}} + y - 5 = 0$$

It intersects x and y axis at $(5\sqrt{3},\ 0)$ and $(0,\ 5)$ so area of the triangle formed $= \frac{5\sqrt{3} \times 5}{2} = \frac{25\sqrt{3}}{2}$

2. Ans (B)

Ans. (B)

$$\frac{y_2 - y_1}{x_2 - x_1} + \frac{y_3 - y_2}{x_3 - x_2} + \ldots\ldots + \frac{y_n - y_{n-1}}{x_n - x_{n-1}}$$

because $x_2 - x_1 = x_3 - x_2 = \ldots\ldots\ x_n - x_{n-1} = d$

$$\frac{y_n - y_1}{d} = \frac{(n-1)(y_n - y_1)}{(n-1)d} = \frac{(n-1)(y_n - y_1)}{(x_n - x_1)}$$

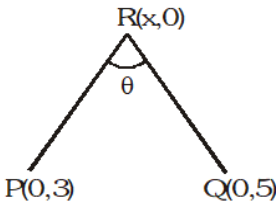
$$= (n-1)m_{A_1A_n}$$

3. Ans (A)

Ans. (A)

$$m_1 = -\frac{3}{x}$$

$$m_2 = -\frac{5}{x}$$



$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1m_2}\right| = \left|\frac{\frac{2}{x}}{1 + \frac{15}{x^2}}\right|$$

$$\tan\theta = 2\left|\frac{x}{15 + x^2}\right|$$

$$\lambda = \frac{x}{15 + x^2}$$

$$\frac{d\lambda}{dx} = \frac{(15 + x^2) - 2x^2}{(15 + x^2)^2} = 0$$

$$x^2 = 15 \quad \Rightarrow \quad |x| = \sqrt{15}$$

4. Ans (B)

Ans. (B)

Let the point of intersection of tangents at A & B is (x1, y1), then equation of chord of contact of (x_1, y_1) w.r.t. given circle is $T = 0$

$\Rightarrow\ xx_1 + yy_1 - 3(x + x_1) - 2(y + y_1) + 3 = 0$

$\Rightarrow\ x(x_1 - 3) + y(y_1 - 2) - 3x_1 - 2y_1 + 3 = 0$

and equation of chord whose mid point is (1, 1) is

$T = S_1 \ \Rightarrow\ x + y - 3(x + 1) - 2(y + 1) + 3 = 1 + 1 - 6 - 4 + 3$

$\Rightarrow\ -2x - y + 3 = 0$

As both equations are representing the same straight line

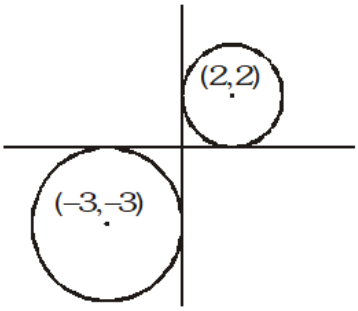
$$\Rightarrow \frac{x_1 - 3}{-2} = \frac{y_1 - 2}{-1} = \frac{-3x_1 - 2y_1 + 3}{3} \Rightarrow$$

$x_1 = -1, \quad y_1 = 0$

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5. Ans (C)

Ans. C



$x^2 + y^2 - 4x - 4y + 4 = 0$ $\quad c = (2, 2) \quad r = 2$

$x^2 + y^2 + 6x + 6y + 9 = 0$ $\quad c = (-3, -3) \quad r = 3$

From figure it is clear that

tangents are $x = 0$; $y = 0$

6. Ans (B)

Ans. (b)

Bisectors of the transverse common tangents of $(x \pm 2)^2 + y^2 = 1$ are x & y–axis

$\therefore$ Let the centre of the required circle be (r, r).

$\therefore$ It touches $(x - 2)^2 + y^2 = 1$

$\therefore \quad c_1c_2 = r_1 + r_2$

$\Rightarrow (r - 2)^2 + r^2 = (r + 1)^2 \quad \Rightarrow \quad r^2 - 6r + 3 = 0$

$\Rightarrow \quad r = \dfrac{6 \pm 2\sqrt{6}}{2}$

$r = 3 \pm \sqrt{6}$

7. Ans (D)

Ans. (D)

(a,3) lies inside $x^2 + y^2 - 16 = 0$ and outside $x^2 + y^2 - 4 = 0$ and above y = x in first quadrant so

$a^2 + 9 - 16 < 0 \Rightarrow -\sqrt{7} < a < \sqrt{7}$ ...(i)

$a^2 + 9 - 4 > 0 \Rightarrow a \in R$ ...(ii)

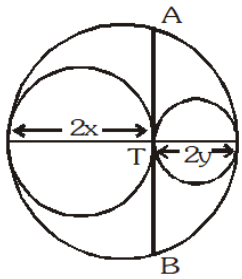
$a - 3 < 0 \Rightarrow a < 3$ ...(iii)

in first quadrant $a > 0$ ...(iv)

from (i), (ii), (iii) & (iv) $\Rightarrow a \in (0, \sqrt{7})$

8. Ans (D)

Ans. (D)



Let T is point of contact

power of point

$= TA \cdot TB = 2.2 = 2x \cdot 2y$

$\Rightarrow \quad xy = 1$

we have to find

$\pi\left[(x + y)^2 - x^2 - y^2\right] = \pi[2xy] = 2\pi$

9. Ans (A)

Ans. (A)

$-f = -g - 2 \qquad -g = -2f$

$f = g + 2 \quad \ldots\ldots (i) \quad g = 2f \quad \ldots\ldots (ii)$

$f = 2f + 2$

$\therefore f = -2 \quad \& \quad g = -4$

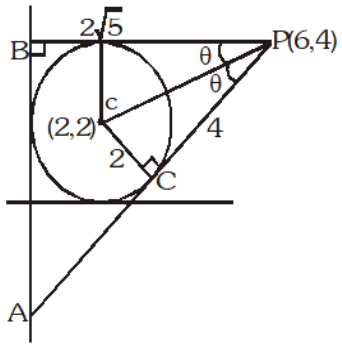
$g^2 - c = 0 \quad \Rightarrow \quad c = 16$

$\therefore$ Equation of the required circle is

$x^2 + y^2 - 8x - 4y + 16 = 0$

10. Ans (B)

Ans. (B)



Circle is $(x - 2)^2 + (y - 2)^2 = 2^2$

BP = 6

$\Rightarrow$ CP = 4

$\tan\theta = \dfrac{1}{2}$

$\tan 2\theta = \dfrac{AB}{BP}$

$\Rightarrow \quad AB = 6\tan 2\theta$

$= \dfrac{12\tan\theta}{1 - \tan^2\theta} = \dfrac{6}{1 - \frac{1}{4}} = 8$

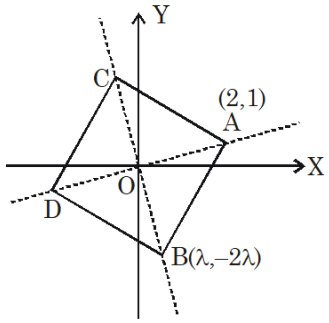
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MATHS

Section–I(ii) Multiple Correct Type

11. Ans (C,D)

Ans. C,D



$2x^2 - 3xy - 2y^2 = 0$

$2x^2 - 4xy + xy - 2y^2 = 0$

$(2x + y)(x - 2y) = 0$

$\therefore$ diagonals are $y = \frac{x}{2}$ and $y = -2x$

A(2, 1) and B(λ, -2λ)

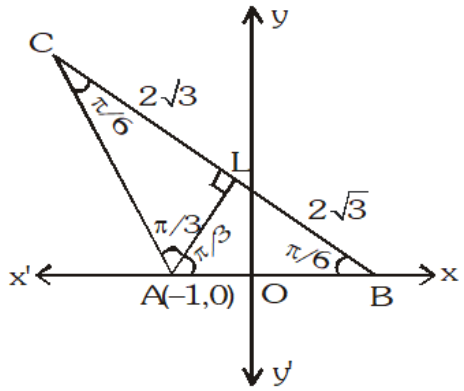
$OA^2 = OB^2$

$\lambda = \pm 1$

$\therefore$ B(1, –2) or C(–1, 2)

12. Ans (A,B,D)

Ans. (A,B,D)



In ΔABC, we have

$BL = AB\cos\frac{\pi}{6}$

$2\sqrt{3} = AB \times \frac{\sqrt{3}}{2} \Rightarrow AB = 4 \Rightarrow OB = 3$

So, the co-ordinates of B are (3, 0)

So the equation of BC is

$y - 0 = \tan\frac{5\pi}{6}(x - 3)$

$\Rightarrow y = -\frac{1}{\sqrt{3}}(x - 3) \Rightarrow x + \sqrt{3}y - 3 = 0$

$\Delta = \frac{1}{2}AB.AC.\sin\frac{2\pi}{3} = 4\sqrt{3}$

13. Ans (B,C,D)

Ans. (B,C,D)

$S_1 : x^2 + y^2 - 4x - 6y - 12 = 0$

$r_1 = \sqrt{4 + 9 + 12} = 5$

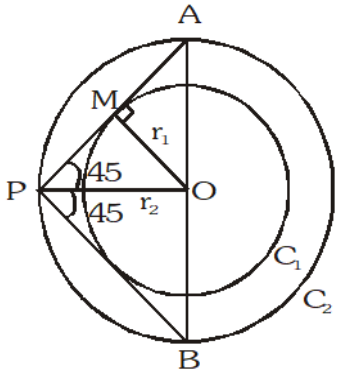
$S_2 : x^2 + y^2 + 6x + 4y - 12 = 0$

$r_2 = \sqrt{9 + 4 + 12} = 5$

$S_1 - S^2 = x + y = 0 \quad C_1C_2 = \sqrt{25 + 25} = 5\sqrt{2}$

14. Ans (B,D)

Ans. (B,D)



In ΔOPM, we have

$\sin 45° = \frac{OM}{OP}$

$\Rightarrow \frac{1}{\sqrt{2}} = \frac{r_1}{r_2}$

$\Rightarrow r_2 = \sqrt{2}r_1$

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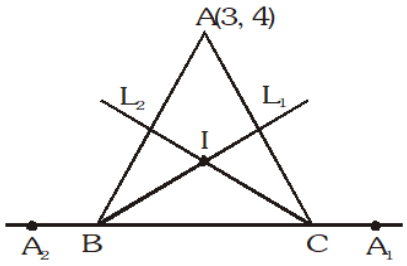
MATHS

Section–I(iii) Comprehension Type

15. Ans (B)

Ans. (B)

Let the image of point A with respect to L_2 is A_2



let A_1 is (α, β)

Now equation of

the line perpendicular to L_1

and passing through (3, 4) will be

$y - 4 = x - 3 \Rightarrow x - y + 1 = 0$

Intersection point of this line & L_1 is $\left(\frac{1}{2}, \frac{3}{2}\right)$

Now $\frac{\alpha+3}{2} = \frac{1}{2}$ & $\frac{\beta+4}{2} = \frac{3}{2}$

$\alpha = -2, \quad \beta = -1$

$\therefore$ $A_1 \equiv (-2, -1)$

Similarly A_2 will be $(-1, 2)$

from geometry A_1, A_2 will lie on BC

so slope of BC will be 3.

16. Ans (C)

Ans. (C)

Intersection point of L_1 & L_2 will be incentre

i.e. $(3, -1)$

equation of BC will be

$y - 2 = 3(x + 1)$

$3x - y + 5 = 0$

Now the inradius will be

$\frac{9+1+5}{\sqrt{10}} = \frac{15}{\sqrt{10}} = \frac{3}{2}\sqrt{10}$

17. Ans (C)

$C_1 (0,0)$; $C_2(6,0)$

$r_1 = 1 \quad r_2 = 2$

Distance between centres $d = 6$

Ans.(C)

Length of direct common tangent

$= \sqrt{d^2 - (r_2 - r_1)^2} = \sqrt{36 - 1} = \sqrt{35}$

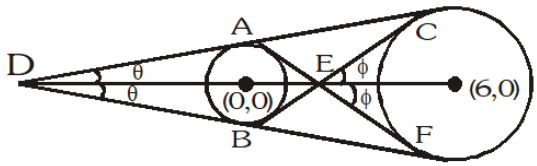
18. Ans (B)

$C_1 (0,0)$; $C_2(6,0)$

$r_1 = 1 \quad r_2 = 2$

Distance between centres $d = 6$

Ans.(B)



Both centres lie on x-axis

$\therefore$ Intersection point of direct common tangent & Intersection point of transverse common tangent lie on x-axis

Slope of tangent DA = $\tan\theta$

Slope of tangent DB = $-\tan\theta$

Slope of tangent CE = $\tan\phi$

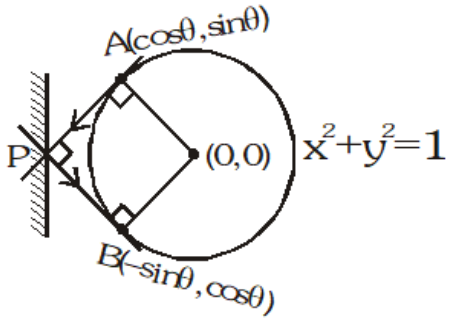
Slope of tangent FE = $-\tan\phi$

$\therefore$ Sum of slope of all common tangents is 0.

MATHS

Section–III Integer Type

1. Ans (2)



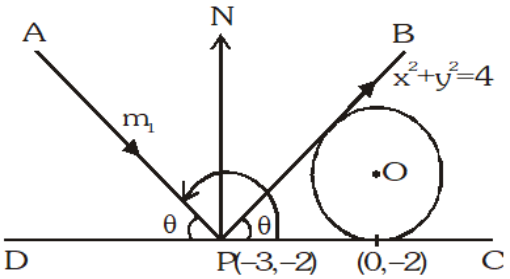
It is obvious that tangents are perpendicular, then locus of P is its director circle $x^2 + y^2 = 2$.

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2. Ans (7)

Ans. 7

Any line through P(−3,−2) is



$y + 2 = m(x + 3)$

It touches the circle if $mx - y + 3m - 2 = 0$

$$\left|\frac{3m-2}{\sqrt{1+m^2}}\right| = 2$$

$9m^2 + 4 - 12 = 4 + 4m^2$

$5m^2 - 12m = 0$

$m = 0, \frac{12}{5}$

Equation of PB is

$y + 2 = \frac{12}{5}(x + 3)$

$5y + 10 = 12x + 36$

$12x - 5y + 26 = 0$

Slope $m = \frac{12}{5}$

$$\tan\theta = \left|\frac{\frac{12}{5}-0}{1+0}\right|$$

$\tan\theta = \frac{12}{5}$

$\angle CPB = \angle DPA$

$\therefore \angle CPA = \pi - \theta$

$\tan(\pi-\theta) = -\tan\theta$

$\tan(\pi-\theta) = -\frac{12}{5}$

Let slope of AP is m_1

$m_1 = -\frac{12}{5}$

So equation of AP is

$y + 2 = -\frac{12}{5}(x + 3)$

$5y + 10 = -12x - 36$

$12x + 5y + 46 = 0$

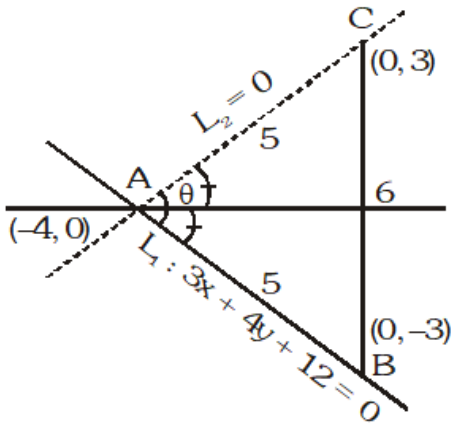
$Jx - ky + 46 = 0$

$J = 12 \quad K = -5$

$J + K = 7$

3. Ans (9)

Ans. 9



$\theta = \tan^{-1}\left(\frac{24}{7}\right)$

$\frac{24}{7} = \left(\frac{m-(-3/4)}{1+m(-3/4)}\right)$

$m = \frac{3}{4}$

$\therefore$ equation of the line $L_2 = 0$ is

$L_2 : 3x - 4y + 12 = 0$

Now, incentre of the ΔABC is

$$\left(\frac{5\times0+6\times(-4)+5\times0}{16}, \frac{5\times3+6\times0+5\times(-3)}{16}\right)$$

$$\equiv \left(-\frac{3}{2}, 0\right) \equiv (4\lambda, 3k)$$

$\therefore \quad \lambda = -\frac{3}{8}, \quad k = 0$

$10k - 24\lambda = 9$

4. Ans (5)

Ans. 5

$y - 1 = m(x - 1) \Rightarrow \frac{mx-y}{m-1} = 1 \quad .....(1)$

Homogenise the curve with (1)

$$2x^2 - y^2 + 2x\left(\frac{mx-y}{m-1}\right) - 4y\left(\frac{mx-y}{m-1}\right) + 2\left(\frac{mx-y}{m-1}\right)^2 = 0$$

Coeff. of xy = 0

$$\Rightarrow -\frac{2}{m-1} - \frac{4m}{m-1} + \frac{2}{(m-1)^2}(-2m) = 0$$

$-2(m - 1) - 4m(m - 1) - 4m = 0$

$-2m + 2 - 4m^2 + 4m - 4m = 0$

$2m^2 + m - 1 = 0$

$2m^2 + 2m - m - 1 = 0$

$\Rightarrow (2m - 1)(m + 1) = 0$

$L : y - 1 = \frac{1}{2}(x - 1) \Rightarrow \quad 2y - 2 = x - 1$

$x - 2y + 1 = 0$

$\therefore$ a = 1 & b = −2

Hence $3a - b = 5$

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MATHS

Section–IV Matrix Match Type

1. **Ans (A->S,B->R,C->P,D->Q)**

Ans. (A) ⟶ (S), (B) ⟶ (R), (C) ⟶ (P), (D) ⟶ (Q)

(A) Equation of common chord

$10x + 4y - b - a = 0$

this chord should pass through the centre of the circle $x^2 + y^2 + 6x + 6y - b = 0$

i.e. $(-3, -3)$

$-30 - 12 - b - a = 0$

$a + b = -42$

$|a + b| = 42$

(B) Equation of chord of contact drawn from point (a, b) will be $ax + by = 10$

comparing it with given chord

$x + y = 2$

$a = 5, b = 5$

so $a^2 + b^2 = 50$

(C) Equation of the family of circles passing through AB will be

$x^2 + y^2 - 5x + \lambda(2x - y) = 0$

$\Rightarrow x^2 + y^2 + x(2\lambda - 5) - \lambda y = 0$

centre $\left(-\frac{(2\lambda - 5)}{2}, \frac{\lambda}{2}\right)$

If AB is a diameter then centre must lie on AB, so

$\frac{\lambda}{2} = -(2\lambda - 5)$

$\lambda = -4\lambda + 10$

$\lambda = 2$

$\therefore$ required circle will be

$x^2 + y^2 - x - 2y = 0$

$a = 1, b = 2$

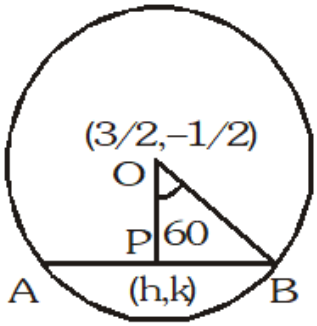
$\therefore$ $2a + 9b = 20$

(D) Let mid point of the chord is (h, k)

In ΔOPB

$\cos 60° = \frac{OP}{OB}$

$\frac{1}{2}$ =



$\frac{\sqrt{(h - 3/2)^2 + (k + 1/2)^2}}{\sqrt{2}}$

$\Rightarrow (h - 3/2)^2 + (k + 1/2)^2 = 1/2 \Rightarrow h^2 + k^2 - 3h + k + 2 = 0$

$\therefore$ Required locus $x^2 + y^2 - 3x + y + 2 = 0 \Rightarrow a = 3, b = -1$

So $a^3 + b^3 = 27 - 1 = 26$.

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GUIDED REVISION

JEE(Main + Advanced) 2025
ENTHUSIAST+LEADER COURSE
SCORE ADVANCED

ANSWER KEY

(Conic Section) GUIDED REVISION # 7

TIME : 60 Min.

M.M. : 86

SOLUTIONS

MATHS

Section–I(i) Single Correct Type

1. Ans (D)

Ans. (D)

$S_1P + S_2P = 2a$ & $S_1P - S_2P = 2A$

$S_1P = a + A$ & $S_2P = (a - A)$

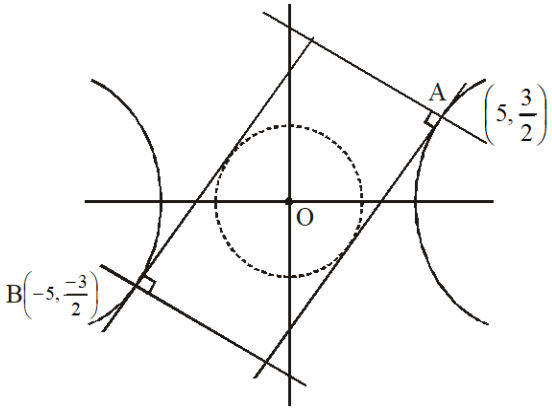
quadratic equation is $x^2 - 2ax + (a^2 - A^2) = 0$

2. Ans (D)

Ans. (D)

$\frac{x^2}{16} - \frac{y^2}{4} = 1$

$\left(5, \frac{3}{2}\right)$ lies on the curve.



tangent at A

$\frac{5x}{16} - \frac{3}{2}\left(\frac{y}{4}\right) = 1 \Rightarrow 5x - 6y = 16$

tangent at B

$5x - 6y = -6$

Normal at A

$6x + 5y = \lambda$

A $\left(5, \frac{3}{2}\right)$ $30 + \frac{15}{2} = 1$

$\Rightarrow \lambda = \frac{75}{2}$ $6x + 5y = \frac{75}{2}$

$|||^{ly}$ normal at B is $6x + 5y = \frac{-75}{2}$ distance between tangents is $2r = \frac{32}{\sqrt{25+36}}$

$\therefore r = \frac{16}{\sqrt{61}}$

Radius of circumcircle of rectangle is

$R = \sqrt{25 + \frac{9}{4}} = \frac{\sqrt{109}}{2}$

3. Ans (B)

Ans. (B)

Let the point be $(\alpha, \beta) \Rightarrow b = \alpha + 2$

Chord of contact of hyperbola is $T = 0$

$\Rightarrow \frac{\alpha x}{6} - \frac{\beta y}{2} = 1 \Rightarrow \frac{\alpha x}{6} - \frac{(\alpha+2)y}{2} = 1 \Rightarrow$

$\alpha\left(\frac{x}{6} - \frac{y}{2}\right) - \left(\frac{2y}{2} + 1\right) = 0$

Since this passes through (λ, μ)

$\therefore \frac{\lambda}{6} - \frac{\mu}{2} = 0$ and $\frac{2\mu}{2} + 1 = 0 \Rightarrow \frac{\lambda}{\mu} = 3$

4. Ans (D)

Ans. (D)

The chord of contact of (2, 3) is $3y = 2(x + 2)$ solving it with $y^2 = 4x$

$y^2 = 2(3y - 4) \Rightarrow y^2 - 6y + 8 = 0 \Rightarrow y = 2, 4$

so $x = 1$ & 4 respectively so points are (1, 2) and (4, 4).

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5. Ans (A)

Ans. (A)

Equation of the normal to the parabola $y^2 = 12x$ is given by

$y = mx - 6m - 3m^3$

It passes through (1, 2) so

$2 = m - 6m - 3m^3$

$\Rightarrow 3m^3 + 5m + 2 = 0$

$m_1m_2 + m_2m_3 + m_3m_1 = 5/3$ ...(i)

$m_1m_2m_3 = -2/3$ ...(ii)

(i) divided by (ii)

$$\frac{1}{m_1} + \frac{1}{m_2} + \frac{1}{m_3} = \frac{-5}{2}$$

6. Ans (C)

Ans. (C)

Any tangent to $y = x^2$ can be written as

$y = mx - \frac{1}{4}m^2$ if it touches 2nd, then

$$mx - \frac{m^2}{4} = x^2 - 2x + 2 \quad \Rightarrow$$

$$x^2 - (2+m)x + \left(2 + \frac{m^2}{4}\right) = 0$$

$$D = 0 \quad \Rightarrow \quad (m+2)^2 - 4\left(2 + \frac{m^2}{4}\right)$$

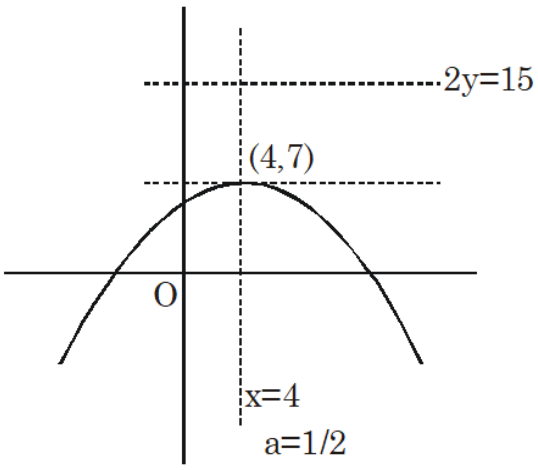
$4m - 4 = 0 \quad \Rightarrow \quad m = 1 \Rightarrow 4x - 4y - 1 = 0$

7. Ans (A)

Ans. (A)

$x^2 - 2.4x + 16 = 16 - 2 - 2y$

$(x-4)^2 = -2(y-7)$



8. Ans (B)

Ans. (B)

Focus ≡ (2, 0), point ≡ $(2t^2, 4t)$

$$\Rightarrow \quad h = \frac{2 + 2t^2}{2}$$

$$k = \frac{4t + 0}{2}$$

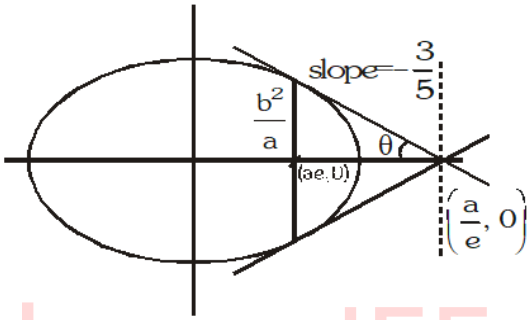
$\Rightarrow \quad h = 1 + t^2 \quad \& \quad k = 2t$

$$\Rightarrow \quad h = 1 + \frac{k^2}{4} \quad \Rightarrow \quad y^2 = 4x - 4$$

9. Ans (D)

Ans. (D)

Point of intersection of tangents is which lie on the corresponding directrix of the ellipse



$$\therefore \left(\frac{20}{3}, 0\right)$$

$$\tan\theta = \frac{b^2}{a\left(\frac{a}{e} - ae\right)} = \frac{b^2e}{a^2(1-e^2)} = e$$

$$\frac{3}{5} = e \quad \therefore \quad a = \frac{20}{3} \times \frac{3}{5} = 4$$

$$b^2 = 16\left(1 - \frac{9}{25}\right) = \frac{16 \times 16}{25}$$

$$\ell_{LR} = \frac{2b^2}{a} = \frac{2 \times 256}{4 \times 25} = 5.12$$

10. Ans (D)

Ans. (D)

The vertices of the square will lie on director's circle of ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$

$\Rightarrow$ vertices of square lies on $x^2 + y^2 = 16 + b^2$

$\therefore$ 2(side of square)^2 = (diameter of director's circle)^2

$\Rightarrow$ Radius of director's circle = 5 $\Rightarrow$ $16 + b^2 = 25$

or $b = \pm 3$

$\therefore$ Area of ellipse = $\pi(4)(3)$ sq. units

$= 12\pi$ sq. units

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MATHS

Section–I(ii) Multiple Correct Type

11. Ans (B,D)

Ans. (B,D)

Let the common tangent be

$y = mx \pm \sqrt{12m^2 - 12}$ solving it with

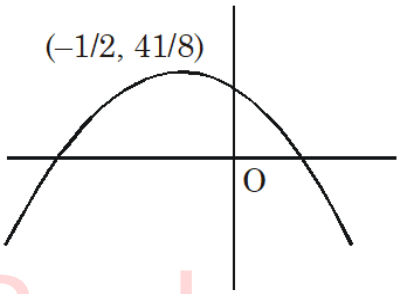
$xy = 8 \Rightarrow mx^2 \pm \sqrt{12(m^2-1)}x - 8 = 0$

$D = 0 \quad \Rightarrow \quad 12(m^2 - 1) + 32m = 0$

$3m^2 + 8m - 3 = 0 \quad \Rightarrow \quad m = -3, \frac{1}{3}$

$y = -3x \pm 4\sqrt{6} \quad m = \frac{1}{3}$ rejected

12. Ans (A,D)



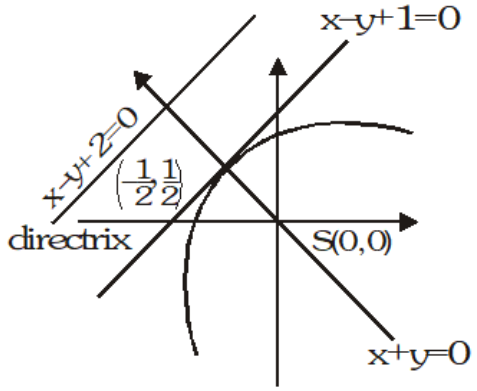
$(f(0) - 5)^2 + (f(1) - 4)^2 + (f(2) - 2)^2 = 0$

$f(0) = 5,\ f(1) = 4,\ f(2) = 2$

$f(x) = -\frac{x^2}{2} - \frac{x}{2} + 5$

13. Ans (A,C)

Ans. (A,C)



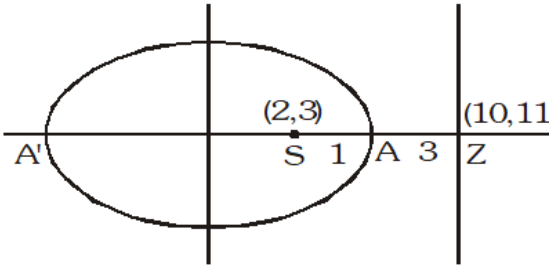
$(x + y)^2 = 4(x - y + 1)$

$2\left(\frac{x+y}{\sqrt{2}}\right)^2 = 4\sqrt{2}\left(\frac{x-y+1}{\sqrt{2}}\right)$

$\left(\frac{x+y}{\sqrt{2}}\right)^2 = 2\sqrt{2}\left(\frac{x-y+1}{\sqrt{2}}\right)$

14. Ans (A,B,C)

Ans. (A,B,C)



$A \equiv \left(\frac{10+6}{4}, \frac{11+9}{4}\right) = (4, 5)$

$A' \equiv \left(\frac{10-6}{1-3}, \frac{11-9}{1-3}\right) = (-2, -1)$

Major axis, $AA' = \sqrt{36 + 36} = 6\sqrt{2}$

Minor axis $2b = 2a\sqrt{1 - e^2} = 2\left(3\sqrt{2}\right)\sqrt{1 - \frac{1}{9}} = 8$

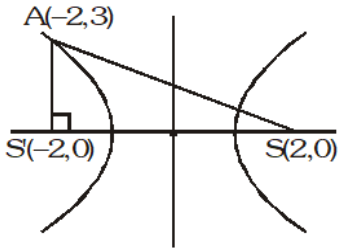
Latus rectum $= \frac{2b^2}{a} = \frac{2 \times 16}{3\sqrt{2}} = \frac{16\sqrt{2}}{3}$

15. Ans (A,B,C,D)

Ans. (A,B,C,D)

$S \equiv (2, 0),\ S' \equiv (-2, 0)$

Using reflection property of hyperbola, S'A is incident ray.



Equation of incident ray S'A is $x = -2$

Equation of reflected ray SP is $3x + 4y = 6$.

Now $2ae = 4 \Rightarrow ae = 2$ ....(i)

Point $(-2, 3)$ lies on hyperbola,

$\therefore \frac{4}{a^2} - \frac{9}{b^2} = 1 \Rightarrow \frac{4}{a^2} - \frac{9}{4 - a^2} = 1$

on solving it we get $a = 4$ (reject), $a = 1$ .....(ii)

$\therefore$ Using (i) & (ii), we get $e = 2$

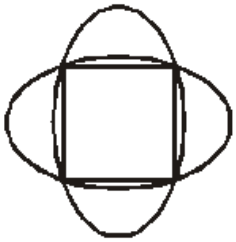
length of latus rectum $= 2a(e^2 - 1) = 6$

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16. **Ans (A,C)**

Ans. (A,C)

The given points lie on



$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ & $\frac{x^2}{b^2}+\frac{y^2}{a^2}=1$

The given points lie on a square & no parabola can pass through all vertices of a square

MATHS

Section–I(iii) Comprehension Type

17. **Ans (B)**

Ans. (B)

Minimum value occurs at x = 2

so $f(2)=|a-2|=1 \Rightarrow a=3,\ 1$

$\Rightarrow a=3 \ (as>2) \frac{x^2}{9}+\frac{y^2}{4}=1$

$\Rightarrow$ Ellipse is

& $e=\sqrt{1-\frac{4}{9}}=\frac{\sqrt{5}}{3}$

18. **Ans (D)**

Ans. (D)

Let tangent AB is $\frac{x\cos\theta}{3}+\frac{y\sin\theta}{2}=1$

AB passes through P(1, 2)

$\Rightarrow \frac{\cos\theta}{3}+\sin\theta=1$

$\theta=\frac{\pi}{2}$ is one solution

also $\frac{1}{3}+\tan\theta=\sec\theta \Rightarrow \sec\theta-\tan\theta=\frac{1}{3}$ ....(i)

& $\sec\theta+\tan\theta=3$ .....(ii)

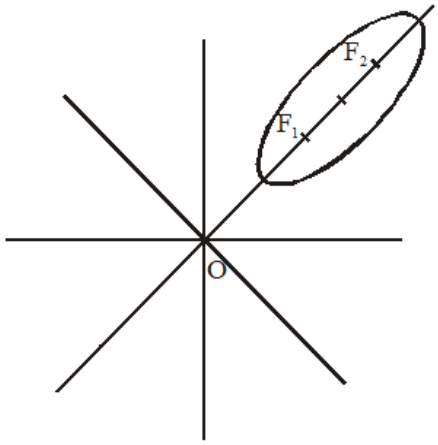
$\Rightarrow \sec\theta=\frac{5}{3}$ & $\tan\theta=\frac{4}{3}$

Hence none of the options necessarily are true, so (D)

19. **Ans (D)**

Ans. (D)

Focus of C_2 is $(\sqrt{2},\ \sqrt{2})$ &



asymptote of C_1 is $y=-x$ & $y=x$.

$\therefore$ Focus of C_3 is F_1 $(\sqrt{2},\ \sqrt{2})$ directrix is $y=-x$

for curve C_3

$a\left(\frac{1}{e}-e\right)=2$

$\frac{4}{3}\left(\frac{1}{e}-e\right)=2$

$\Rightarrow e=\frac{1}{2}$

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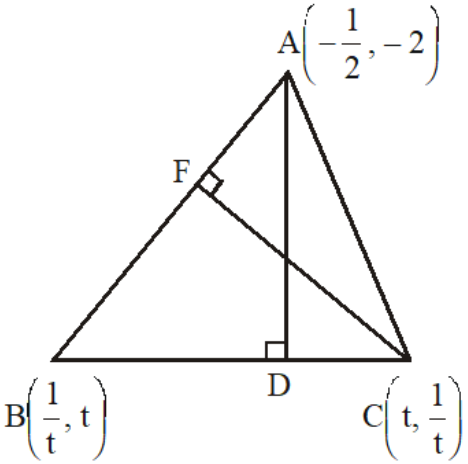
20. Ans (A)

Ans. (A)

$A\left(-\frac{1}{2}, -2\right)$

Let $B\left(\frac{1}{t}, t\right)$,

then



$C\left(t, \frac{1}{t}\right)$

Slope of BC = −1 ⇒ Slope of AD = 1

Equation of AD : $y + 2 = x + \frac{1}{2}$

or $2x - 2y = 3$ .....(1)

Now slope of AB = 2t

⇒ Slope of CF $= -\frac{1}{2t}$

⇒ equation of CF : $y - \frac{1}{t} = -\frac{1}{2t}(x - t)$

or $x + 2ty = t + 2$ .....(2)

Solving (1) & (2), we get

∴ orthocentre is $\left(2, \frac{1}{2}\right)$

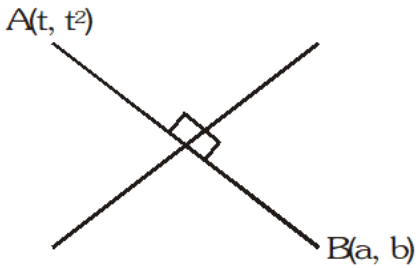
MATHS

Section–III Integer Type

1. Ans (2)

Ans. 2

General point on $y = x^2$ in (t, t^2) let us find the image of (t, t^2) in x + y = 1



$\frac{a + t}{2} + \frac{b + t^2}{2} = 1$

$a + t + b + t^2 = 2$ ...(1)

and $\frac{b - t^2}{a - t} = 1$ ...(2)

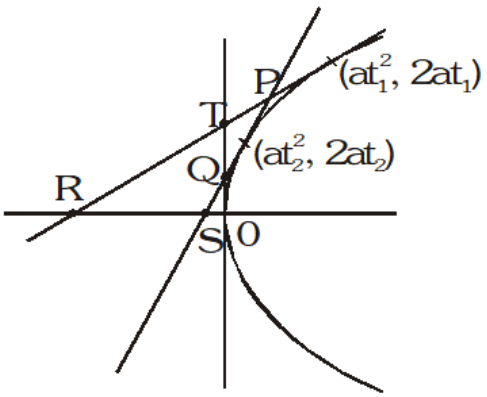
eliminate t from (1) & (2)

$a + b^2 = 2b$

⇒ $x + y^2 = 2y$ ⇒ k = 2

2. Ans (1)

Ans. 1



The equation of first tangent is $t_1y = x + at_1^2$

$T(0, at_1)$ & $R(-at_1^2, 0)$

For second tangent

$Q(0, at_2)$ & $S(-at_2^2, 0)$

For concyclic points (OT) (OQ) = (OR) (OS)

⇒ $t_1t_2 = 1$

Point $x = at_1t_2$

⇒ x = a

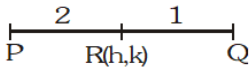
⇒ $\lambda_1 = 1$

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3. Ans (9)

Ans. 9

Equation of tangent is $\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$

$P\left(\frac{a}{\cos\theta}, 0\right)$ & 

$Q\left(0, \frac{b}{\sin\theta}\right)$

$h = \frac{a}{3\cos\theta}$ & $k = \frac{2b}{3\sin\theta}$

$\cos\theta = \frac{a}{3h}$ & $\sin\theta = \frac{2b}{3k}$

$\frac{a^2}{h^2} + \frac{4b^2}{k^2} = 9 \Rightarrow \frac{a^2}{x^2} + \frac{4b^2}{y^2} = 9$

4. Ans (1)

Ans. 1

A tangent to $\frac{x^2}{16} - \frac{y^2}{9} = 1$ is

$$y = m_1x \pm \sqrt{16m_1^2 - 9} \quad .....(i)$$

The other hyperbola is $\frac{x^2}{-9} - \frac{y^2}{-16} = 1$

then tangent to it is

$$y = m_2x \pm \sqrt{-9m_2^2 - (-16)} \quad .....(ii)$$

(i) & (ii) are same.

$\Rightarrow m_1 = m_2$ & $16m_1^2 - 9 = -9m_2^2 + 16$

$\Rightarrow 25m_1^2 = 25 \Rightarrow m_1^2 = 1 \Rightarrow m_1 = \pm 1$

$\therefore m = 1$

5. Ans (5)

Ans. 5

Assume (h,k) be the middle point of chord

$\therefore$ Its equation is

$hx - ky = h^2 - k^2 \Rightarrow y = \frac{h}{k}x + \frac{k^2 - h^2}{k}$

Now using $c = \frac{a}{m}$, we get

$\frac{k^2 - h^2}{k} = \frac{5}{h}.k$

$\Rightarrow x^3 = y^2(x - 5)$

$\therefore a = 5$

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GUIDED REVISION

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ANSWER KEY — **(Conic Section) GUIDED REVISION # 8**

TIME : 60 Min. — **M.M. : 76**

SOLUTIONS

MATHS

Section–I(i) Single Correct Type

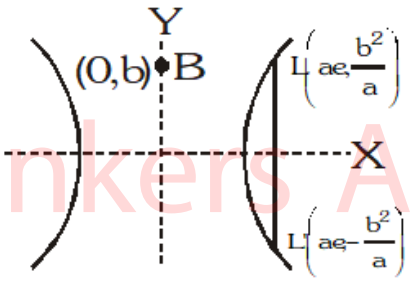
1. Ans (C)

Ans.(C)

Given that $m_{BL}.m_{BL'} = -1$

$$\left(\frac{\frac{b^2}{a} - b}{ae}\right).\left(\frac{-\frac{b^2}{a} - b}{ae}\right) = -1$$

$\Rightarrow$



$$\frac{b^4}{a^2} - b^2 = a^2e^2 \quad \Rightarrow \quad \frac{b^4}{a^4} - \frac{b^2}{a^2} = e^2$$

$\Rightarrow (e^2 - 1)^2 - (e^2 - 1) = e^2 \quad \Rightarrow \quad e^4 - 4e^2 + 2 = 0$

$\Rightarrow e^2 = 2 \pm \sqrt{2}$

$e^2 = 2 + \sqrt{2}$ for hyperbola.

2. Ans (B)

Ans. (B)

$\because$ slope of chord joining $A(t_1)$, $B(t_2) = \frac{2}{t_1 + t_2}$

and slope of nomral at $A(t_1) = -t_1$

$\Rightarrow m = \frac{2}{t_1 + t_2} = -t_1$

Also $t_1t_2 = -4$

$\therefore t_1 = \pm\sqrt{2}$

3. Ans (B)

Ans. (B)

$\because$ focus (2,0) lies on the line $\ell x - y - 2\ell = 0$ hence it is focal chord.

Tangents drawn at end points of a focal chord intersect at right angle.

4. Ans (A)

Ans. (A)

For tangent use : $C^2 = A^2m^2 + B^2$

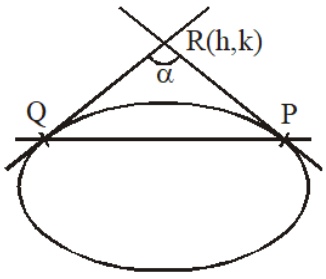
$c^2 = 4c^2 + 1$

$\Rightarrow 3c^2 + 1 = 0$

Not possible

5. Ans (D)

Ans. (D)



Chord of contact of R

$hx + 2ky = 6$

compare with $\frac{x}{2}\cos\theta + y\sin\theta = 1$

$\frac{2h}{\cos\theta} = \frac{2k}{\sin\theta} = 6$

$h^2 + k^2 = 9$

which is director circle of $x^2 + 2y^2 = 6$.

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6. **Ans (A)**

Ans. (A)

Equation of chord with given middle point (x, y) is

$T = S_1 \Rightarrow \frac{xx_1}{4} + \frac{yy_1}{9} = \frac{x_1^2}{4} + \frac{y_1^2}{9}$ ...(1)

Compare (1) with $9x - 4y = 13$

$x_1 = 1$ and $y_1 = -1$ (i.e. a = 1 & b = −1)

Hence, $(\alpha + \beta) = 0$

MATHS

Section–I(ii) Multiple Correct Type

7. **Ans (A,B,C,D)**

Ans. (A,B,C,D)

Given minor axis $\Rightarrow$ $2y + x = 3$

Equation of major axis = $2x - y = k$

Passes throu (2, 3) $k = 1$

hence major axis $2x - y = 1$

centre of ellipse (1, 1)

$\frac{a}{e} = \sqrt{(3-1)^2 + (2-1)^2} = \sqrt{5}$

$\because b^2 = a^2(1 - e^2)$

$\therefore \frac{5}{4} = 5e^2(1 - e^2) \quad \Rightarrow e = \frac{1}{\sqrt{2}}$

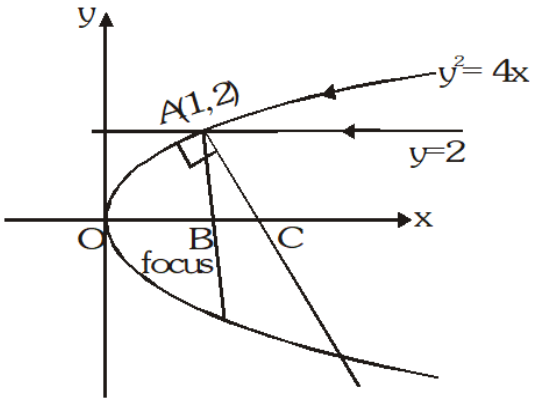
$a = \frac{\sqrt{5}}{\sqrt{2}}$

equation of ellipse

$4(2x - y - 1)^2 + 2(x + 2y - 3)^2 = 25$

8. **Ans (B,D)**

Ans.(B,D)



Let A $(t^2, 2t)$

$\because$ y = 2 $\therefore$ 2t = 2

t = 1

again meet C $(t_1^2, 2t_1)$

equation of normal at A

$y + x = 3$

put y = 0, x = 3

C ≡ (3, 0)

B is focus (1, 0)

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9. Ans (B,C,D)

Ans. (B,C,D)

Let $P_1(a\sec\theta, b\tan\theta)$ be a point on

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Now chord of contact w.r.t. P_1 to the ellipse is

$\frac{x\sec\theta}{a} + \frac{y\tan\theta}{b} = 1$

Now solve with $y = \pm\frac{b}{a}x$

$\frac{x}{a}\sec\theta + y\frac{\tan\theta}{a} = 1$

$Q(a(\sec\theta - \tan\theta), b(\sec\theta - \tan\theta))$

Similarly $\frac{x\sec\theta}{a} + \frac{y\tan\theta}{b} = 1$

and $y = -\frac{b}{a}x$

$x = a(\sec\theta + \tan\theta)$

$R(a(\sec\theta + \tan\theta), -b(\sec\theta + \tan\theta))$

Mid point of Q and R is

$h = a\sec\theta$ and $k = -b\tan\theta$

$\Rightarrow \frac{h^2}{a^2} - \frac{k^2}{b^2} = 1$

(C) For mirror image w.r.t. x axis

Replace $y \longrightarrow -y$

Equation remains same.

(D) Locus of foot of perpendicular to the line QR from (ae,0) is auxilliary circle. As line QR itself is tangent to the given hyperbola.

10. Ans (B,C,D)

Ans. (B,C,D)

$\Rightarrow (\log b)^2 = (\log a)^2\left[1 - \frac{1}{2}\right]$

$2 = (\log_b a)^2$

$e^2 = 1 + (\log_a b)^2 = 1 + \frac{1}{2} = \frac{3}{2}$

11. Ans (B,D)

Ans. (B,D)

C is $\frac{x^2}{7} - \frac{y^2}{3} = 21$ $(y > 0)$

$\because$ Equation of L will be $y = x - 2$ and A $\left(\frac{7}{2}, \frac{3}{2}\right)$

Equation of circle

$\left(x - \frac{7}{2}\right)^2 + \left(y - \frac{3}{2}\right)^2 + \lambda(x - y - 2) = 0$

passing through (0, 0) $\Rightarrow \lambda = \frac{29}{4}$

12. Ans (C,D)

Ans. (C,D)

Let $P\left(\frac{\sqrt{325}}{2}, y_1\right)$

$\frac{325}{25\times 4} - \frac{y_1^2}{16} = 1$

$\frac{y_1^2}{16} = \frac{225}{100}$

$y_1 = \pm\frac{15\times 4}{10} = \pm 6$

Let $P\left(\frac{\sqrt{325}}{2}, -6\right)$

Tangents is $\frac{x.5.\sqrt{13}}{2\times 25} + \frac{6y}{16} = 1$

$\frac{x.\sqrt{13}}{10} + \frac{3y}{8} = 1$

$x = \left(1 - \frac{3y}{8}\right)\times\frac{10}{\sqrt{13}}$

Solve with $x^2 + y^2 = 25$

$\frac{100}{13}\left(1 - \frac{3y}{8}\right)^2 + y^2 = 25$

$100\left(1 - \frac{3y}{8}\right)^2 + 13y^2 = 325$

$y^2\left(13 + \frac{900}{64}\right) - \frac{600y}{8} - 225 = 0$

$H.M. = \left|\frac{2\delta\gamma}{(\gamma+\delta)}\right| = \left|\frac{2(-225)}{\left(\frac{600}{8}\right)}\right|$

$= \frac{16\times 15}{40} = |-6| = 6$

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MATHS

Section–I(iii) Comprehension Type

13. Ans (A)

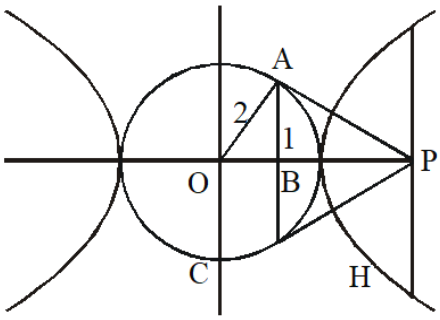
$\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1 \therefore \quad e_1 = e_2$

$\Rightarrow \quad e_1 = \sqrt{2} \quad \Rightarrow \quad a = b$

foot of perpendicular drawn from focus upon any tangent always lies on auxiliary circle.

$x^2 + y^2 = a^2 \quad \Rightarrow \quad a = 2 \Rightarrow \quad b = 2$

C : $x^2 + y^2 = 4$ (which is auxiliary circle itself)



Ans. (A)

Number of common tangent of H & C is 2

14. Ans (C)

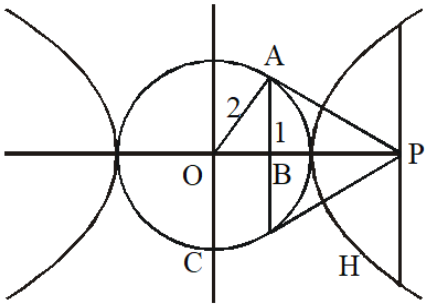
$\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1 \quad \therefore \quad e_1 = e_2$

$\Rightarrow \quad e_1 = \sqrt{2} \quad \Rightarrow \quad a = b$

foot of perpendicular drawn from focus upon any tangent always lies on auxiliary circle.

$x^2 + y2 = a^2 \quad \Rightarrow \quad a = 2 \quad \Rightarrow \quad b = 2$

C : $x^2 + y^2 = 4$ (which is auxiliary circle itself)



Ans. (C)

If ellipse & hyperbola cuts orthogonally

$\Rightarrow$ confocal.

foci of H $\Rightarrow$ $(\pm 2\sqrt{2}, 0)$

$\Rightarrow (4e_e, 0) = (2\sqrt{2}, 0)$

$e_e = \frac{1}{\sqrt{2}}$

$\beta^2 = 16\left(1 - \frac{1}{2}\right) = 8 \Rightarrow \beta = 2\sqrt{2}$

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15. **Ans (D)**

Ans. (D)

Let A $(t_1^2, 2t_1)$ and B$(t_2^2, 2t_2)$

$\because$ AB is focal chord $\Rightarrow t_1t_2 = -1$

Now $t_1 = \frac{y_1}{2}, t_2 = \frac{y_2}{2}$

$\Rightarrow \frac{y_1}{2}\cdot\frac{y_2}{2} = -1$ & $x_0 = \frac{x_1+x_2}{2}$, $y_0 = \frac{y_1+y_2}{2}$

A $\left(\frac{y_1^2}{4}, y_1\right)$

B $\left(\frac{y_2^2}{4}, y_2\right) \equiv \left(\frac{4}{y_1^2}, \frac{-4}{y_1}\right)$

C $\left(\frac{y_3^2}{4}, y_3\right)$

D $\left(\frac{y_4^2}{4}, y_4\right)$

$M_{BC} = \frac{y_3-y_2}{x_3-x_2}$

$m_{AE} = \frac{y_1-0}{x_1-2}$

$\therefore \frac{y_3-y_2}{x_3-x_2} = \frac{y_1}{x_1-2}$

$\frac{4}{y_3+y_2} = \frac{4y_1}{y_1^2-8}$

$\Rightarrow 4\left(y_1^2-8\right) = 4y_1y_3 + 4(-4)$

$\Rightarrow y_3 = \frac{4y_1^2-16}{4y_1} = y_1 - \frac{4}{y_1} = y_1 + y_2 = 2y_0$

16. **Ans (B)**

Ans. (B)

Parabola $y^2 = 4x$ & $y_1y_4 = 4$

tangent at D : $yy_4 = 2(x + x_4)$

normal at A : $y - y_1 = \frac{-y_1}{2}(x - x_1)$

$\Rightarrow yy_4 - y_1y_4 = \frac{-y_1y_4}{2}(x - x_1)$

abssissa of intersection point is

$x = 1 + \frac{x_1 - x_4}{2}$

$= 1 + \frac{x_1 - \frac{1}{x_1}}{2}$ $\left\{\text{Use : } \frac{1}{x_1} = x_2,\ 2x_0 = x_1 + x_2\right\}$

$= 1 + x_0 - x_2$

MATHS

Section–I(v) Matching list type

17. **Ans (A)**

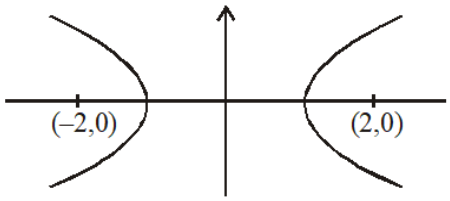
(A)

For hyperbola 'H'

$ae = 2$, $e = 2 \Rightarrow$

$a = 1$

$b^2 = 3$



$\frac{x^2}{1} - \frac{y^2}{3} = 1$, LR = 6

18. **Ans (D)**

Ans. (D)

Let $y = mx + c$ is common tangent

$\therefore c = -m^2 = \pm\sqrt{\frac{m^2}{4} + \frac{3}{4}} = \frac{a}{m}$

$\Rightarrow m = -1, a = 1, c = -1$

common tangent $y = -x - 1$

$x + y + 1 = 0$

$\ell = 1, m = 1$

MATHS

Section–II(iii) Numerical Type

1. **Ans (4.50)**

Ans. 4.50

Tangent to parabola $y = mx + \frac{2}{m}$

is also tangent to hyperbola.

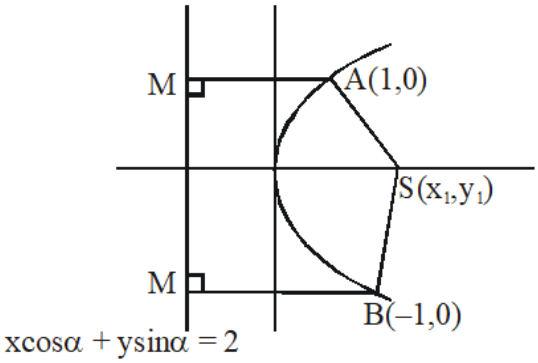
$\therefore c^2 = a^2m^2 - b^2 \Rightarrow \frac{4}{m^2} = m^2 - 3 \Rightarrow m = \pm 2$

$\therefore$ Area $= \frac{1}{2}.6.\frac{3}{2} = \frac{9}{2}$

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2. Ans (1.33 or 1.34)

Ans. 1.33 or 1.34



Let directrix touches the circle at $(2\cos\alpha, 2\sin\alpha)$

$\Rightarrow$ equation of directix is $x\cos\alpha + y\sin\alpha = 2$

Let focus be $S(x_1, y_1)$

A(1,0), B(−1,0) lies on parabola so

$(x_1 - 1)^2 + y_1^2 = (\cos a - 2)^2$ ...(1) (for parabola, PS = PM, P is considered as A(1,0))

$(x_1 + 1)^2 + y_1^2 = (\cos a + 2)^2$ ...(2) (for parabola, PS = PM, P is considered as B(−1,0))

$x_1 = 2\cos\alpha \quad y_1 = \sqrt{3}\sin\alpha$

Locus of focus $\Rightarrow \frac{x^2}{4} + \frac{y^2}{3} = 1$

3. Ans (4.00)

Ans. 4.00

$AS_1 + AS_2 \geq 2\sqrt{AS_1 . AS_2}$

$4 \geqslant 2\sqrt{AS_1 . AS_2}$ ($\because AS_1 + AS_2 = 2a = 4$)

$AS_1 . AS_2 \leqslant 4$

4. Ans (2.00)

Ans. 2.00

Let point on line $\left(h, \frac{1}{t+2} - h\right)$

chord of contact $4xh + 16y\left(\frac{1}{t+2} - h\right) = 1$

it passes through fixed point so $x = 4y$ &

$\frac{16y}{t+2} - 1 = 0$

fixed point = $\left(\frac{t+2}{4}, \frac{t+2}{16}\right)$ lies inside ellipse

So $S_1 < 0$

$-\frac{4}{\sqrt{5}} < (t+2) < \frac{4}{\sqrt{5}}$

$t = -1, -3, \quad (t \neq -2)$

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ANSWER KEY

(Determinant and Matrix) GUIDED REVISION # 9

TIME : 60 Min.

M.M. : 63

SOLUTIONS

MATHS

Section–I(i) Single Correct Type

1. Ans (C)

Ans. (C)

$P^2 = I - P$

$P^3 = P - P^2 = P - I + P = 2P - I$

$P^4 = 2P^2 - P = 2(I - P) - P = 2I - 3P$

$P^5 = 2P - 3P^2 = 2P - 3(I - P)$

$= 5P - 3I$

$P^6 = 5P^2 - 3P = 5(I - P) - 3P = 5I - 8P$

2. Ans (D)

Ans.(D)

$$\Delta = \begin{vmatrix} 1+\sin^2 x & \cos^2 x & \sin 2x \\ \sin^2 x & 1+\cos^2 x & \sin 2x \\ \sin^2 x & \cos^2 x & 1+\sin 2x \end{vmatrix}$$

$c_1 + c_2 \longrightarrow c_1$

$$\Delta = \begin{vmatrix} 2 & \cos^2 x & \sin 2x \\ 2 & 1+\cos^2 x & \sin 2x \\ 1 & \cos^2 x & 1+\sin 2x \end{vmatrix}$$

$R_1 - R_2 \longrightarrow R_1$

$$\Delta = \begin{vmatrix} 0 & -1 & 0 \\ 2 & 1+\cos^2 x & \sin 2x \\ 1 & \cos^2 x & 1+\sin 2x \end{vmatrix}$$

$\Delta = 2 + \sin 2x$

Maximum value = 3 minimum value = 1

$\alpha = 3, \beta = 1, \alpha + \beta^{99} = 4, \alpha^3 - \beta^{17} = 26$

$\alpha^{2n} - \beta^{2n}$ is always even but triangle with side $\alpha = 3$, $\beta = 1$ & $5\alpha - 13\beta = 2$ is not possible.

3. Ans (A)

Ans. (A)

$C_2 \longrightarrow C_2 - C_3$

$$\begin{vmatrix} a_1 a_2 & d & a_0 \\ a_2 a_3 & d & a_1 \\ a_3 a_4 & d & a_2 \end{vmatrix}$$

$R_3 \longrightarrow R_3 - R_2$ & then $R_2 \longrightarrow R_2 - R_1$

$$\begin{vmatrix} a_1 a_2 & d & a_0 \\ a_2(2d) & 0 & d \\ a_3(2d) & 0 & d \end{vmatrix} = -2d^3(a_2 - a_3) = 2d^4$$

4. Ans (C)

Ans. (C)

trace (adj(|A||B| AB))

= trace($|A|^{n-1}|B|^{n-1}$(AdjB)(AdjA))

trace(Adj(|AB|BA))

= trace($|A|^{n-1}|B|^{n-1}$(AdjA).(AdjB))

5. Ans (A)

Ans. (A)

$\because BC = I$

$\therefore B^2C^2 = I$

$A(B^2C^2)A^T = AA^T$

which is a symmetric matrix.

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6. Ans (C)

Ans. (C)

$2A + B = B^T - A^T$ ...(i)

take transpose both side

$2A^T + B^T = B - A$ ...(ii)

equation (i) + equation (ii)

$\Rightarrow 3A = -3A^T \Rightarrow A^T = -A$

$\Rightarrow 2A + B = B^T + A$

$\Rightarrow A + B = B^T$

$\Rightarrow |A + B| = |B^T| = |B| = 1$

7. Ans (A)

Ans. (A)

$|adjA| = 3^{16} \Rightarrow |A|^{n-1} = 3^{16}$

$\Rightarrow 3^{2(n-1)} = 3^{16}$

$\Rightarrow n = 9$

$\Rightarrow$ number of elements $= n^2 = 81$

8. Ans (D)

Ans. (D)

$R_3 \longrightarrow R_3 + R_2$

$$\begin{vmatrix} \sin\theta & \cos\theta & \sin 2\theta \\ \cos\left(\theta+\frac{\pi}{6}\right) & \cos\left(\theta+\frac{2\pi}{3}\right) & -\sin\left(2\theta+\frac{\pi}{3}\right) \\ -\sin\theta & -\cos\theta & -\sin 2\theta \end{vmatrix}$$

so determinant is identically zero as two rows are identical $\Rightarrow$ It will have infinite solutions

MATHS

Section–I(ii) Multiple Correct Type

9. Ans (B,C,D)

Ans. (B,C,D)

$$D = \begin{vmatrix} 1 & 1 & a \\ 2 & 3 & 0 \\ 3 & 4 & a^2 \end{vmatrix} = a(a-1)$$

for $a = 0 \Rightarrow D = 0, D_x = 0, D_y = 0, \quad D_z = 2 - b$

so for $a = 0, b = 2 \Rightarrow$ Infinite Solutions.

for $a = 0, b = 1 \Rightarrow$ No solutions.

for $a = 1, D = 0 \Rightarrow D_x = 0, D_y = 0, D_z = 0$

$\Rightarrow$ Infinite solutions.

If $a \neq 0$ but $a = 1 \Rightarrow$ does't have unique solutions.

10. Ans (B,C)

Ans. (B,C)

Characteristic equation of A is

$x^3 - 2x^2 - 4x + 1 = 0$

$\Rightarrow \lambda = -4$

$\because A^3 - 2A^2 - 4A + I = 0$

$\therefore A^2 + A^{-1} = 2A + 4I$

$\Rightarrow |A^2 + A^{-1}| = |2A + 4I| = 56$

MATHS

Section–I(iii) Comprehension Type

11. Ans (C)

Ans. (C)

$\because |A^T|A|B| = |A|B|B^T|$

$\therefore |A|\,|A|\,|B| = |B|\,|A|\,|B|$

$\Rightarrow |A| = 2$

$\Rightarrow |AB^{-1}\ adj\ (A^TB)^{-1}|$

$= \frac{|A|}{|B|}\ \frac{1}{|A^T|^2|B|^2} = \frac{1}{16}$

$= \frac{1}{|A||B|^3} = \frac{1}{2 \times 2^3} = \frac{1}{16}$

12. Ans (C)

Ans. (C)

$B = B^T$

option (A) $adj(B) = adj(B^T)$

option (B) $adj(B^{-1})^T = (B^T)^{-1} = B^{-1}$

option (C) $A^T|A|B = A|B|B^T$

$\Rightarrow A^TB = AB^T$

$\Rightarrow A^TB = AB$ or $A = AT$

but $A^{2015}B^{2016}$ is symmetric if $AB = BA$

option (D) $(AB^{-1})^T = (B^{-1})^T A^T = B^{-1}A^T = B^{-1}A$

$= AB^{-1}$ ($\because AB = BA$)

13. Ans (A)

Ans. (a)

Constant term (absolute term) will be same as g(0)

since $g(y) = Ay^3 + By^2 + Cy + D$

$g(0) = (p + q + r)(p^2 + q^2 + r^2 - pq - qr - rp) > 0$

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14. Ans (D)

Ans. (D)

$g'(y) = 3Ay^2 + 2By + C$

$\Rightarrow C = g'(0)$

$g'(0) = C = 3qr - 3p^2$

15. Ans (A)

Ans. (a)

$y_1 = 1, y_2 = w, y_2 = w_2$

$g(0) = p^3 + q^3 + r^3 - 3pqr$

$= (p + q + r)(p + qw + rw^2)(p + qw^2 + rw)$

$\Rightarrow g(0) = f(y_1) \cdot f(y_2) \cdot f(y_3)$

MATHS

Section–III Numerical Type

1. Ans (7.00)

Ans. 7.00

$$\begin{vmatrix} 1 & \sin\alpha & \sin^2\alpha \\ 1 & \cos\alpha & \cos^2\alpha \\ 1 & \sin 2\alpha & \sin^2 2\alpha \end{vmatrix} = 0$$

$\Rightarrow (\sin\alpha - \cos\alpha)(\cos\alpha - \sin 2\alpha)(\sin 2\alpha - \sin\alpha) = 0$

$\Rightarrow \tan\alpha = 1, \cos\alpha = 0, \sin\alpha = \frac{1}{2}, \sin\alpha = 0, \cos\alpha = \frac{1}{2}$

$\Rightarrow \alpha = 0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}, \frac{5\pi}{6}$ & π

2. Ans (6.00)

Ans. 6.00

Put x = 1

$a_4 + a_3 + a_2 + a_1 + a_0$

$$= \begin{vmatrix} 2 & -1 & 1 \\ 5 & 1 & 3 \\ -4 & -1 & 2 \end{vmatrix} = 10 + 22 - 1 = 31$$

3. Ans (25.00)

Ans. 25.00

$$A_2 = \begin{vmatrix} a & b+3c & c+4a \\ b & c+3a & a+4b \\ c & a+3b & b+4c \end{vmatrix} + 2\begin{vmatrix} b & b+3c & c+4a \\ c & c+3a & a+4b \\ a & a+3b & b+4c \end{vmatrix}$$

Assume $A_2 = B_1 + 2B_2$

In B_1 : $C_3 \to C_3 - 4C_1$ & In B_2 : $C_2 \to C_2 - C_1$

$$A_2 = \begin{vmatrix} a & b+3c & c \\ b & c+3a & a \\ c & a+3b & b \end{vmatrix} + 6\begin{vmatrix} b & c & c+4a \\ c & a & a+4b \\ a & b & b+4c \end{vmatrix}$$ Assume

$A_2 = B_3 + 6B_4$

In $B_3 : C_2 \to C_2 - 3C_3$ & In $B_4 : C_3 \to C_3 - C_2$

$$A_2 = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} + 6\begin{vmatrix} b & c & 4a \\ c & a & 4b \\ a & b & 4c \end{vmatrix}$$

$$= \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} + 24\begin{vmatrix} b & c & a \\ c & a & b \\ a & b & c \end{vmatrix}$$

$= 25A_1$

4. Ans (162.00)

Ans. 162.00

$\because$ Trace of matrix is 1 $\Rightarrow$ diagonal elements are 1,1,−1 or 1,0,0

$\therefore$ Number of possible symmetric matrices

$= (3 \times 3^3) \times 2 = 162$

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GUIDED REVISION

JEE(Main + Advanced) 2025
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ANSWER KEY — **(Function & ITF) GUIDED REVISION # 10**

TIME : 60 Min. — **M.M. : 62**

SOLUTIONS

MATHS

Section–I(i) Single Correct Type

1. Ans (B)

Ans. (B)

$f(x) = 1 + x^n$ {put x = 3}

$1 + 3^n = -26$ (Rejected)

or

$f(x) = 1 - x^n$ {put x = 3}

$1 - 3^n = -26$

$3^n = 27$

$n = 3$

So $f(x) = 1 - x^3$

$f(4) = 1 - 4^3 = -63$

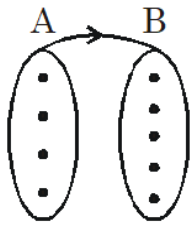
2. Ans (C)

Ans. (C)

$f(x) + (x - 6) = (x-1)(x-2)(x-3)(x-4)(x-5) \Rightarrow f(6) = 120$

3. Ans (C)

Ans. (C)



* $x = 5 \cdot 5 \cdot 5 \cdot 5 = 625$
* $y = {}^5C_4 . 4! = 120$
* $\therefore x + y = 625 + 120 = 745$

4. Ans (B)

Ans. (B)

$y = \frac{2|x| - 1}{3|x| + 2} \Rightarrow |x| = \frac{-2y - 1}{3y - 2} \geqslant 0 \Rightarrow \frac{2y + 1}{3y - 2} \leqslant 0$

$\Rightarrow y \in \left[-\frac{1}{2}, \frac{2}{3}\right)$

5. Ans (B)

Ans. (B)

$x^2 - 4x > 0, \quad x > 0$

$x(x - 4) > 0$

$\Rightarrow x > 4$

6. Ans (C)

Ans. (C)

$\cot^{-1}\frac{n}{\pi} > \frac{\pi}{6}$

$\frac{n}{\pi} < \sqrt{3} \Rightarrow n < (\sqrt{3}\pi) \approx 5.44$

$n = 5$

7. Ans (B)

Ans. (B)

* $\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$
* $\cos^{-1}\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right) = \frac{\pi}{12}$
* $\tan^{-1}(2 - \sqrt{3}) = \frac{\pi}{12}$

$\therefore \sin^{-1}\frac{1}{2} + \cos^{-1}\frac{\sqrt{3}+1}{2\sqrt{2}} + \tan^{-1}(2 - \sqrt{3}) = \frac{\pi}{3}$

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MATHS

Section–I(ii) Multiple Correct Type

8. Ans (A,C,D)

Ans. (A,C,D)

(A) $f(x) = \dfrac{x^3}{(2^{|x|} + \sec x)}$ ...(1)

$f(-x) = \dfrac{-x^3}{(2^{|x|} + \sec x)}$ ...(2)

$f(x) + f(-x) = 0$ (odd function)

(B) Even function

(C) $f(x) = \dfrac{(e^x - 1)}{(e^x + 1)}$ $(g(x) + g(-x))$ ...(1)

$f(x) = \dfrac{-(e^x - 1)}{(e^x + 1)}$ $(g(x) + g(-x))$ ...(2)

(1) + (2) $\Rightarrow$ $f(x) + f(-x) = 0$

(D) odd function.

9. Ans (A,C,D)

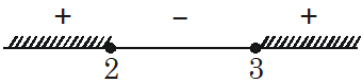
Ans. (A,C,D)

Let $|x| = t$

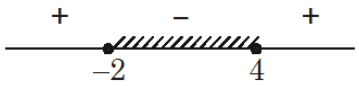
$\therefore \sqrt{t^2 - 5t + 6} = \sqrt{-t^2 + 2t + 8}$

$\Rightarrow \sqrt{(t-3)(t-2)} = \sqrt{-(t-4)(t+2)}$

$\because (t-3)(t-2) \geq 0$ & $-(t-4)(t+2) \geq 0 \Rightarrow (t-4)(t+2) \leq 0$

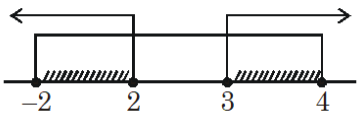


...(1)



...(2)

(1) $\cap$ (2)



$-2 \leq t \leq 2 \cup 3 \leq t \leq 4$

$\Rightarrow |x| \leq 2 \cup 3 \leq |x| \leq 4$

$\Rightarrow x \in [-2, 2] \cup [3, 4] \cup [-4, -3]$

10. Ans (A,B,D)

Ans. (A,B,D)

$2\cot^{-1}7 + \cos^{-1}\dfrac{3}{5} \Rightarrow 2\tan^{-1}\dfrac{1}{7} + \tan^{-1}\dfrac{4}{3}$

$\Rightarrow \tan^{-1}\left(\dfrac{7}{24}\right) + \tan^{-1}\left(\dfrac{4}{3}\right)$

$\{\because x > 0, y > 0, xy < 1\}$

$\Rightarrow \tan^{-1}\left[\dfrac{\frac{7}{24} + \frac{4}{3}}{1 - \frac{7}{24}\cdot\frac{4}{3}}\right]$

$\Rightarrow$

$\tan^{-1}\dfrac{117}{44} = \cot^{-1}\dfrac{44}{117} = \operatorname{cosec}^{-1}\dfrac{125}{117} = \cos^{-1}\dfrac{44}{125}$

11. Ans (A,D)

Ans. (A,D)

$\cos\left(\dfrac{\pi}{2} + \cos^{-1}x\right)$

$= -\sin\left(\cos^{-1}x\right) = -\sqrt{1 - x^2} = -\dfrac{\sqrt{24}}{5}$

MATHS

Section–I(iii) Comprehension Type

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12. Ans (B)

Ans. (B)

$\dfrac{g(f(x))}{h(f(x))} = y = \dfrac{x^2 - 4x + 5}{x^2 - 4x - 12}$

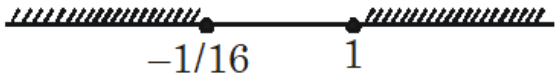
$\Rightarrow (y-1)x^2 + (4 - 4y)x - (12y + 5) = 0$

$D \geq 0$

$\Rightarrow (4 - 4y)^2 + 4(y-1)(12y + 5) \geq 0$

$\Rightarrow 16y^2 - 15y - 1 \geq 0$

$(y - 1)(16y + 1) \geq 0$



But at $y = 1$

$1 = \dfrac{x^2 - 4x + 5}{x^2 - 4x - 12}$

$\Rightarrow -12 = 5$ Not possible

$\therefore$ Range of y : $\left(-\infty, -\dfrac{1}{16}\right] \cup (1, \infty)$

$\therefore$ Integral values : $\{0, 1\}$

13. Ans (C)

Ans. (C)

$g(f(x)) = x^2 - 4x + 5.$

Largest interval for invertible : $\left[-\frac{B}{2A}, \infty\right)$

$\therefore\ a = \frac{-(-4)}{2.1} = 2$

Least value of $h(f(x)) = x^2 - 4x - 12$

$b = -\frac{(16+48)}{4} = -16$

$\therefore$

$$\sin^{-1}\left(\frac{8}{-16}\right) + \cos^{-1}\left(\frac{-8}{-16}\right) + \tan^{-1}\left(\frac{-32}{2\times(-16)}\right)$$

$$= \sin^{-1}\left(-\frac{1}{2}\right) + \cos^{-1}\left(\frac{1}{2}\right) + \tan^{-1}(1)$$

$$= -\frac{\pi}{6} + \frac{\pi}{3} + \frac{\pi}{4} = \frac{\pi}{4} + \frac{\pi}{6} = \frac{5\pi}{12}$$

14. Ans (C)

Ans. (C)

$$\lim_{x\to 7}\lim_{n\to\infty}\frac{(x-7)^{2n}+1}{x^2-4x-12}$$

$$\lim_{x\to 7}\lim_{n\to\infty}\frac{((x-7)^2)^n+1}{x^2-4x-12} = \lim_{x\to 7}\frac{1}{(x-6)(x+2)}$$

$$= \frac{1}{1.9} = \frac{1}{9}$$

MATHS

Section–III Integer Type

1. Ans (2)

Ans. 2

$f(x) = \text{sgn}[\ell n(\underbrace{(x-1)^2+1}_{1 \text{ to } \infty})]$

$= \text{sgn}[\ell n(1 \text{ to } \infty)]$

$= \text{sgn}[0 \text{ to } \infty]$

$\Rightarrow \text{sgn}(0) = 0$ & $\text{sgn}(+ve) = 1$

Number of integers in the range is 2.

2. Ans (2)

Ans. 2

$$f(x) = \frac{1}{(1+\cot x)} = \frac{\tan x}{(1+\tan x)};$$

$$f\left(\frac{\pi}{2} - x\right) = \frac{1}{(1+\tan x)};$$

$$f\left(\frac{\pi}{2} + x\right) = \frac{1}{1+\cot\left(\frac{\pi}{2}+x\right)} = \frac{1}{(1-\tan x)}$$

$$\text{and } f(\pi - x) = \frac{1}{(1-\cot x)}$$

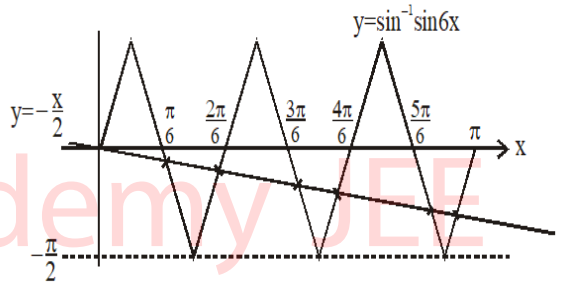
$$\Rightarrow f(x) + f\left(\frac{\pi}{2}-x\right) + f\left(\frac{\pi}{2}+x\right) + f(\pi - x)$$

$$= \left[\frac{\tan x}{1+\tan x} + \frac{1}{1+\tan x}\right] + \left[\frac{1}{1-\tan x} + \frac{1}{1-\cot x}\right]$$

$= 1 + 1 = 2$

3. Ans (7)

Ans. 7



4. Ans (2)

Ans. 2

$-1 \le x - 1 \le 1$ and $2 - x > 0$

$0 \le x \le 2$ and $x < 2$

integral values = {0, 1}

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ANSWER KEY

(Limits) GUIDED REVISION # 11

TIME : 60 Min.

M.M. : 57

SOLUTIONS

MATHS

Section–I(i) Single Correct Type

1. Ans (A)

Ans. (A)

$$R = \tan\left[\lim_{x\to\infty}\sum_{r=1}^{n}\tan^{-1}\left[\frac{1+2r}{1+r^2(r+1)^2}\right]\right]$$

$$T_r = \tan^{-1}\frac{(r+1)^2-r^2}{1+r^2(r+1)^2}$$

$$\therefore T_r = \tan^{-1}(r+1)^2 - \tan^{-1}r^2$$

$$T_1 = \tan^{-1}2^2 - \tan^{-1}1^2$$

$$T_2 = \tan^{-1}3^2 - \tan^{-1}2^2$$

$$\vdots$$

$$T_n = \tan^{-1}(n+1)^2 - \tan^{-1}n^2$$

$$s = T_1 + T_2 \ldots . T_n$$

$$s = \tan^{-1}(n+1)^2 - \frac{\pi}{4}$$

$$R = \tan\left(\lim_{x\to\infty}\left\{\tan^{-1}(n+1)^2 - \frac{\pi}{4}\right\}\right) = \tan\left(\frac{\pi}{2}-\frac{\pi}{4}\right) = 1.$$

2. Ans (D)

Ans. (D)

$$f\left(\frac{2}{x-2}\right) = \frac{1}{\frac{4}{(x-2)^2}-\frac{34}{(x-2)}+66} = \frac{(x-2)^2}{4-34(x-2)+66(x-2)^2} = \frac{(x-2)^2}{2(3x-7)(11x-24)}$$

$\Rightarrow f\left(\frac{2}{x-2}\right)$ is continuous in its domain

3. Ans (B)

Ans. (B)

$$f(x) = \frac{\pi}{2} - \tan^{-1}\left(\frac{2x}{1-x^2}\right)$$

put $\tan^{-1}x = a$ and $\alpha \in \left(0, \frac{\pi}{4}\right)$

$$\therefore f(x) = \frac{\pi}{2} - 2a = \frac{\pi}{2} - 2\tan^{-1}x$$

similarly, $g(x) = 2\tan^{-1}x$

$$\lim_{x\to t}\frac{f(x)-f(t)}{g(x)-g(t)} = \lim_{x\to t}\frac{\left(\frac{\pi}{2}-2\tan^{-1}x\right)-\left(\frac{\pi}{2}-2\tan^{-1}t\right)}{2\tan^{-1}x-2\tan^{-1}t}$$

$$= \lim_{x\to t}\frac{2(\tan^{-1}t-\tan^{-1}x)}{2(\tan^{-1}x-\tan^{-1}t)} = -1$$

4. Ans (B)

Ans.(B)

$$f'(x) = \lim_{h\to0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to0}\frac{f(x-h)-f(x)}{h} = -f'(x)$$

$\Rightarrow f'(x) = 0$

$\Rightarrow f(x) = k$ (constant)

5. Ans (D)

Ans. (D)

Let $y = \sqrt{x+\sqrt{x-\sqrt{x+\ldots\ldots\infty}}}$ &

$z = \sqrt{x-\sqrt{x+\sqrt{x+\ldots\ldots\infty}}}$

$\Rightarrow y = \sqrt{x+z}$ & $z = \sqrt{x-y}$

$\Rightarrow y^2 = x+z$ & $z^2 = x-y$

$\Rightarrow y^2 - z^2 = (z+y) \Rightarrow y - z = 1$

$\Rightarrow y^2 = x+(y-1) \Rightarrow y = \frac{1+\sqrt{4x-3}}{2}$

Now $\lim_{x\to1^+}\frac{f(x)-\left(\frac{\sqrt{x}+1}{2}\right)}{(x-1)} = \lim_{x\to1^+}\frac{\sqrt{4x-3}-\sqrt{x}}{2(x-1)} = \frac{3}{4}$

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6. Ans (B)

Ans. (B)

$$f(x) = \begin{cases} a + (x-b)^2 & b-k \leqslant x \leqslant b+k \\ c + b - x & x < b-k \\ c + x - b & x > b+k \end{cases}$$

continuous at $x = b-k$ and $x = b+k$

$\Rightarrow a + k^2 = c + k$

differentiable at $x = b-k$ and $x = b+k$

$\Rightarrow -2k = -1$ and $2k = 1$

$\Rightarrow k = \frac{1}{2}$

7. Ans (D)

Ans.(D)

$$f(1) = \frac{e + e^{-1}}{e + e^{-1}} = 1$$

$$\text{L.H.L} = \lim_{\substack{x\to 1^- \\ n\to\infty}} \frac{e^{2x} + x^{2n}}{e^{2x}x^{2n} + 1} = \frac{e^2 + 0}{1} = e^2$$

$$\text{R. H. L.} = \lim_{x\to 1^+} \frac{e^{2x} + x^{2n}}{e^{2x}x^{2n} + 1} = \frac{1}{e^2}$$

MATHS

Section–I(ii) Multiple Correct Type

8. Ans (A,B,C)

Ans.(A,B,C)

$$\text{Let } L = \lim_{x\to 0} \left(\frac{a^x + b^x + c^x}{3} \right)^{1/x} \quad (1^\infty \text{ form})$$

$$= e^{\lim_{x\to 0} \frac{a^x+b^x+c^x-3}{3x}} = e^{\lim_{x\to 0} \frac{1}{3}\left(\frac{a^x-1}{x} + \frac{b^x-1}{x} + \frac{c^x-1}{x}\right)} = (abc)^{1/3}$$

since $\frac{a+b+c}{3} \geqslant (abc)^{1/3}$ and $a + b + c = 15$

$\Rightarrow \quad L \leq 5$

$\therefore$ L cannot be 6.

9. Ans (B,C,D)

Ans. B,C,D

$$f(x) = \frac{\sin\frac{\pi}{4}}{1} = \frac{1}{\sqrt{2}} \quad ; \quad 1 \leq x < 2$$

$$= \frac{\sin\frac{\pi}{2}}{2} = \frac{1}{2} \quad ; \quad 2 \leq x < 3$$

Hence f(x) is continuous at $\frac{3}{2}$, differentiable at $\frac{4}{3}$ & discontinuous at 2.

10. Ans (A,C,D)

Ans. (A,C,D)

If f (x) is continuous at x = 0

$$f(0) = \lim_{x\to 0} f(x)$$

$$0 = \lim_{x\to 0} x^m \left(\cos x \cos\frac{1}{x} + \sin x \sin\frac{1}{x} \right)$$

$$0 = \lim_{x\to 0} \left(x^m \cos x \cos\frac{1}{x} + x^{m+1} \left(\frac{\sin x}{x}\right) \sin\frac{1}{x} \right)$$

true when m > 0

If f (x) is differentiable at x = 0

$$\lim_{h\to 0} \frac{f(h) - f(0)}{h} \quad \text{exist} \quad \Rightarrow$$

$$\lim_{h\to 0} \frac{h^m \left(\cosh\cos\frac{1}{h} + \sinh\sin\frac{1}{h}\right) - 0}{h} \quad \text{exist}$$

MATHS

Section–I(iii) Comprehension Type

11. Ans (B)

Ans. (B)

P(x) = g(f (x)) =

$$\begin{cases} (1-x)^4 + (1-x)^2 + 1, & 1 \leqslant x \leqslant 2 \\ (3-x)^4 + (3-x)^2 + 1, & 2 < x \leqslant 3 \end{cases}$$

$P(2^-) = P(2^+) = 3 \Rightarrow$ P is continuous at x = 2

$$P'(2) = \begin{cases} 4(x-1)^3 + 2(x-1), & 1 < x < 2 \\ 4(x-3)^3 + 2(x-3), & 2 < x < 3 \end{cases}$$

$P'(2^-) = 6$

$P'(2^+) = -6 \Rightarrow$ P is non-derivable at x = 2

12. Ans (A)

Ans. (A)

$$h(f(x)) = \begin{cases} (1-x)^3, & 1 \leqslant x \leqslant 2 \\ (3-x)^3, & 2 < x \leqslant 3 \end{cases}$$

$h(f(2^-)) = -1$

$h(f(2^+)) = 1$

$\therefore$ h(f (x)) is discontinuous at x = 2

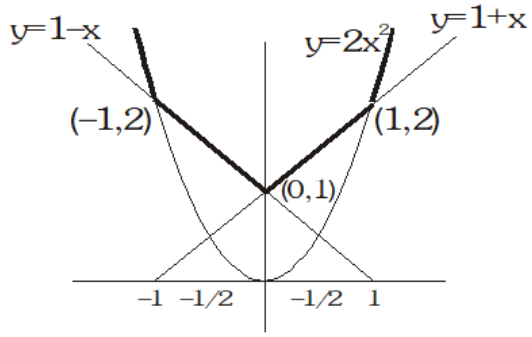
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MATHS

Section–III Integer Type

1. Ans (3)

Ans. 3



from graph

$$f(x)=\begin{cases} 2x^2 & x<-1 \\ 1-x & -1\le x<0 \\ 1+x & 0\le x<1 \\ 2x^2 & x\ge 1 \end{cases}$$

hence function is continuous every where but non-differentiable at 3 points

$\Rightarrow$ $p=0$ & $q=3$ $\Rightarrow$ $p+q=3$

2. Ans (5)

Ans. 5

$|f(x)\,f(x-y)|-|f(y)\,f(x-y)|\le (x-y)^2$

$\Rightarrow |f(x-y)|\,(|f(x)|-|f(y)|)\le (x-y)^2$

$\Rightarrow f(x-y)\,(f(x)-f(y))\le (x-y)^2$ ($\because f(x)\in R^+$)

Case-I : $x>y$ $\left\{\dfrac{f(x)-f(y)}{x-y}\right\}\le\{x-y\}$

$f(x-y)$

$\Rightarrow \underset{y\to x}{\text{Lim}}\, f(x-y)\left\{\dfrac{f(y)-f(x)}{y-x}\right\}\le \underset{y\to x}{\text{Lim}}\{x-y\}$

$\Rightarrow f(0).\,Lf'(x)\le 0$ ($Lf'(x)$ denotes LHD of $f(x)$)

$\Rightarrow Lf'(x)\le 0$

Case-II : $x<y$

$f(x-y)\left\{\dfrac{f(x)-f(y)}{x-y}\right\}\ge\{x-y\}$

$\Rightarrow \underset{y\to x}{\text{Lim}}\, f(x-y)\left\{\dfrac{f(y)-f(x)}{y-x}\right\}\ge \underset{y\to x}{\text{Lim}}+\{x-y\}$

$\Rightarrow f(0).\,Rf'(x)\ge 0$

$\Rightarrow Rf'(x)\ge 0$ ($Rf'(x)$ denotes RHD of $f(x)$)

$\because f(x)$ is differentiable everywhere

$\therefore f'(x)=0$

$\Rightarrow f(x)=$ constant

$\Rightarrow f(5)=5$ ($\because f(0)=5$)

3. Ans (2)

Ans. 2

$\because$ $f(x)$ is differentiable in $\left(0,\dfrac{1}{\sqrt{2}}\right)$

$\therefore$ $f(x)$ is continuous in $\left[0,\dfrac{1}{\sqrt{2}}\right]$

$\Rightarrow f\left(\dfrac{1}{2}\right)=\lim\limits_{x\to\frac{1}{2}} f(x)$ $\Rightarrow$

$\lim\limits_{x\to\frac{1}{2}^-}\tan^{-1}\left(\dfrac{ax+b}{c}\right)=0$

and $\lim\limits_{x\to\frac{1}{2}^+}\log\left(2+bx^2\right)=0$

$\Rightarrow \dfrac{a}{2}+b=0$ and $2+\dfrac{b}{4}=1$

$\Rightarrow a+2b=0$ and $b=-4$ $\Rightarrow$ $a=8$

Now, $\because f(x)$ is differentiable at $x=\dfrac{1}{2}$

$\Rightarrow \dfrac{1}{1+\left(\frac{ax+b}{c}\right)^2}\cdot\dfrac{a}{c}\Big|_{x=\frac{1}{2}}=\dfrac{1}{2+bx^2}.2bx\Big|_{x=\frac{1}{2}}$

$\Rightarrow \dfrac{8}{c}=\dfrac{-4}{2-4\left(\frac{1}{4}\right)}$ $\Rightarrow$ $c=-2$

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MATHS

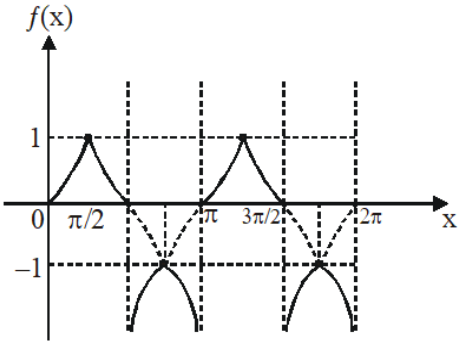
Section–IV Matrix Match Type

1. **Ans (A->R,B->P,C->S,D->Q)**

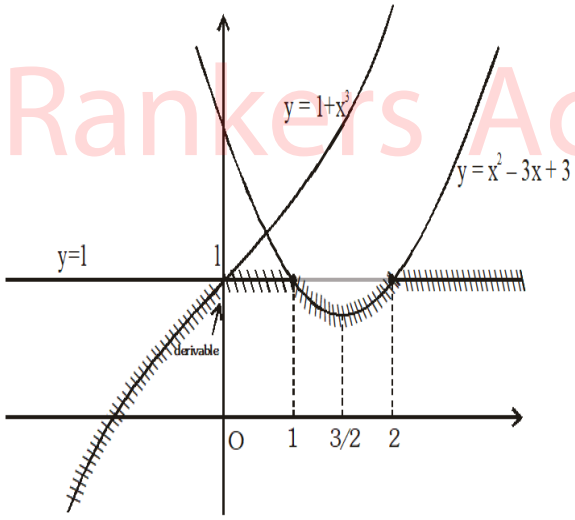
(A) $f(x) = \dfrac{\tan x + \cot x}{2} - \left|\dfrac{\tan x - \cot x}{2}\right|$

$$f(x) = \begin{cases} \cot x & , \quad \tan x \geqslant \cot x \\ \tan x & , \quad \tan x < \cot x \end{cases}$$

There are 4 points where the function is continuous but not differentiable in $(0, 2\pi)$



(B)



This hatched graph is for $f(x)$.

$\therefore$ $f(x)$ is non-derivable at two pints

Ans = option (P)

(C) $f'(x) = (x+4)^{1/3}$

$$f'(x) = \frac{1}{3}(x+4)^{-2/3}$$

Not derivable at $x = -4$

(D) $f\left(\dfrac{\pi}{2}\right) = \dfrac{\pi}{2}$

$$f\left(\frac{\pi^-}{2}\right) = \lim_{x \to \frac{\pi^-}{2}} \left(-\frac{\pi}{2}\ell n\left(\frac{2x}{\pi}\right) + \frac{\pi}{2}\right)$$

$$= \frac{\pi}{2}$$

$$f\left(\frac{\pi^+}{2}\right) = \lim_{x \to \frac{\pi^+}{2}} \left(\sin^{-1}(\sin x)\right) = \frac{\pi}{2}$$

$\Rightarrow f(x)$ is continuous at $x = \dfrac{\pi}{2}$

$$f(x) = \begin{cases} -\frac{\pi}{2}\ell n\left(\frac{2x}{\pi}\right) + \frac{\pi}{2} & ; \quad 0 < x \leqslant \frac{\pi}{2} \\ \pi - x & ; \quad \frac{\pi}{2} < x < \frac{3\pi}{2} \end{cases}$$

$$f'(x) = \begin{cases} -\frac{\pi}{2x} & ; \quad 0 < x \leqslant \frac{\pi}{2} \\ -1 & ; \quad \frac{\pi}{2} < x < \frac{3\pi}{2} \end{cases}$$

$$f'\left(\frac{\pi^-}{2}\right) = f'\left(\frac{\pi^+}{2}\right)$$

$\Rightarrow f(x)$ is differentiable at $x = \dfrac{\pi}{2}$

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2. Ans (A->Q,B->P,C->S,D->R)

(A) $f(x)=\begin{cases} 1-1=0 & ; \quad 1<x\leqslant 2 \\ 0 & ; \quad x=1 \\ 1-x & ; \quad 0\leqslant x<1 \\ -\sin\pi x & ; \quad -1\leqslant x<0 \end{cases}$

at x = 0, f(x) is not continuous & not differentiable

at x = 1, f(x) is continuous & not differentiable

at x = 2 and − 1, f(x) is continuous & differentiable

(B) $\text{LHD}=f'(0^-)=\lim\limits_{h\to 0}\dfrac{f(0-h)-f(0)}{-h}$

$=\lim\limits_{h\to 0}\dfrac{h^2e^{-\frac{1}{h}}-0}{-h}$

$=\lim\limits_{h\to 0}\left(-h.\,e^{-\frac{1}{h}}\right)=0$

Ans. = Option (P)

(C) $f(x)=\dfrac{x}{x+1}$, not defined at x = − 1

$g(x)=\dfrac{f(x)}{f(x)+2}$

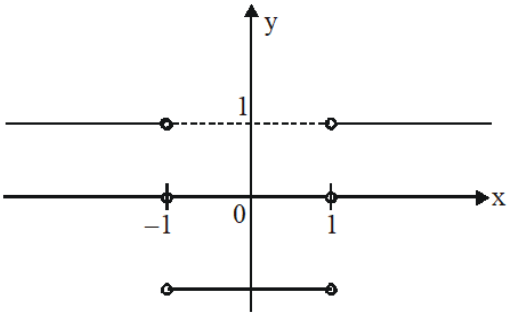
g(x) is not defined at f(x) = − 2

$\dfrac{x}{x+1}=-2 \quad \Rightarrow x=\dfrac{-2}{3}$

Also x = 0 is not in the domain of f(x)

So, at 3 points g(x) is not differentiable.

(D) $y=\begin{cases} 0 & ; \quad x=\pm 1 \\ -1 & ; \quad |x|<1 \\ 1 & ; \quad |x|>1 \end{cases}$



not continuous at x = 1,−1

∴ not differentiable at x = 1,−1

so no tangent exist at x = 1,−1(2 Points)

Ans. = option (r)

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SOLUTIONS

MATHS

Section–I(i) Single Correct Type

1. Ans (B)

Ans. (B)

$g(x) = f^{-1}(x) \Rightarrow fog(x) = x$

$\Rightarrow f'(g(x))g'(x) = 1 \Rightarrow g'(x) = \frac{1}{f'(g(x))}$

$\Rightarrow g'(2) = \frac{1}{f'(g(2))} = \frac{1}{f'(a)} = \frac{1+a^2}{a^{10}}$

2. Ans (C)

Ans.(C)

By eliminating θ, we get

$y^2 - x^2 = 4 \Rightarrow 2y\frac{dy}{dx} - 2x = 0$

$\Rightarrow y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 1 \Rightarrow y\frac{d^2y}{dx^2} + \frac{x^2}{y^2} = 1$

$y^3\frac{d^2y}{dx^2} = y^2 - x^2 = 4$

3. Ans (B)

Ans. (B)

$f(x) = \sqrt{(x-1)+4-4\sqrt{x-1}} + \sqrt{(x-1)+9-6\sqrt{x-1}}$

$= |\sqrt{x-1}-2| + |\sqrt{x-1}-3|$

at x=1.5

$f(x) = (2-\sqrt{x-1}) + (3-\sqrt{x-1}) = 5 - 2\sqrt{(x-1)}$

$f'(x) = \frac{-2}{2\sqrt{x-1}}$

$f'(1.5) = -\frac{1}{\sqrt{\frac{3}{2}-1}} = -\sqrt{2}$

4. Ans (A)

Ans. (A)

Slope of normal = 3

$\Rightarrow$ slope of tangent $f'(0) = -\frac{1}{3}$

$L = \lim_{x\to 1} \frac{(x-1)^2}{f(x-1)^2 - 5f\{4(x-1)^2\} + 4f\{7(x-1)^2\}}$ $\left(\frac{0}{0} \text{ form}\right)$

$= \lim_{x\to 1} \frac{2(x-1)}{2(x-1)f'(x-1) - 40f'\{4(x-1)^2\}(x-1) + 56f'\{7(x-1)^2\}(x-1)}$

$= \lim_{x\to 1} \frac{2}{2f'(x-1) - 40f'\{4(x-1)^2\} + 56f'\{7(x-1)^2\}}$

$= \frac{1}{f'(0) - 20f'(0) + 28f'(0)} = \frac{1}{-\frac{1}{3} + \frac{20}{3} - \frac{28}{3}} = -\frac{1}{3}$

5. Ans (D)

Ans. (D)

$\because$ Function is continuous at x = 0 then

$f(0) = \lim_{x\to 0}(x+1)^{\cot x}$ (1^∞ for m)

$= e^{\lim_{x\to 0} \cot x(x+1-1)}$

$= e^{\lim_{x\to 0} \frac{x}{\tan x}} = e$

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6. Ans (A)

Ans. (A)

Let $x = 1 + t$

$$\lim_{t\to0}\frac{\ell n(\sin^{-1}(\ell n(1+t))}{1/t}\to\frac{\infty}{\infty}$$

$$\lim_{t\to0}\frac{1}{\sin^{-1}(\ell n(1+t))}\frac{1}{\sqrt{1-\ell n^2(1+t)}}\cdot$$

$$\frac{1}{(1+t)\left(-\frac{1}{t^2}\right)} = -\lim_{t\to0}\frac{t^2}{\sin^{-1}\ell n(1+t)} = -\lim_{t\to0}\frac{t^2}{\sin^{-1}\ell n(1+t)}\frac{\ell n(1+t)}{\ell n(1+t)}$$

$$= -\lim_{t\to0} t.\frac{t}{\ell n(1+t)} = 0$$

7. Ans (B)

$\because g(x) = g(-x) \Rightarrow g'(x) = -g'(-x)$

$\Rightarrow g'(0) = 0$

$$f'(0+) = \lim_{h\to0}\frac{g(0+h)\cos\left(\frac{1}{0+h}\right) - f(0)}{h}$$

$$= \lim_{h\to0}\frac{g(h)\cos\left(\frac{1}{h}\right)}{h} = \lim_{h\to0}\frac{g(h)-g(0)}{h}\cos\frac{1}{h} \ (\because g(0) = 0)$$

= g'(0)x a finite value

$$f'(0^-) = \lim_{h\to0} = \frac{g(0-h)\cos\left(\frac{1}{0-h}\right) - f(0)}{-h}$$

$$= \lim_{h\to0}\frac{g(-h)\cos\left(\frac{-1}{h}\right)}{-h} = 0 \text{ (similarly)}$$

$\therefore f'(0) = 0$

8. Ans (C)

Ans. (C)

Numerator =

$$\lim_{x\to\infty}\cos^{-1}\left(\frac{\sqrt{x+\sqrt{x}}-\sqrt{x}}{\sqrt{x+\sqrt{x+\sqrt{x}}+\sqrt{x+\sqrt{x}}}}\right)$$

=

$$\lim_{x\to\infty}\cos^{-1}\left(\frac{\sqrt{x}}{\left(\sqrt{x+\sqrt{x+\sqrt{x}}+\sqrt{x+\sqrt{x}}}\right).\left(\sqrt{x+\sqrt{x}}+\sqrt{x}\right)}\right)$$

$= \cos^{-1} 0 = \frac{\pi}{2}$

$$\text{Denominator} = \lim_{x\to\infty}\sec^{-1}\left(-\ell n\left(\frac{x^2+2}{x^2+1}\right)^{x^2}\right) =$$

$$\sec^{-1}\left(-\ell n e^{\lim_{x\to\infty}\frac{x^2}{x^2+1}}\right) = \sec^{-1}(-1) = \pi$$

$$\therefore \text{Required limit} = \frac{\pi/2}{\pi} = \frac{1}{2}$$

9. Ans (D)

Ans. (D)

$$\lim_{n\to\infty}\frac{g\left(2009+\frac{1}{n}\right)-g(2009)}{\frac{1}{n}} = 27 \Rightarrow g'(2009) = 27$$

$$\lim_{\alpha\to\frac{\pi}{2}} g(2009+\cos\alpha) = 3 \Rightarrow g(2009) = 3$$

$$\lim_{t\to0^+}\left(\frac{g(2009)}{g(2009+t)}\right)^{1/t} \text{ is } 1^\infty \text{ form}$$

$$e^{\lim_{t\to0}\frac{1}{t}\left(\frac{g(2009)-g(2009+t)}{g(2009+t)}\right)} \Rightarrow$$

$$e^{-\lim_{t\to0}\left(\frac{g(2009+t)-g(2009)}{t}\right)\frac{1}{g(2009+t)}} = e^{\frac{-g'(2009)}{g(2009)}} \Rightarrow e^{-9}$$

10. Ans (D)

Ans.(D)

function is always continuous at every point in its domain.

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MATHS

Section–I(ii) Multiple Correct Type

11. Ans (A,C)

Ans. A,C

$f^{-1}(x) = g(x) \Rightarrow x = f(g(x))$

Differentiating both sides,

$1 = f'(g(x))\, g'(x) \Rightarrow g'(x) = \dfrac{1}{f'(g(x))}$

Now $f'(x) = 2x + 3$

So $g'(x) = \dfrac{1}{2g(x)+3} \Rightarrow g'(1) = \dfrac{1}{2(g(1))+3}$

$\because f(x) = x^2 + 3x - 3 \Rightarrow f(1) = 1 \Rightarrow g(1) = 1$

$\Rightarrow g(1) = 1$

$\Rightarrow g'(1) = \dfrac{1}{5}$

12. Ans (A,B,C)

Ans. (A,B,C)

(A) for a = 0

$$L = \lim_{x\to\infty}\left(\frac{2}{\sqrt{4+x^2}}\right)^{\frac{1}{x^2+1}} \quad (0^\circ \text{ from})$$

$$\log L = \lim_{x\to\infty}\frac{1}{x^2+1}\log\left(\frac{2}{\sqrt{4+x^2}}\right)$$

$$\log L = \lim_{x\to\infty}\frac{\log 2 - \frac{1}{2}\log(4+x^2)}{x^2+1} \quad \left(\frac{0}{0}\text{ form}\right)$$

$$= \lim_{x\to\infty}\frac{-\frac{x}{4+x^2}}{2x} \text{ (applying L'Hopital's rule)}$$

$= 0$

$\Rightarrow L = e^0 = 1$

(B) For a = 1

$$L = \lim_{x\to\infty}\left(\frac{x+2}{\sqrt{4+x^2}}\right)^{\frac{x^3+1}{x^2+1}} \quad (1^\infty \text{ form})$$

$$L = e^{\lim_{x\to\infty}\left(\frac{x+2-\sqrt{4+x^2}}{\sqrt{4+x^2}}\right)\frac{x^3+1}{x^2+1}}$$

$$L = e^{\lim_{x\to\infty}\left(\frac{4x}{\sqrt{4+x^2}\left(x+2+\sqrt{4+x^2}\right)}\right)\frac{x^3+1}{x^2+1}}$$

$$L = e^{\lim_{x\to\infty}\left(\frac{4}{\sqrt{\frac{4}{x^2}+1}\left(1+\frac{2}{x}+\sqrt{\frac{4}{x^2}+1}\right)}\right)\left(\frac{1+\frac{1}{x^3}}{1+\frac{1}{x^2}}\right)}$$

$L = e^{4/2} = e^2$

(C) For $0 < a < 1$ $L = \lim_{x\to\infty}(a)^\infty = 0$

13. Ans (B,D)

Ans. B,D

Since $\sin^{-1} x$ and $\cos\dfrac{1}{x}$ are continuous & differentiable in $x \in [-1, 1] - \{0\}$

Now at x = 0

$$f'(0^-) = \lim_{h\to0}\frac{(\sin^{-1}(0-h))^2\cos\left(\frac{-1}{h}\right)-0}{-h} = 0$$

$$f'(0^+) = \lim_{h\to0}\frac{(\sin^{-1}h)^2\cos\left(\frac{1}{h}\right)-0}{h} = 0$$

Hence LHD = RHD

so f(x) is continuous & differentiable every where in $-1 \le x \le 1, x \ne 0$

MATHS

Section–I(iii) Comprehension Type

15. Ans (B)

Ans. (B)

$F'(x) = -1$ (constant function) in $(-4,-1)$

$\Rightarrow F(x) = -x + c$ i.e. linear function

16. Ans (C)

Ans. (C)

From graph, it is clear that F'(x) is not defined at x = −4,−1 and 4

MATHS

Section–III Integer Type

1. Ans (4)

Ans. 4

$f'(x) = 3x^2 + 2xf'(1) + f''(2)$ .....(i)

$f''(x) = 6x + 2f'(1)$ .....(ii)

Now put x = 1 in(i)

$f'(1) = 3 + 2f'(1) + f''(2)$

$f'(1) + f''(2) + 3 = 0$ .....(iii)

and x = 2 in (ii)

$f''(2) = 12 + 2f'(1)$

$2f'(1) - f''(2) + 12 = 0$ .....(iv)

by (iii) & (iv)

$f'(1) = -5,\ f''(2) = 2$

also $f'''(x) = 6 \Rightarrow f'''(4) = 6$ .....(v)

$\Rightarrow f(x) = x^3 - 5x^2 + 2x + 6$

$f(1) = 1 - 5 + 2 + 6 = 4$

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2. Ans (0)

Ans. 0

$y(x) = \sin(x - \sin x)$

$\Rightarrow \quad y'(x) = \cos(x - \sin x).(1 - \cos x)$

$\Rightarrow \quad y''(x) = (1 - \cos x)[-\sin(x - \sin x)(1 - \cos x)] + \cos(x - \sin x).\sin x$

put $x = \frac{\pi}{2}$

$\Rightarrow \quad y''\left(\frac{\pi}{2}\right) = -\sin\left(\frac{\pi}{2} - 1\right) + \cos\left(\frac{\pi}{2} - 1\right) = -\cos 1 + \sin 1$

$\Rightarrow \quad y''\left(\frac{\pi}{2}\right) + \cos 1 = \sin 1 \quad \Rightarrow \quad \left[y''\left(\frac{\pi}{2}\right) + \cos 1\right] = [\sin 1] = 0$

3. Ans (2)

Ans. 2

$$b = \lim_{x\to 1} \frac{(7+x^3)^{1/a} - (3+x^2)^{1/2}}{(x-1)}$$

if limit exist, then $8^{1/a} = 4^{1/2}$

$$b = \lim_{x\to 1} \frac{(7+x^3)^{1/3} - (3+x^2)^{1/2}}{(x-1)} = -\frac{1}{4}$$

$$b = \lim_{x\to 1} \frac{(8+x^3-1)^{\frac{1}{3}} - (4+x^2-1)^{\frac{1}{2}}}{x-1}$$

$$= \lim_{x\to 1} \frac{2\left[1+\frac{x^3-1}{8}\right]^{\frac{1}{3}} - 2\left[1+\frac{x^2-1}{4}\right]^{\frac{1}{2}}}{x-1}$$

$$= 2\lim_{x\to 1} \frac{\left[1+\frac{1}{3}.\left(\frac{x^3-1}{8}\right)+...\right] - \left[1+\frac{1}{2}\left(\frac{x^2-1}{4}\right)+..\right]}{x-1}$$

$$= 2\left(\frac{1}{8} - \frac{1}{4}\right) = -\frac{1}{4}$$

$\Rightarrow a + 4b = 2$

4. Ans (3)

Ans. 3

$$f(x) = \sin\left([x]\frac{\pi}{2}\right).\text{sgn}(x-1)(x-2)$$

$$= \begin{cases} \sin\left([x]\frac{\pi}{2}\right) & x < 1 \text{ or } x > 2 \\ -\sin\left([x]\frac{\pi}{2}\right) & 1 < x < 2 \\ 0 & x = 1,\ 2 \end{cases}$$

$$= \begin{cases} 0 & x \in [0,\ 1) \\ 0 & x = 1 \\ -1 & x \in (1,2) \\ 0 & x = 2 \\ 0 & x \in (2,3) \\ -1 & x = 3 \end{cases} = \begin{cases} 0 & 0 \leqslant x \leqslant 1 \\ -1 & 1 < x < 2 \\ 0 & 2 \leqslant x < 3 \\ -1 & x = 3 \end{cases}$$

$\Rightarrow$ function is discondinuous at x = 1, 2, 3

MATHS

Section–IV Matrix Match Type

1. Ans (A->R,B->QS,C->R,D->T)

Ans. (A) $\longrightarrow$ (R); (B) $\longrightarrow$ (Q,S); (C) $\longrightarrow$ (R); (D) $\longrightarrow$ (T)

$$f(x) = \begin{cases} 1, & x = 0 \\ 0, & 0 < x < \frac{\pi}{2} \\ 1, & x = \frac{\pi}{2} \\ -1, & \frac{\pi}{2} < x \leqslant \pi \\ 0, & \pi < x < \frac{3\pi}{2} \\ -1, & \frac{3\pi}{2} \leqslant x \leqslant 2\pi \end{cases}$$

Number of points where $f(x)$ is discontinuous & nonderivable = 4

at $x = 0,\ \frac{\pi}{2},\ \pi,\ \frac{3\pi}{2}$

$\lim_{x\to a} f(x)$ does not exist

for $x = \frac{\pi}{2},\ \pi,\ \frac{3\pi}{2}$

$f(x) = -1$ (Minimum) for $x \in \left(\frac{\pi}{2},\ \pi\right] \cup \left[\frac{3\pi}{2},\ 2\pi\right]$

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GUIDED REVISION

JEE(Main + Advanced) 2025
ENTHUSIAST+LEADER COURSE
SCORE ADVANCED

ANSWER KEY

(Indefinite & Definite Integration) GUIDED REVISION # 13

SOLUTIONS

MATHS

Section–I(i) Single Correct Type

1. Ans (C)

Ans. (C)

$$\int_{-T/2}^{3T/2} f(x)dx = 18 \Rightarrow \int_{0}^{2T} f(x)dx = 18 \Rightarrow \int_{0}^{T} f(x)dx = 9$$

$$\int_{-a}^{a+5T} f(x)dx = \int_{-a}^{a} f(x)dx + \int_{a}^{a+5T} f(x)dx = 2$$

$$\int_{0}^{a} f(x)dx + 5\int_{0}^{T} f(x)dx = 2 \times 3 + 5 \times 9 = 51$$

2. Ans (C)

Ans. (C)

$$\int_{3/2}^{15} [\ell n[x]]dx = \int_{3/2}^{3} 0.dx + \int_{3}^{8} 1.dx + \int_{8}^{15} 2.dx = 0 + (8 - 3) + 2(15 - 8) = 19$$

3. Ans (D)

Ans. (D)

$$\int_{-2}^{4} f(x)dx = \int_{-2}^{-1} f(x)dx + \int_{-1}^{0} f(x)dx + \int_{0}^{1} f(x)dx + \int_{1}^{2} f(x)dx + \int_{2}^{3} f(x)dx + \int_{3}^{4} f(x)dx = 2^2 + 1^2 + 0 + 1^2 + 2^2 + 3^2 = 19$$

4. Ans (B)

Ans. (B)

$$I = \int_{-1}^{1} \frac{x\tan^{-1}x}{e^{\tan^{-1}x} + 1} dx \quad \text{.....(i)}$$

$$\text{Also } I = \int_{-1}^{1} \frac{x\tan^{-1}x}{1 + e^{-\tan^{-1}x}} dx \quad \text{.....(ii)}$$

add (i) & (ii)

$$2I = \int_{-1}^{1} x\tan^{-1}x\, dx$$

$$I = \int_{0}^{1} x\tan^{-1}x = \frac{x^2}{2}\tan^{-1}x\Big|_0^1 - \frac{1}{2}\int_{0}^{1} \frac{x^2}{x^2+1} dx$$

$$I = \frac{\pi}{8} - \frac{1}{2}\int_{0}^{1} \frac{x^2+1-1}{x^2+1} = \frac{\pi}{8} - \frac{1}{2} + \frac{1}{2}(\tan^{-1}x)_0^1$$

$$I = \frac{\pi}{4} - \frac{1}{2}$$

5. Ans (D)

Ans. (D)

$x^2 - 4x + 6 = A(x^2 - 2x + 4) + B(2x - 2) + C$

$\Rightarrow A = 1, B = -1, C = 0$

$$\Rightarrow \int \frac{x^2 - 4x + 6}{x^2 - 2x + 4}dx = \int Adx + B\int \frac{dt}{t} + 0$$

where $t = x^2 - 2x + 4$

$= x - \ell n|x^2 - 2x + 4| + C$

6. Ans (A)

Ans. (A)

$$I = \int \frac{\left(1 - \frac{\sin^2 x}{\cos^2 x}\right)}{\frac{1}{\cos^2 x}} \frac{dx}{\sin 2x}$$

$$= \int \cot 2x dx = \frac{\ell n|\sin 2x|}{2} + C$$

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7. Ans (D)

Ans.(D)

$I = \int e^x (\ell nx + x\ell nx + 1)\,dx$

$= \int e^x (x\ell nx + (1 + \ell nx))\,dx \Rightarrow e^x f(x) + c$

$I = e^x.x\ \ell n\ x + C$

$\Rightarrow f(x) = e^x.x\ \ln x + C$

$f(1) = 0 \ \Rightarrow \ C = 0$

$\Rightarrow f(e) = e^{e+1}$

MATHS

Section–I(ii) Multiple Correct Type

8. Ans (A,B,C,D)

Ans. (A,B,C,D)

$\text{Let } I = \int_0^{\pi} |\sin \lambda x|\,dx, \ \lambda \in N$

put $\lambda x = t \ \Rightarrow \ \lambda dx = dt$

$\Rightarrow I = \frac{1}{\lambda}\int_0^{\lambda\pi} |\sin t|dt = \int_0^{\pi} |\sin t|dt = 2$

Using this

$I_{(2)} = \int_0^{\pi} (1 - |\sin x|)dx = \pi - 2$

$I_{(3)} = \int_0^{\pi} (1 - |\sin x| + |\sin 2x|)dx$

$= \int_0^{\pi} 1dx - \int_0^{\pi} |\sin x|dx + \int_0^{\pi} |\sin 2x|dx \ = \pi - 2 + 2$

$= \pi$

Similarly $I_{(2008)} = \pi - 2$ & $I_{(2009)} = \pi$

9. Ans (A,B,C)

Ans. (A,B,C,)

$I_n = \int_0^1 (1 - x^2)^n dx = (1 - x^2)^n \cdot x|_0^1 + 2n\int_0^1 (1 - x^2)^{n-1} x^2 dx = 2n\int_0^1 (1 - x^2)^{n-1} \{1 - (1 - x^2)\}\,dx$

$I_n = 2nI_{n-1} - 2nI_n \ \Rightarrow I_n = \frac{2n}{2n+1} I_{n-1}$

$I_1 = \int_0^1 (1 - x^2)dx = \frac{2}{3}$

$I_2 = \frac{4}{5}\cdot\frac{2}{3}, \ I_3 = \frac{6.4.2}{7.5.3}$

$\vdots$

$I_n = \frac{2.4.6.....2n}{3.5.7.....(2n+1)} = \frac{2^n n!}{3.5.7.....(2n+1)}$

10. Ans (A,B,C)

Ans.(A,B,C)

$\int \frac{3e^x + 5e^{-x}}{4e^x - 5e^{-x}} dx$

$(3e^x + 5e^{-x}) = P(4e^x - 5e^{-x}) + Q(4e^x + 5e^{-x})$

$4P + 4Q = 3 \ \& \ 5Q - 5P = 5 \Rightarrow P = -\frac{1}{8}, Q = \frac{7}{8}$

$\int \left(-\frac{1}{8} + \left(\frac{7}{8}\right)\frac{4e^x + 5e^{-x}}{4e^x - 5e^{-x}}\right)dx = -\frac{x}{8} + \frac{7}{8}$

$\ell n|4e^x - 5e^{-x}| + C$

$= -\frac{x}{8} + \frac{7}{8}\ell n|4e^{2x} - 5| - \frac{7}{8}x + C$

$= -x + \frac{7}{8}\ell n|4e^{2x} - 5| + C$

$A = -1, B = \frac{7}{8}$, C is an arbitrary constant.

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11. **Ans (A,C,D)**

Ans. (A,C,D)

$$f^3(x) = \int_0^x t f^2(t)\,dt$$

Diffentiate both sides

$$3f^2(x)f'(x) = xf^2(x)$$

$$\Rightarrow f'(x) = \frac{x}{3} \quad \text{As } x \geq 0$$

$\Rightarrow f(x)$ is increasing

$$\Rightarrow \boxed{f(x) = \frac{x^2}{6}} \quad \text{as } f(0) = 0$$

MATHS

Section–III Integer Type

1. **Ans (2)**

Ans. 2

$$\frac{f'(x)}{f(x)} = \frac{f'(-x)}{f(-x)}$$

Integrate both the sides

$$\ell n f(x) = -\ell n f(-x) + \ln c$$

$$f(x)\,f(-x) = c$$

Put $x = 0 \Rightarrow c = 1$

$$f(x)\,f(-x) = 1$$

$$I = \int_{-2}^{2} \frac{dx}{1+f(x)} \qquad \ldots.(1)$$

$$I = \int_{-2}^{2} \frac{dx}{(1+f(-x)}$$

$$I = \int_{-2}^{2} \frac{dx}{\left(1+\frac{1}{f(x)}\right)} = \int_{-2}^{2} \frac{f(x)dx}{1+f(x)} \qquad \ldots(2)$$

$(1) + (2) \Rightarrow 2I = 4 \Rightarrow I = 2$

2. **Ans (4)**

Ans. 4

$$I = \int_{1/2}^{2} e^{|x-1/x|}dx \qquad \ldots(i)$$

Let $x = \frac{1}{t} \Rightarrow dx = -\frac{dt}{t^2}$

$$I = -\int_{2}^{1/2} \frac{1}{t^2} e^{\left|\frac{1}{t}-t\right|}dt$$

$$I = \int_{1/2}^{2} \frac{1}{t^2} e^{\left|t-\frac{1}{t}\right|}dt \qquad \ldots(ii)$$

adding equation (i), (ii), we get

$$2I = \int_{1/2}^{2} \left(1+\frac{1}{t^2}\right) e^{\left|t-\frac{1}{t}\right|}dt$$

$$= \int_{1/2}^{1} \left(1+\frac{1}{t^2}\right) e^{-t+\frac{1}{t}}dt + \int_{1}^{2} \left(1+\frac{1}{t^2}\right) e^{t-\frac{1}{t}}dt$$

Put $e^{-t+\frac{1}{t}} = k$ & $e^{t-\frac{1}{t}} = m$

$$\left(e^{-t+\frac{1}{t}}\right)\left(-1-\frac{1}{t^2}\right)dt = dk \quad \&$$

$$\left(1+\frac{1}{t^2}\right) e^{\left(t-\frac{1}{t}\right)}dt = dm$$

$$-\int_{e^{3/2}}^{1} dk + \int_{1}^{e^{3/2}} dm = 2I$$

$$2I = 2\int_{1}^{e^{3/2}} dm = 2\left[e^{3/2}-1\right]$$

$$I = \left(e^{3/2}-1\right) = \lambda_1\left(e^{\lambda_2/2}-1\right)$$

$\lambda_1 = 1$ and $\lambda_2 = 3$

$\therefore \lambda_1 + \lambda_2 = 4$

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3. Ans (4)

Ans. 4

$f(x) = \sin(\cos x) + \cos(\sin x)$ is periodic with period 2π

$\Rightarrow$ From Jack property

$$\Rightarrow I_1 = 50\int_0^{2\pi}((\sin(\cos x) + \cos(\sin x))dx$$

$$\Rightarrow I_1 = 100\int_0^{\pi}(\sin(\cos x) + \cos(\sin x)dx \quad .....(i)$$

(From queen property $\because f(x) = f(2\pi - x)$)

$$\Rightarrow I_1 = 100\int_0^{\pi}(-\sin(\cos x) + \cos(\sin x)dx \quad .....(ii)$$

(From king property)

Add (i) & (ii)

$$\Rightarrow 2I_1 = 200\int_0^{\pi}\cos(\sin x)dx \quad \Rightarrow \quad I_1 = 100\int_0^{\pi}\cos(\sin x)dx$$

$$I_1 = 200\int_0^{\pi/2}\cos(\sin x)dx$$

(Queen property)

Put sinx = t

cosxdx = dt

$$\Rightarrow dx = \frac{dt}{\sqrt{1-t^2}} \quad \Rightarrow \quad I_1 = 200\int_0^1\frac{\cos t dt}{\sqrt{1-t^2}}$$

$$\Rightarrow I_1 = 200I_2 \Rightarrow \frac{I_1}{50I_2} = 4$$

4. Ans (7)

Ans. 7

Limit of a sum

$$\Rightarrow \frac{r}{n} \equiv x \ \& \ \frac{1}{n} \equiv dx$$

$$L = \lim_{n\to\infty}\sum_{r=n}^{2n}\left(\frac{1}{\frac{r^3}{n^3}\sqrt{\frac{r^4}{n^4}+1}} \times \frac{1}{n}\right) = \int_1^2\frac{dx}{x^3\sqrt{x^4+1}}$$

$$= \int_1^2\frac{dx}{x^5\sqrt{1+\frac{1}{x^4}}}$$

Let $1 + \frac{1}{x^4} = t^2$

$$\Rightarrow \frac{dx}{x^5} = -\frac{tdt}{2}$$

$$\Rightarrow L = -\frac{1}{2}\int_{\sqrt{2}}^{\sqrt{17}/4}\frac{tdt}{t} = -\frac{1}{2}\left(\frac{\sqrt{17}}{4} - \sqrt{2}\right) =$$

$$\frac{1}{\sqrt{2}} - \frac{\sqrt{17}}{8} = \sqrt{\frac{1}{2}} - \sqrt{\frac{17}{64}}$$

$$\Rightarrow \frac{32b}{a} - 10 = 7$$

5. Ans (3)

Ans. 3

$$I = \int\left(x^{12} + x^8 + x^4\right)\left(2x^8 + 3x^4 + 6\right)^{1/4}dx$$

$$= \int\left(x^{11} + x^7 + x^3\right)\left(2x^{12} + 3x^8 + 6x^4\right)^{1/4}dx$$

Put $2x^{12} + 3x^8 + 6x^4 = t$

$$\Rightarrow 24(x^{11} + x^7 + x^3)dx = dt$$

$$\Rightarrow I = \int\frac{t^{1/4}}{24}dt = \frac{t^{5/4}}{24.\frac{5}{4}} + C$$

$$= \frac{1}{30}(2x^{12} + 3x^8 + 6x^4)^{5/4} + C$$

$\Rightarrow (60a + 1) = (2+1) = 3$

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MATHS

Section–IV Matrix Match Type

1. Ans (A->Q,B->S,C->P,D->P)

(A) $I = \int_4^{10} \frac{[x^2]dx}{[(14-x)^2]+[x^2]}$ --------- (i)

$I = \int_4^{10} \frac{[(14-x)^2]}{[x^2]+[(14-x)^2]}dx$ --------- (ii)

add (i) & (ii)

$2I = \int_4^{10} dx$

$\Rightarrow 2I = 6 \Rightarrow I = 3$

(B) $\int_{-1}^{2} \frac{|x|}{x}dx = \int_{-1}^{0}(-1)dx + \int_0^2 (1)\,dx = 1$

(C) $\lim_{n\to\infty} \sum_{r=1}^{n} \frac{r^{99}}{n^{100}} = \int_0^1 x^{99}dx = \left[\frac{x^{100}}{100}\right]_0^1 = \frac{1}{100}$

(D) $5050\int_{-1}^{1}\sqrt{x^{200}}dx = 5050\times 2\int_0^1 |x^{100}|\,dx$

$= 5050 \times 2\int_0^1 x^{100}dx$

$= 10100 \times \left[\frac{x^{101}}{101}\right]_0^1 = 100 = \frac{1}{\alpha}$

$\Rightarrow \alpha = \frac{1}{100}$

2. Ans (A->PT,B->T,C->PT,D->R)

Ans. (A) $\longrightarrow$ (P,T); (B) $\longrightarrow$ (T); (C) $\longrightarrow$ (P,T); (D) $\longrightarrow$ (R)

(A) $\int \cos^5 x dx = \int (1-\sin^2 x)^2 \cos x dx$

Let sinx = t

$\Rightarrow \cos x\,dx = dt = \int (1-t^2)^2 dt = \int (1+t^4-2t^2)dt$

$= t - \frac{2t^3}{3} + \frac{t^5}{5} + c = \sin x - \frac{2}{3}\sin^3 x + \frac{\sin^5 x}{5} + c$

$\Rightarrow n + 2m = 5 - \frac{3}{2}\times 2 = 2$

(B) $I = \int \frac{xdx}{(x-3)(x-5)}$

$\Rightarrow 2I = \int \frac{(5(x-3)-3(x-5))dx}{(x-3)(x-5)}$

$= \ln|x-5|^5 - \ln|x-3|^3$

$I = \ell n\left|\frac{(x-5)^{5/2}}{(x-3)^{3/2}}\right| + c$

$\Rightarrow a + b = 4$

(C) $\frac{dy}{dx} = \frac{f'(t)}{g'(t)}$

$\frac{d^2y}{dx^2} = \frac{(g'(t).f''(t) - f'(t).g''(t))}{(g'(t))^2}\cdot\frac{1}{g'(t)}$

Put t = 1

$\frac{d^2y}{dx^2} = \frac{2}{1} = 2$

(D) $L = \lim_{x\to 0^+}(-\ell nx)^{\frac{1}{\ell nx}}$

$e^{\lim_{x\to 0^+}\frac{\ell n(-\ell nx)}{\ell nx}}$

Applying L'Hospital rule

$L = e^{\lim_{x\to 0^+}\frac{1}{-\ell nx}\cdot\frac{1}{x}.x} = e^0 = 1$

$\Rightarrow \ell nL = 0$

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SOLUTIONS

MATHS

Section–I(i) Single Correct Type

1. Ans (A)

Ans. (A)

$\log(\sin^2 x.\cos x) = t$

$\Rightarrow (2\sin x.\cos^2 x - \sin^3 x)dx = dt.\sin^2 x.\cos x$

$\Rightarrow (2\sin x(1-\sin^2 x) - \sin^3 x)dx = dt.\sin^2 x.\cos x$

$\Rightarrow (2\sin x - 3\sin^3 x)dx = dt\dfrac{\sin x.\sin 2x}{2}$

$\Rightarrow \dfrac{(2-3\sin^2 x)}{\sin 2x}dx = \dfrac{dt}{2}$

$I = \dfrac{1}{2}\int tdt = \dfrac{t^2}{4} + C$

$= \dfrac{\log^2(\sin^2 x.\cos x)}{4} + C$

$= \log^2\left(\dfrac{1}{2}\sin 2x.\sin x\right) + C$

2. Ans (A)

Ans. (1)

$\int \ell n\left(\sqrt{2+x} + \sqrt{2-x}\right)dx$ (By parts)

$= x\ell n\left(\sqrt{2+x} + \sqrt{2-x}\right),$

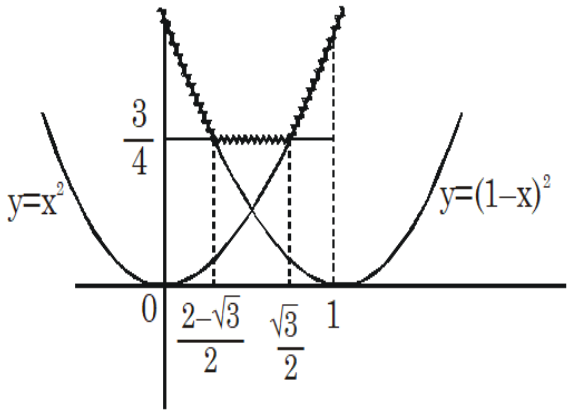
$-\int x.\dfrac{\left(\frac{1}{2\sqrt{2+x}} - \frac{1}{2\sqrt{2-x}}\right)}{\sqrt{2+x}+\sqrt{2-x}}dx$

$= x\log\left(\sqrt{2+x}+\sqrt{2-x}\right) - \dfrac{1}{4}\int \dfrac{4-2\sqrt{4-x^2}}{\sqrt{4-x^2}}dx$

$= xf(x) + \sin^{-1}\dfrac{x}{2} - \dfrac{x}{2} + C$

3. Ans (A)

Ans. (A)



$\int_0^1 f(x) = \int_0^{\frac{2-\sqrt{3}}{2}} (1-x)^2dx + \int_{\frac{2-\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} \dfrac{3}{4}dx + \int_{\frac{\sqrt{3}}{2}}^{1} x^2dx$

$= \dfrac{6\sqrt{3}-1}{12}$

4. Ans (A)

Ans. (A)

$I = \int_{\pi/3}^{\pi/6} \ell n\sin x dx + \int_{\frac{1}{2}\ell n\frac{3}{4}}^{-\ell n2} \sin^{-1}e^x dx$

$= -\dfrac{\pi}{6}\ell n2 - \dfrac{\pi}{6}\ell n\dfrac{3}{4} = \dfrac{\pi}{6}\ell n\dfrac{2}{3}$

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5. Ans (A)

Ans. (A)

$$\int_0^y e^t dt + \int_0^x f(t)dt = g(x) \qquad .....(1)$$

diff. w.r.t. x

$$e^y \frac{dy}{dx} + f(x) = g'(x)$$

$$\Rightarrow \frac{dy}{dx} = \frac{g'(x) - f(x)}{e^y} \qquad .....(2)$$

Again (1)

$$\Rightarrow e^y - 1 = g(x) - \int_0^x f(t)dt$$

$$\Rightarrow e^y = 1 + g(x) - \int_0^x f(t)dt$$

$$\therefore (2) \Rightarrow \frac{dy}{dx} = \frac{g'(x) - f(x)}{1 + g(x) - \int_0^x f(t)dt}$$

6. Ans (A)

Ans. (A)

$$\int e^x \left(\frac{1}{\ell nx} - \frac{1}{(x)(\ell nx)^2} \right) dx$$

$$\int \frac{e^x}{\ell nx} dx - \int \frac{e^x}{x(\ell nx)^2} dx$$

$$\left[\frac{1}{\ell nx} e^x - \int \frac{-1}{(\ell nx)^2 x} e^x dx \right] - \int \frac{e^x}{x(\ell nx)^2} dx + C$$

(Applying integration by parts)

$$= \frac{e^x}{\ell nx} + C$$

7. Ans (A)

Ans. (A)

$$\lim_{x \to 0} \frac{\ell n(1+x) + \frac{x}{1+x}}{f(x).\sin x} = 5$$

$$\lim_{x \to 0} \frac{\frac{\ell n(1+x)}{x} + \frac{1}{1+x}}{f(x).\frac{\sin x}{x}} = 5$$

$$\frac{2}{5} = f(0)$$

8. Ans (A)

Ans. (A)

$$I_2 = \int_0^1 \frac{x^2}{e^{x^3}(2 - x^3)} dx = \frac{1}{3} \int_0^1 \frac{dz}{e^{1-z}(z+1)} , (z = 1 - x^3)$$

$$= \frac{1}{3} \int_0^1 \frac{e^z}{e(z+1)} dz = \frac{1}{3e} I_1$$

$$\frac{I_1}{I_2} = 3e$$

MATHS

Section–I(ii) Multiple Correct Type

9. Ans (A,C,D)

Ans. (A,C,D)

$$\int \frac{\sec^2 x + 3}{\sqrt{\sec^2 x + 3}} dx = \int \frac{\sec^2 x dx}{\sqrt{4 + \tan^2 x}} + \int \frac{3\cos x}{\sqrt{4 - 3\sin^2 x}} dx$$

$$= \int \frac{dt}{\sqrt{4 + t^2}} + \int \frac{\sqrt{3}\, d\theta}{\sqrt{4 - \theta^2}}$$

$$= \ell n \left| t + \sqrt{4 + t^2} \right| + \sqrt{3} \sin^{-1} \left(\frac{\theta}{2} \right) + C$$

$$= \ell n \left| \tan x + \sqrt{4 + \tan^2 x} \right| + \sqrt{3} \sin^{-1} \left(\frac{\sqrt{3} \sin x}{2} \right) + C$$

$f(x) = \tan x$, $g(x) = \sin x$, $k = \sqrt{3}$

10. Ans (A,B,C,D)

Ans. (A,B,C,D)

Let $I = \int (2x^6 + 15x^4 + 2x^2 + 3)\cos 2x \, dx$

$$= \int (2x^6 + 2x^2).\cos 2x + \int (15x^4 + 3).\cos 2x dx$$

$\therefore$ Apply integrate by part, we get

$$= (x^6 + x^2 + 1)\sin(2x) + (3x^5 + x).\cos(2x) + k$$

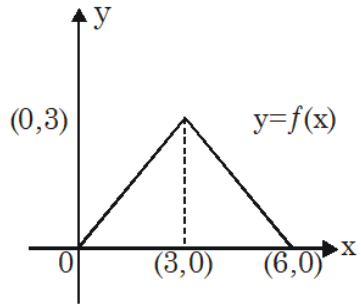
$\therefore f(x) = (x^6 + x^2 + 1)$; $g(x) = (3x^5 + x)$.

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11. Ans (A,C)

Ans. (A,C)

$f(x)$ is periodic with period '6'



$$\therefore \int_0^6 f(x)\,dx = \frac{1}{2} \times 6 \times 3 = 9$$

12. Ans (A,C)

Ans. (A,C)

$$\int_0^x f(t)\,dt + \int_1^{f(x)} g(t)\,dt = f'(x) \qquad \text{Using :}$$

$$\int_a^b f(x)\,dx + \int_{c=f(a)}^{d=f(b)} f^{-1}(x)\,dx = bd - ac$$

$$\Rightarrow x.f(x) - 0 \times 1 = f'(x) \qquad \text{(think !)}$$

$$\Rightarrow \frac{f'(x)}{f(x)} = x \qquad (\because f(x) \neq 0)$$

$$\Rightarrow \ell n f(x) = \frac{x^2}{2} + C$$

put $x = 0 \Rightarrow C = 0$

$$\Rightarrow f(x) = e^{\frac{x^2}{2}}$$

$$f(\sqrt{2}) = e \Rightarrow g(e) = \sqrt{2}$$

$$f(1) = \sqrt{e}$$

Section–I(iii) Matching list type

13. Ans (B)

Ans. (B)

(P) $$\lim_{n\to\infty} \int_0^2 \left(1 + \frac{t}{n+1}\right)^n dt$$

$$\frac{\left[\left(1 + \frac{t}{n+1}\right)^{n+1}\right]_0^2}{(n+1)\left(\frac{1}{n+1}\right)}$$

$$\lim_{n\to\infty} \left(1 + \frac{2}{n+1}\right)^{n+1} - 1 = \boxed{e^2 - 1} \quad \boxed{P \to 3}$$

(Q) $f'(x) = f(x)$, $f(0) = 1$

$f(x) + g(x) = x^2$

$f(x) = e^x$

$g(x) = x^2 - e^x$

$$\int_0^1 e^x(x^2 - e^x)\,dx$$

$$\int_0^1 (x^2e^x - e^{2x})\,dx$$

$$\left[x^2e^x - 2\int xe^x - \frac{e^{2x}}{2}\right]_0^1$$

$$\left[x^2e^x - 2\left(xe^4 - e^x\right) - \frac{e^{2x}}{2}\right]_0^1$$

$$\left(e - 2(e - e) - \frac{e^2}{2}\right) - \left(2 - \frac{1}{2}\right)$$

$$e - \frac{e^2}{2} - \frac{3}{2} \quad \boxed{Q \to 1}$$

(R) $$\int_0^1 e^{e^x}(1 + xe^x)\,dx$$

$e^x = t \Rightarrow e^x dx = dt \Rightarrow dx = \frac{dt}{t}$

$$\int_1^e e^t(1 + \ell nt.\,t)\,\frac{dt}{t}$$

$$\int_1^e e^t\left(\ell nt + \frac{1}{t}\right)dt$$

$$\left[e^t\,\ell nt\right]_1^e \Rightarrow \boxed{e^e - 0} \quad \boxed{R \to 4}$$

(S) $$\lim_{K\to 0} \frac{1}{K} \int_0^K (1 + \sin 2x)^{\frac{1}{x}}\,dx$$

$$\lim_{K\to 0} \left((1 + \sin 2K)^{\frac{1}{K}}\right)$$

$$e^{\lim_{K\to 0}\left(\frac{\sin 2K}{2K}\right)^2} = e^2 \quad \boxed{S \to 2}$$

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MATHS

Section–III Integer Type

1. Ans (2)

Ans. 2

$$\alpha = \int_{2017}^{2019} (x-2017)^2 f(x-2017)\,dx$$

$x - 2017 = t$

$$\alpha = \int_0^2 t^2 f(t)\,dt$$

$$\beta = \int_{-2019}^{-2015} (x+2017)^2 f(x+2017)\,dx$$

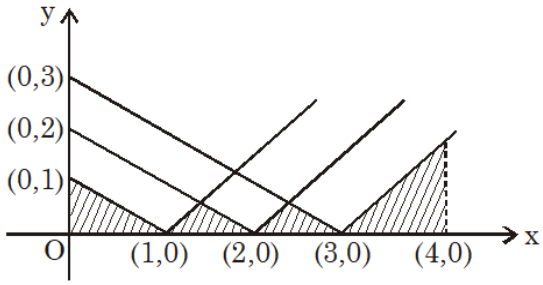
$x + 2017 = z$

$$\beta = \int_{-2}^{2} z^2 f(z)\,dz = 2\alpha$$

$$\frac{\beta}{\alpha} = 2$$

2. Ans (5)

Ans. 5



Required value

$$= \frac{1}{2}(1)(1) + 2\left(\frac{1}{2}\times 1\times\frac{1}{2}\right) + \frac{1}{2}\times 1\times 1 = \frac{3}{2}$$

4. Ans (9)

Ans. 9

$$f(x) = -\int_0^1 f^2(t)dt - 2x\int_0^1 f(t)dt$$

$$\therefore f'(x) = -2\int_0^1 f(t)dt \quad .....(i)$$

$= \text{constant}$

$\Rightarrow$ f must be linear.

So, let $f(x) = (ax + b)$

$$\therefore a = -2\int_0^1 (at+b)dt$$

$$\& \quad b = -\int_0^1 (at+b)^2 dt$$

$\Rightarrow a = 3, b = -3$

$\Rightarrow f(x) = 3(x - 1)$

So, $f(4) = 3(4 - 1) = 9$

5. Ans (2)

Ans. 2

$$\frac{1}{n\pi}.2n\int_0^1 \frac{dx}{\sqrt{\{x\}\{-x\}}} = \frac{2}{\pi}\int_0^1 \frac{dx}{\sqrt{x(1-x)}}$$

$$= \frac{4}{\pi}\left(\sin^{-1}\sqrt{x}\right)_0^1 = 2$$

6. Ans (7)

Ans. 7

$$I_7 = \int_0^1 \underset{I}{\left(1-x^4\right)}.\underset{II}{1}\,dx$$

$$I_7 = \left[x\left(1-x^4\right)^7\right]_0^1 - \int_0^1 7\left(1-x^4\right)^6\left(-4x^3\right).x\,dx$$

$$I_7 = 0 + 28\int_0^1 \left(1-x^4\right)^6 x^4 dx$$

$$I_7 = 28\int_0^1 \left(1-x^4\right)^6\left[1-\left(1-x^4\right)\right]dx$$

$I_7 = 28I_6 - 28I_7 \Rightarrow 29I_7 = 28I_6$

$$\Rightarrow \frac{29I_7}{4I_6} = 7$$

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GUIDED REVISION

JEE(Main + Advanced) 2025
ENTHUSIAST+LEADER COURSE
SCORE ADVANCED

ANSWER KEY

(Application of Derivatives) GUIDED REVISION # 15

SOLUTIONS

MATHS

Section–I(i) Single Correct Type

1. Ans (B)

Ans. (B)

$f'(a) = f'(b)$

$3a^2 + a = 3b^2 + a \Rightarrow \quad a = -b$

so $f(1) = 1 + a + b = 1$

2. Ans (C)

Ans. (C)

$f'(x) = 3(x^2 - 2ax + a^2 - 1)$

Clearly roots of $f'(x) = 0$ must be distinct and lie in the interval $(-2, 4)$.

$\therefore \quad D > 0 \quad \Rightarrow \quad a \in R \qquad .....(1)$

$f'(-2) > 0 \quad \Rightarrow \quad a^2 + 4a + 3 > 0$

$\Rightarrow \quad a < -3 \text{ or } a > -1 \qquad .....(2)$

$f'(4) > 0 \quad \Rightarrow \quad a^2 - 8a + 15 > 0$

$\Rightarrow \quad a > 5 \text{ or } a < 3 \qquad .....(3)$

and $-2 < -\frac{B}{2A} < 4 \quad \Rightarrow -2 < a < 4 \qquad .....(4)$

From (1), (2), (3) & (4), $a \in (-1, 3)$

3. Ans (B)

Ans. B

Using LMVT in [2, 4]

$f'(c) = \frac{f(4) - f(2)}{4 - 2} = \frac{f(4) + 4}{2}$

$f'(x) \geq 6 \quad \Rightarrow \quad \frac{f(4) + 4}{2} \geqslant 6 \quad \Rightarrow \quad f(4) \geq 8$

4. Ans (C)

Ans. (C)

Let $x = 3\sqrt{3}\cos\theta, y = 3\sqrt{3}\sin\theta$ where $\theta \in [-\pi, \pi]$

Then $A = x - y = 3\sqrt{3}(\cos\theta - \sin\theta)$

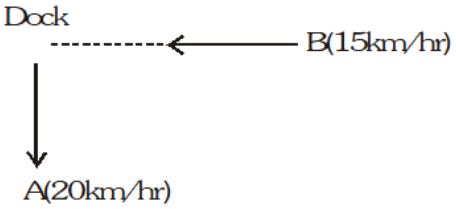
Now $-\sqrt{2} \leqslant \cos\theta - \sin\theta \leqslant \sqrt{2}$

$\therefore \quad A_{max} = 3\sqrt{3}\sqrt{2} = \sqrt{54}$

5. Ans (A)

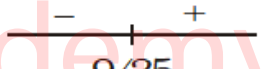
Ans. (A)

At 2 PM boat B is 15 km away from Dock. At time t hr boat A is 20t km away & boat B is 15 – 15t km away from dock.



$(AB)^2 = f(t) = 400t^2 + 15^2(1 - t)^2$

$f'(t) = 800t + 15^2 \cdot 2(t - 1) = 1250t - 450$



AB is minimum at $t = \frac{9}{25}$ hrs. = 21 min 36 sec.

$\therefore$ time = 2:21:36 PM

6. Ans (D)

Ans. D

$f(-2) = f(3) = 0$

f(x) is continuous in $[-2, 3]$ & derivable in $(-2, 3)$ so Rolle's theorem is applicable.

so $\exists\ c \in (-2, 3)$ such that $f'(c) = 0$

$\Rightarrow \quad \frac{2c^3 - 5c^2 + 4c - 1}{(c - 1)^2} = 0 \quad \Rightarrow \quad c = 1/2$

7. Ans (C)

Ans.(C)

$y = \frac{1}{x}$

so $x^2 + y^2 = x^2 + \frac{1}{x^2}$

minimum value = 2

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MATHS

Section–I(ii) Multiple Correct Type

8. Ans (C,D)

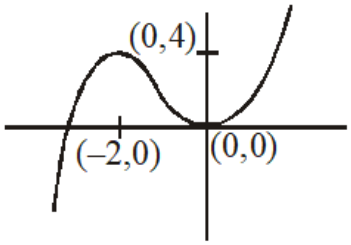
Ans. (C,D)

$$f(x) = 3x^2 + e^{-x}\int_0^x e^t f(t)dt$$

$$f'(x) = 6x - e^{-x}\left(\frac{f(x) - 3x^2}{e^{-x}}\right) + e^{-x}e^x f(x)$$

$f'(x) = 3x^2 + 6x$

$f(x) = x^3 + 3x^2 \quad [\because \quad f(0) = 0]$



9. Ans (B,C)

Ans. (B,C)

By LMVT for $f(x)$ in $x \in [1, 2]$

$\exists\ c_1 \in (1, 2)$ where

$$f'(c_1) = \frac{f(2) - f(1)}{2 - 1}$$

$\because f'(x) \geq 1\ \forall\ x \in R$

$\Rightarrow f(2) - f(1) \geq 1 \quad \Rightarrow \quad f(2) \geq 2$

By LMVT for $f(x)$ in $[2, 3]$

$\exists\ c_2 \in (2, 3)$ where

$$f'(c_2) = \frac{f(3) - f(2)}{3 - 2}$$

$\Rightarrow 4 - f(2) \geq 1 \ \Rightarrow \ f(2) \leq 3 \Rightarrow \quad 2 \leq f(2) \leq 3$

10. Ans (B,C,D)

Ans. (B,C,D)

(B) $F(x) = f(3 - 4x) \ \Rightarrow \ F'(x) = f'(3 - 4x)(-4)$

$\Rightarrow \quad F'(x) < 0$

(C) $G(x) = f(x^2 - x)$

$\Rightarrow G'(x) = f'(x^2 - x)(2x - 1)$

$G'(x) > 0 \ \Rightarrow \quad 2x - 1 > 0$

(D) $H(x) = (f(x))^3 \ \Rightarrow \quad H'(x) = 3(f(x))^2 f'(x)$

MATHS

Section–I(iii) Comprehension Type

11. Ans (B)

$$f(x) = x + 2\sqrt{25 + (\ell - x)^2} \quad \text{..... (i)}$$

$0 \leqslant x \leqslant \ell$

Ans. (B)

$$f'(x) = 1 - \frac{2(\ell - x)}{\sqrt{25 + (\ell - x)^2}} \quad \text{.....(ii)}$$

$$= \frac{\sqrt{25 + (\ell - x)^2} - 2(\ell - x)}{\sqrt{25 + (\ell - x)^2}}$$

$25 + (\ell - x)^2 = 4(\ell - x)^2$

$\sqrt{3}(\ell - x) = \pm 5$

$$x = \ell \pm \frac{5}{\sqrt{3}}$$

$$x = \ell - \frac{5}{\sqrt{3}}$$

If $\ell = \frac{185}{\sqrt{3}}$ then $x = \frac{180}{\sqrt{3}} = 60\sqrt{3}$ kms.

12. Ans (D)

$$f(x) = x + 2\sqrt{25 + (\ell - x)^2} \quad \text{..... (i)}$$

$0 \leqslant x \leqslant \ell$

Ans. (D)

$\ell = 10$ kms, then maximum value of $f(x)$ lies at $x = 0$ or $x = 10$

$f(x) = x + 2\sqrt{25 + (10 - x)^2}$

$f(0) = 10\sqrt{5}$

$f(10) = 20$

$\Rightarrow$ maximum freight $= 10\sqrt{5}$ at $x = 0$

13. Ans (B)

Ans. (B)

$$f(x) = 2\cos^2 2x + \frac{3}{4}\sin 4x + ax$$

$f'(x) = -4\sin 4x + 3\cos 4x + a \geq 0\ \forall\ x \in R$

$a \geq 4\sin 4x - 3\cos 4x$

$a \geq 5$

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14. Ans (D)

Ans. (D)

$f'(x) \neq 0$ for any $x \in R$

$a \notin 4\sin 4x - 3\cos 4x$ for any $x \in R \Rightarrow$

$a \notin [-5, 5]$

$a \in (-\infty, -5) \cup (5, \infty)$

MATHS

Section–III Integer Type

1. Ans (4)

Ans. 4

Let a point on $y^3 = x^4$ be (t^3, t^4)

$3y^2y' = 4x^3$

$\Rightarrow y' = \frac{4x^3}{3y^2} \Rightarrow y' = \frac{4}{3}t$

Equation of tangent is

$y - t^4 = \frac{4t}{3}(x - t^3)$

$\because$ it is a normal to $x^2 + y^2 - 2x = 0$

$\therefore$ it must pass through $(1, 0)$

$\Rightarrow -\frac{3}{4}t^3 = 1 - t^3 \Rightarrow \frac{t^3}{4} = 1$

$\Rightarrow t^3 = 4$

Now $m = \frac{4t}{3}$

$\Rightarrow \left(\frac{3m}{4}\right)^3 = t^3 = 4$

2. Ans (1)

Ans. 1

$R = \frac{abc}{4\Delta} = \frac{r.r.\left(2r\sin\frac{\theta}{2}\right)}{4\left(\frac{1}{2}r^2\sin\theta\right)} = \frac{r\sin\frac{\theta}{2}}{\sin\theta} = \frac{r}{2\cos\frac{\theta}{2}}$

$= \frac{r}{2\cos 15^\circ}$

$\therefore \frac{dR}{dr} \times 2\cos 15^\circ = 1$

3. Ans (9)

Ans. 9

Let $\frac{dV}{dt} = kV^2$

$\Rightarrow \int \frac{dV}{V^2} = \int kdt \Rightarrow -\frac{1}{V} = kt + C$

at $t = 0$, $V = V_0$

$\Rightarrow C = -\frac{1}{V_0} \Rightarrow \frac{1}{v} = \frac{1}{V_0} - kt$

at $t = 6$, $V = 2V_0$

$\Rightarrow \frac{1}{2V_0} \quad 6k = \Rightarrow k = \frac{1}{12V_0}$

when $V = 4V_0$

$\Rightarrow kt = \frac{3}{4V_0} \Rightarrow t = \frac{3 \times 12V_0}{4V_0 \times 1}$

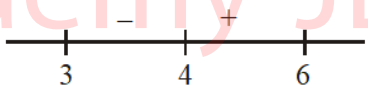
$t = 9$ hours

4. Ans (1)

Ans. 1

$f(x) = \int_3^x t\left(t^2 - 7t + 12\right) dt$,

$f'(x) = x(x-3)(x-4)$

$\therefore$ 

minimum at $x = 4$

maximum at $x = 3$ or 6.

$\therefore \quad f(3) = 0, \quad f(6) = \int_3^6 (t^3 - 7t^2 - 12t)dt$

$= \left(\frac{t^4}{4} - \frac{7t^3}{3} + \frac{12t^2}{2}\right)_3^6 = \frac{99}{4}$

$f(x) = \int_3^4 (t^3 - 7t^2 + 12t)dt = -\frac{7}{12}$

$\therefore \quad 3a + \frac{b}{9} = 1$

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ANSWER KEY

(Application of Derivatives) GUIDED REVISION # 16

SOLUTIONS

MATHS

Section–I(i) Single Correct Type

1. Ans (A)

Ans. (A)



$f'(x) \le 0$

$\Rightarrow (a+2)x^2 - 2ax + 3a \le 0 \ \forall \ x \in R$

$\Rightarrow a + 2 < 0 \ \& \ D \le 0$

2. Ans (B)

Ans. (B)

$f'(x) = m - e^{-x} < 0 \ \forall \ x \in R$

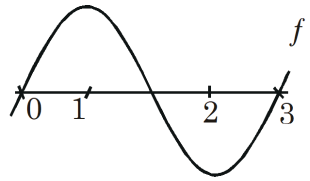
$\Rightarrow \quad m \le 0$

3. Ans (B)

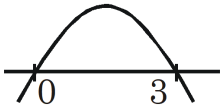
$\left|y\dfrac{dy}{dx}\right| = k \quad \Rightarrow ydy = \pm kdx$

$\Rightarrow \dfrac{y^2}{2} = \pm kx + C$

4. Ans (A)



Min. no.of roots of $f(x) = 0$ is 3.



Min. no.of roots of g(x) = 0 is 2.

given equation is $f''g + f'g' = 0$

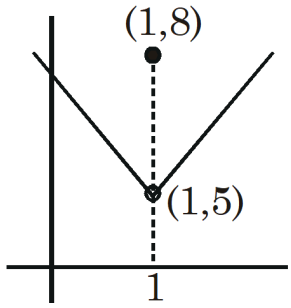
i.e. $(f'(x).g(x))' = 0$

$\because$ $f'(x) = 0$ has minimum 2 roots

$\therefore$ $f'(x).g(x) = 0$ has minimum 4 roots. (so using Rolles theorem)

$\Rightarrow (f'(x).g(x))' = 0$ has minimum 3 roots.

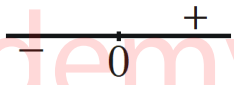
5. Ans (A)



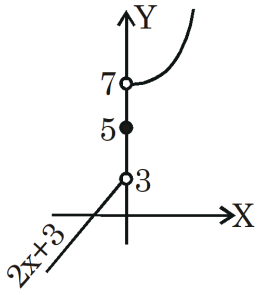
Clearly x = 1 is a point of local maxima.

6. Ans (C)

$f'(x) = \dfrac{x(x^2+4)}{(1+x^2)^{3/2}}$,

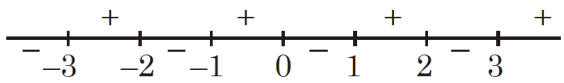


7. Ans (B)



8. Ans (B)

$f' = 2x(x^2-1)(x^2-4)(x^2-9)$



$f(x)$ have 7 critical points and $f'' = 0$ have 3 positive and 3 negative solutions

f' is odd function $\Rightarrow$ f''' is also an odd function so $f'''(0) = 0$

as there are 5 roots of $f'''(x) = 0 \Rightarrow$ it will have 2 positive and 2 negative roots.

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9. Ans (D)

$\because f'(x) = -3x^2 - 2$

$\therefore f(x)$ is decreasing

$\Rightarrow f(f(x)) < f(-x)$

$\Rightarrow f(x) > -x \Rightarrow 30 - 2x - x^3 > -x$

$\Rightarrow x^3 + x - 30 < 0$

$\Rightarrow (x-3)(x^2+3x+10) < 0 \Rightarrow x < 3$

$\therefore$ Non negative x = 0,1,2

10. Ans (C)

Let length = breadth = x and height = y

Given, $x^2 + 4xy = 48$ ...(i)

and volume $V = x^2y = x^2\left(\frac{48-x^2}{4x}\right)$

$\Rightarrow V = \frac{1}{4}(48x - x^3) \Rightarrow \frac{dv}{dx} = \frac{1}{4}(48 - 3x^2) = 0$

$\Rightarrow x = \pm 4$

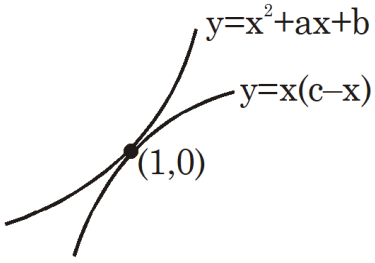
Now (i) $\Rightarrow y = \frac{48-16}{16} = 2$

$\therefore V_{max} = 4^2.2 = 32m^3$.

MATHS

Section–I(ii) Multiple Correct Type

11. Ans (A,B,C,D)



0 = 1 + a + b .....(i)

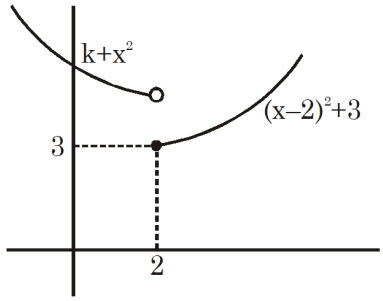
0 = c – 1 .....(ii)

& $\left(\frac{dy}{dx}\right)_{(1,0)} = \left(\frac{dy}{dx}\right)_{(1,0)}$

$\Rightarrow$ 2 + a = c – 2 .....(iii)

a = –3, b = 2, c = 1

12. Ans (B,C)



13. Ans (A,D)

(A) $f(x) > 0, f'(x) > 0$ & $f''(x) > 0$

(B) $f(x) > 0, f'(x) < 0$ & $f''(x) > 0$

(C) $f(x) < 0, f'(x) < 0$ & $f''(x) < 0$

(D) $f(x) < 0, f'(x) > 0$ & $f''(x) < 0$

14. Ans (A,D)

From given condition

$f(x) = e^x$

Now $g(x) = \frac{3x}{e^x}$

$g'(x) = \frac{3(1-x)}{e^x}$

(A) $g'\left(\ell n\sqrt{e}\right) = g'\left(\frac{1}{2}\right) = \frac{3}{2}.e^{-\frac{1}{2}}$

(B) $g'(x) = 0 \Rightarrow x = 1$

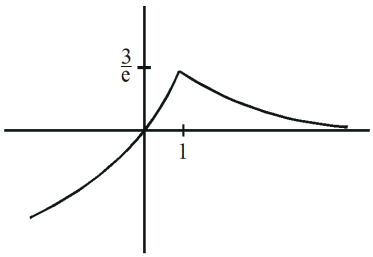
and g'(1–) = +ve

$g'(1^+)$ = –ve

$\Rightarrow$ g(x) has maxima at x = 1

(C) x = 0 is not a critical point

(D) Graph of $g(x) = \frac{3x}{e^x}$

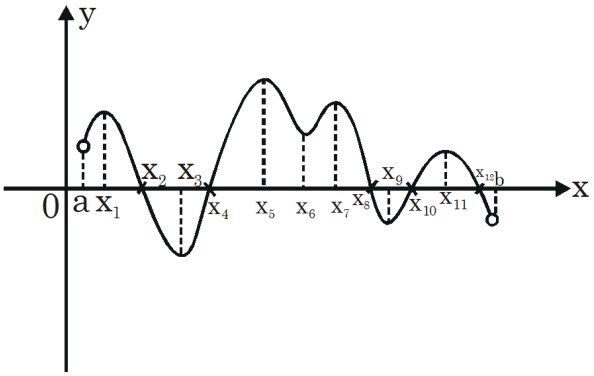


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MATHS

Section–I(v) : Matching list type

15. **Ans (D)**



$f'(x)$ +ve to −ve at x_2, x_8, x_{12}

$\Rightarrow$ point of maxima

$f'(x)$ −ve to +ve at x_4, x_{10}

$\Rightarrow$ point of minima

$f''(x) = 0$ at $x_1, x_3, x_5, x_6, x_7, x_9, x_{11}$

$\Rightarrow$ Point of inflection

$f'(x)$ = zero at

$x_2, x_4, x_8, x_{10}, x_{12}$ $\Rightarrow$ Stationary points

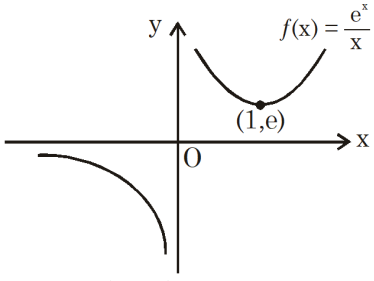
MATHS

Section–(II): (iii) Integer Type

1. **Ans (0.50)**

$a = 6, b = 2, c = 1, d = 2$

2. **Ans (3.00)**



$$f'(x) = \frac{(x-1)e^x}{(x-0)^2}$$

$\therefore$ f is increasing in $(1, \infty)$

f is decreasing in $(-\infty, 1) - \{0\}$ & $x = 0$ is asymptote of curve

$\therefore$ $f_{min} = e$

$\therefore$ Answer = 3

3. **Ans (0.73)**

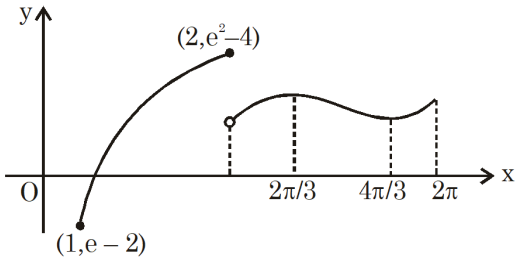
$$f'(x) = \begin{cases} e^x - 2, & 1 < x < 2 \\ 2\cos x + 1, & 2 < x < 2\pi \end{cases}$$

In $[1, 2]$ $\quad f'(x) > 0$

in $(2, 2\pi)$ $f'(x) = 0$ gives

$$\frac{2\pi}{3}, \quad \frac{4\pi}{3}$$

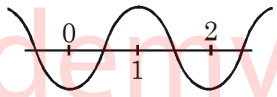
local max $\qquad$ local min



$\therefore$ least value occur at $x = 1$

$\therefore$ $\lambda = e - 2$

4. **Ans (0.25)**



$f(x) = 2x^4 - 8x^3 + 8x^2 - 1$

$f'(x) = 8x^3 - 24x^2 + 16x$

$= 8x(x^2 - 3x + 2)$

$= 8x(x - 1)(x - 2)$

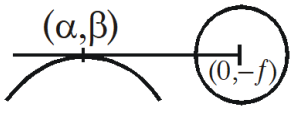
$f(0) = -1$, $f(1) = 1$, $f(2) = -1$

so $f(x)$ has 4 distinct roots.

5. **Ans (0.50)**

$xy' + y = 2x$

$$\Rightarrow \left.\frac{dy}{dx}\right|_{(\alpha,\beta)} = \frac{\beta + f}{\alpha - 0}$$



$$\Rightarrow \frac{2\alpha - \beta}{\alpha} = \frac{\beta + f}{\alpha} \Rightarrow 2(\alpha - \beta) = f$$

$\Rightarrow f.\ \alpha = 2\alpha^2 - 2\alpha\beta$

$= 2(\alpha\beta - 1) - 2\alpha\beta \quad (\because \alpha\beta = \alpha^2 + 1)$

$= -2$

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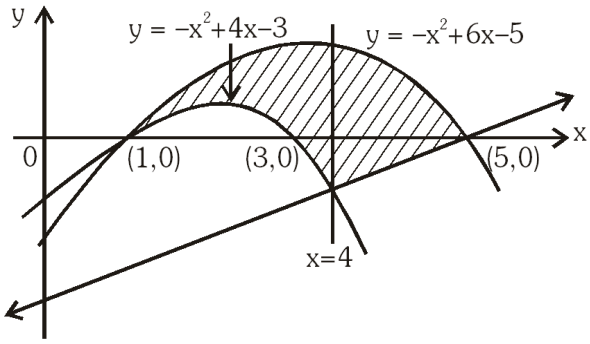
(Differential Equation + Area Under Curve) GUIDED REVISION # 17

SOLUTIONS

MATHS

Section–I(i) Single Correct Type

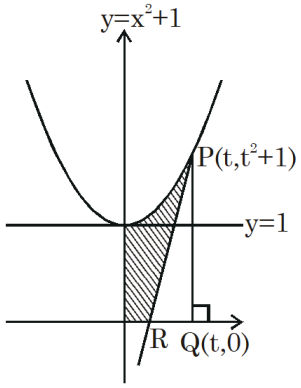
1. Ans (C)



$$A = \int_1^4 \{(-x^2+6x-5)-(-x^2+4x-3)\}\,dx$$

$$+\int_4^5 (-x^2+6x-5)-(3x-15)dx = \frac{73}{6}$$

2. Ans (B)



Equation of tangent

$y = (t^2+1) + 2t(x-t)$

$y = 0 \;;\; x = \dfrac{t^2-1}{2t}$

$R\left(\dfrac{t^2-1}{2t}, 0\right)$

$$= \int_0^t (x^2+1)\,dx - \Delta PQR = \frac{11}{3}$$

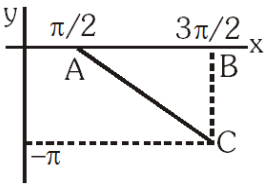
$= t^4 + 6t^2 - 44t - 3 = 0$

$(t-3)(t^3+3t^2+15t+1) = 0$

$t = 3$ is positive solution

$\Rightarrow$ x coordinate is 3

3. Ans (D)



$$f(x) = \int_0^{\pi/2} [\cos t]\,dt + \int_{\pi/2}^{x} [\cos t]dt,\; x \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$$

$$\Rightarrow f(x) = 0 + \int_{\pi/2}^{x} -1\,dt = \frac{\pi}{2} - x \quad \Rightarrow \text{ Required}$$

$$\text{area} = \frac{1}{2} \times AB \times BC = \frac{\pi^2}{2}$$

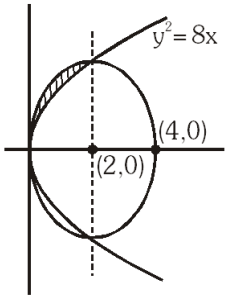
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4. **Ans (D)**

Area of shaded portion

$= \frac{1}{4}(\pi.2.4) - \int_0^2 \sqrt{8x}\,dx$

$= 2\pi - \frac{16}{3}$



5. **Ans (B)**

$\frac{1}{1+y^2}.\frac{dy}{dx} + 2x(\tan^{-1}y) = x^3$

Put $\tan^{-1}y = z$

$\therefore \quad \frac{1}{1+y^2}.\frac{dy}{dx} = \frac{dz}{dx}$

$\frac{dz}{dx} + (2x)z = x^3 \quad \Rightarrow \quad z\,.\,e^{x^2} = \frac{1}{2}\int 2e^{x^2}.x^3dx + c$

Put $x^2 = t$

$\Rightarrow \quad (\tan^{-1}y)\,e^{x^2} = \frac{1}{2}\int e^t.\,t\,dt + c \quad \Rightarrow \quad e^{x^2}(\tan^{-1}y) = \frac{1}{2}\{e^t.\,t - e^t\} + c$

$\Rightarrow \quad 2e^{x^2}\tan^{-1}y = x^2e^{x^2} - e^{x^2} + 2c \quad \Rightarrow \quad 2\tan^{-1}y = x^2 - 1 + 2ce^{-x^2}$

6. **Ans (C)**

$(K+1)^2\,e^{(K+1)x} - 4(K+1)e^{(K+1)x} + 4e^{(K+1)x} = 0$

$(K+1)^2 - 4(K+1) + 4 = 0$

$K + 1 = 2 \quad \Rightarrow \quad K = 1$

7. **Ans (C)**

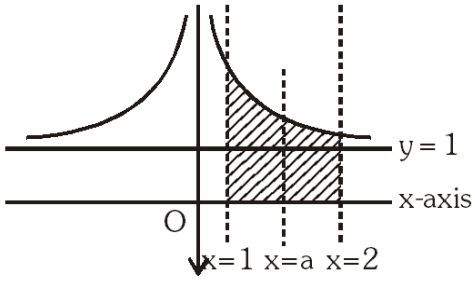
$\left(\frac{dy}{dx}\right)^{\frac{1}{3}} = 4\frac{d^2y}{dx^2} + 7x$

$\left(\frac{dy}{dx}\right) = \left(4\frac{d^2y}{dx^2} + 7x\right)^3$

$\Rightarrow$ order = 2

degree = 3

8. **Ans (D)**



$\int_1^a (1+x^{-2})dx = \int_a^2 (1+x^{-2})dx$

$\Rightarrow x - \frac{1}{x}\Big|_1^a = x - \frac{1}{x}\Big|_0^2$

$\Rightarrow 2\left(a - \frac{1}{a}\right) = \frac{3}{2}$

$\Rightarrow 4a^2 - 3a - 4 = 0$

$\Rightarrow a = \frac{3+\sqrt{73}}{8}$

MATHS

Section–I(ii) Multiple Correct Type

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9. **Ans (A,B,D)**

$\frac{dy}{dx} = \frac{e^x}{\sqrt{1-e^{2x}}}$

$\therefore \int dy = \int \frac{d(e^x)}{\sqrt{1-(e^x)^2}}$

$y = \sin^{-1}(e^x) + c \;\; = \cos^{-1}\sqrt{1-e^{2x}} + c$

$y = \frac{\pi}{2} + c - \cos^{-1}(e^x) \Rightarrow y = k - \cos^{-1}e^x$

10. **Ans (A,B)**

$\int \frac{bdy}{y^2+b^2} + \int \frac{adx}{x^2+a^2} = 0$

$\Rightarrow \tan^{-1}\left(\frac{x}{a}\right) + \tan^{-1}\left(\frac{y}{b}\right) = c$

Apply tan on both sides

$\frac{\frac{x}{a}+\frac{y}{b}}{1-\frac{xy}{ab}} = k \quad \Rightarrow bx + ay = k(ab - xy)$

11. **Ans (B,C,D)**

d(xy) = dx

integrate both sides xy = x + c

c = 1 ⇒ $y = 1 + \frac{1}{x}$

(1) Non periodic function

(2) $f(x_1) = f(x_2)$ ⇒ $x_1 = x_2$

(3) Domain R – {0}

(4) Neither even nor odd

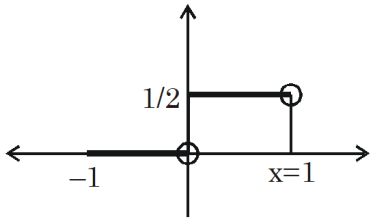
12. **Ans (A,B,C)**

$2\{y\} = [x] + 1$ .....(1)

∵ $0 \le y < 1$

⇒ $\{y\} = y$

∴ (1) ⇒ $2y = [x] + 1$



$0 \le [x] + 1 < 2$

$-1 \le [x] < 1$

but [x] is an integer

⇒ [x] = –1 ⇒ $-1 \le x < 0$

[x] = 0 ⇒ $0 \le x < 1$

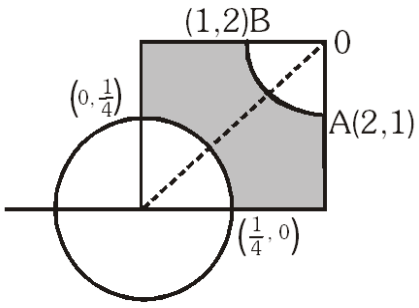
when $x \in [-1, 0)$ ⇒ y = 0

when $x \in [0, 1)$ ⇒ $y = \frac{1}{2}$ ∴ Area = $\frac{1}{2}$

MATHS

Section–II Integer Type

1. **Ans (4.00)**



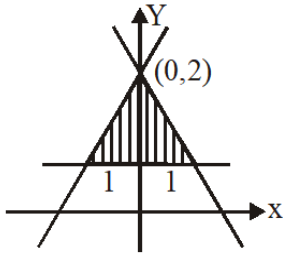
xy = 2 is rectangular hyperbola & $8(x^2 + y^2) > 2$ is exterior of circle; shaded portion is given by figure.

Area of portion OAB = $2 - \int_1^2 \frac{2}{x}dx = (2 - \ell n4)$

Total area = $4 - \left(\frac{\pi}{16} + 2 - \ell n4\right)$

$2 + \ell n4 - \frac{\pi}{16}$

2. **Ans (1.00)**



$x^2y - x^2 - y^3 + y^2 + 4y^2 - 4y - 4y + 4 = 0$

$\Rightarrow x^2(y-1) - y^2(y-1) + 4y(y-1) - 4(y-1) = 0$

$\Rightarrow (y - 1)(x^2 - y^2 + 4y - 4) = 0$

$\Rightarrow$ y = 1 or $x^2 = (y - 2)^2$

$\Rightarrow$ y = 1, x = y – 2 or x = 2 – y

Shaded area = $\frac{1}{2} \times 2 \times 1 = 1$ sq. units

3. **Ans (2.00)**

$(\tan y + y\sec^2 y)dy = e^x(\sin x + \cos x)dx$

$y\tan y = e^x \sin x + C$

(0,0) gives C = 0

$y\tan y = e^x \sin x$

$\pi \tan \pi = e^{x_1} \sin x_1$

$\Rightarrow \sin x_1 = 0 \Rightarrow x_1 = 2\pi$

$\frac{x_1}{\pi} = 2$

4. **Ans (7.00)**

$y = ax^p + bx^{-q}$

$\Rightarrow y' = apx^{p-1} - bqx^{-q-1}$

$\Rightarrow x^2y' = apx^{p+1} - bqx^{1-q}$

$\Rightarrow x^2y'' + 2xy' = (p+1)\,apx^p - bq(1-q)x^{-q} = 12y$

$\Rightarrow p(p + 1) = 12$ & $q(q - 1) = 12$

$\Rightarrow p = 3$ & $q = 4$

5. **Ans (12.00)**

$d(x^2y) = d(xy) + d(xy^2)$

$\Rightarrow x^2y = xy + xy^2 + c$

at x = 1, y = 1 ⇒ c = –1

$\Rightarrow x^2y = xy + xy^2 - 1 \Rightarrow y = -y - y^2 - 1$

$\Rightarrow y^2 + 2y + 1 = 0 \Rightarrow y = -1$

∴ $12|y(-1)| = 12$

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GUIDED REVISION

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ANSWER KEY

(Quadratic Equation and Sequence Series) GUIDED REVISION # 18

SOLUTIONS

MATHS

Section–I(i) Single Correct Type

1. Ans (B)

$\alpha + \beta = \alpha\beta$

$\Rightarrow \quad \alpha + \beta - \alpha\beta - 1$

$\Rightarrow \quad (\alpha - 1)(1 - \beta) = -1$

$\therefore \ \alpha = 0 = \beta$ or $\alpha = 2 = \beta$

but $\alpha\beta = \alpha^2 + \beta \Rightarrow \alpha = 0 = \beta$

$\Rightarrow$ only one quadratic equation

2. Ans (B)

$(x - \alpha)(x - \beta) + (x - \gamma)(x - \delta)$

$= 2(x - m)(x - n)$

$\Rightarrow \quad 2(x - m)(x - n) - (x - \alpha)(x - \beta)$

$= (x - \gamma)(x - \delta)$

So roots are γ, δ.

3. Ans (D)

$x^2 - 4x - a^2 + 2a + 3 = 0$

given equation

$\Rightarrow \quad (x - 2)^2 = (a - 1)^2$

$\Rightarrow \quad x = 2 \pm (a - 1)$

$\Rightarrow \quad x = 1 + a \ \text{ or } \ 3 - a.$

for equation $x^2 - 4x + 4 - a^2 = 0$

$f(1 + a) < 0$ and $f(3 - a) < 0$

$1 + a^2 + 2a - 4 - 4a + 4 - a^2 < 0$

and $9 + a^2 - 6a - 12 + 4a + 4 - a^2 < 0$

$\Rightarrow \frac{1}{2} < a \quad \text{and} \quad \frac{1}{2} < a$

$\therefore a \in \left(\frac{1}{2}, 10\right)$

4. Ans (B)

$\log_3 x = 3 \pm \sqrt{9 - k}$

$\Rightarrow \alpha = 3^{3+\sqrt{9-k}}$ & $\beta = 3^{3-\sqrt{9-k}}$

$\because \ \frac{\alpha + \beta}{2} \geqslant \sqrt{\alpha\beta}$

$\Rightarrow \quad \alpha + \beta \geqslant 2\sqrt{3^6}$

$\alpha + \beta \geqslant 54$

5. Ans (B)

$a_n - a_{n-1} = n^2$

Put $n = 2,3,4,....,k$

$\Rightarrow \quad a_k - a_{k-1} = k^2$

$a_{k-1} - a_{k-2} = (k - 1)^2$

:

:

$a_2 - a_1 = 2^2$

Adding, $a_k - a_1 = 2^2 + 3^2 + ...+ k^2$

$\Rightarrow \quad 12 + 22 + k^2 = 204$

$\Rightarrow \quad \frac{k(k + 1)(2k + 1)}{6} = 204$

$\Rightarrow \quad k = 8$

6. Ans (B)

Let the first 3 terms be $\frac{a}{r}$, a, ar

Hence $\frac{a}{a\left(r + \frac{1}{r} + 1\right)} = \frac{1}{1 + r + \frac{1}{r}} \leqslant \frac{1}{3}$ (as $r + \frac{1}{r} \geqslant 2$)

7. Ans (C)

AM $\geq$ GM $\Rightarrow \quad \frac{\frac{12}{x} + \frac{18}{y} + xy}{3} \geqslant \left(\frac{12}{x} \cdot \frac{18}{y} \cdot xy\right)^{\frac{1}{3}}$

$\Rightarrow \quad \frac{12}{x} + \frac{18}{y} + xy \geq 18$

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8. Ans (C)

$$\sum_{i=1}^{6} A_i = 6 \times \left(\frac{2+72}{2}\right) = 3 \times 74$$

$$\prod_{i=1}^{6} G_i = \left(\sqrt{2 \times 72}\right)^6 = 2^3 \times 72^3$$

$$\sum_{i=1}^{6} \frac{1}{H_i} = 6\frac{\left(\frac{1}{2}+\frac{1}{72}\right)}{2} = \frac{6 \times 74}{4 \times 72}$$

$G = 12$

$$\Rightarrow \frac{\prod (G_i)\sum\left(\frac{1}{H_i}\right)}{\sum A_i} = 2^2 72^2 = G^4$$

MATHS

Section–I(ii) Multiple Correct Type

9. Ans (A,B,D)

Equations are $(ax - b)(x - 1) = 0$

& $(bx - c)(x + 1) = 0$

$\Rightarrow$ Roots are $\frac{b}{a}, 1$ and $\frac{c}{b}, -1$

Case-I : $\frac{b}{a} = \frac{c}{b} \Rightarrow b^2 = ac$

Case-II : $\frac{b}{a} = -1$ & $\frac{b}{a} = -1 \Rightarrow -a = b \neq c$

Case-III : $\frac{c}{b} = 1$ & $\frac{b}{a} \neq -1 \Rightarrow c = b \neq -a$

10. Ans (A,B,D)

Here

$p(x) = x^6 - x^5 - x^3 - x^2 - x + x^4 - x^4$

$= x^2\left(x^4 - x^3 - x^2\right) + \left(x^4 - x^3 - x^2\right) - x$

$= x^2 - x + 1$

$\therefore\ p(\alpha) + p(\beta) + p(\gamma) + p(\delta)$

$= \sum \alpha^2 - \sum \alpha + 4 = (\alpha + \beta + \gamma + \delta)^2$

$-2(\alpha\beta + \beta\gamma + \gamma\delta + \alpha\delta + \alpha\gamma + \beta\delta) - \sum \alpha + 4$

$= 6$

11. Ans (B,C)

A.M of 1,2,3,.....,n = $\frac{n+1}{2}$

If k, k + 1 are removed then

$$AM = \frac{\frac{n(n+1)}{2} - (2k+1)}{n-2}$$

$$\Rightarrow 1 = \frac{n+1}{2} - \frac{n(n+1) - 2(2k+1)}{2(n-2)}$$

$\Rightarrow n - k = 1$

12. Ans (B,C)

Sol. Since x, y, z are in A.P., therefore $x + z = 2y$

Now $\frac{x+y}{2y-x} + \frac{y+z}{2y-z} = \frac{x+y}{z} + \frac{y+z}{x} = \frac{x}{z} + \frac{y}{z} + \frac{y}{x} + \frac{z}{x}$

$= \frac{x}{z} + \frac{x+z}{2z} + \frac{x+z}{2x} + \frac{z}{x} = 1 + \frac{3}{2}\left(\frac{x}{z} + \frac{z}{x}\right) \geq 1 + \frac{3}{2}(2)$ [$\because$ AM $\geq$ G.M.]

$\Rightarrow$ Minimum value of given expression is 4.

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MATHS

Section-I(iii) Comprehension Type

1. Ans (D)

(i) $D \geq 0 \Rightarrow 4((2\lambda + 1)^2 - 2\lambda^2 - 2\lambda) \geq 0$

$\Rightarrow 4(2\lambda^2 + 2\lambda + 1) \geq 0$

true $\forall\ \lambda \in R$

(ii) $f(-1) > 0$ & $f(1) > 0$

$\Rightarrow \lambda^2 + 5\lambda + 4 > 0$ & $\lambda^2 - 3\lambda > 0$

$\Rightarrow \lambda < -4$ or $\lambda > -1$ & $\lambda < 0$ or $\lambda > 3$

$\Rightarrow \lambda \in (-\infty, -4) \cup (-1, 0) \cup (3, \infty)$

(iii) $-1 < -\frac{b}{2a} < 1 \Rightarrow -1 < \frac{2\lambda+1}{2} < 1 \Rightarrow -\frac{3}{2} < \lambda < \frac{1}{2}$

from (i), (ii) & (iii)

$-1 < \lambda < 0$

$\Rightarrow |10\lambda| \in (0, 10)$

2. **Ans (C)**

if both roots are common than

$$\frac{a}{2} = \frac{b}{-2(2\lambda+1)} = \frac{c}{\lambda(\lambda+1)}$$

$\because$ if a, b, c are sides of a triangle than

$a + b > c$

$2 - 2(2\lambda + 1) > \lambda(\lambda + 1)$

$2 - 4\lambda - 2 > \lambda^2 + \lambda$

$\lambda^2 + 5\lambda < 0$

$\lambda \in (-5, 0)$ ...(i)

Similarly when $b + c > a$

$\lambda^2 - 3\lambda - 4 > 0$

$(\lambda - 4)(\lambda + 1) > 0$

$\lambda \in (-\infty, -1) \cup (4, \infty)$ ...(ii)

When $a + c > b$

$\lambda^2 + 5\lambda + 4 > 0$

$\lambda \in (-\infty, -4) \cup (-1, \infty)$ ...(iii)

from (i), (ii) & (iii)

$\lambda \in (-5, -4)$

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MATHS

Section–II (iii) : Integer Type

1. **Ans (5.00)**

$x^2 + ax + b = 0$ has roots α, β

Replace x by $\frac{1}{x}$ then equation $bx^2 + ax + 1 = 0$ has roots $\frac{1}{\alpha}, \frac{1}{\beta}$ similarly replacing x by $\frac{1}{x}$ in the equation $x^2 - ax + b - 2 = 0$, we get

equation $(b - 2)x^2 - ax + 1 = 0$ has roots $\frac{1}{\gamma}, \frac{1}{\delta}$

Now, $\alpha\beta = b$, $\gamma\delta = b - 2$ &

$\frac{1}{\alpha} + \frac{1}{\beta} = -\frac{a}{b}, \frac{1}{\gamma} + \frac{1}{\delta} = \frac{a}{b-2}$

using given conditions we have

$b(b - 2) = 24$ & $-\frac{a}{b} + \frac{a}{b-2} = \frac{5}{12}$

$$a\left[\frac{1}{b-2} - \frac{1}{b}\right] = \frac{5}{12}$$

$$a\left[\frac{b-b+2}{24}\right] = \frac{5}{12}$$

$\Rightarrow a = 5$

2. **Ans (1.00)**

$$f(x) = \frac{x^2 - 3x + 2}{x^2 + x - 6}$$

$$f(x) = \frac{(x-2)(x-1)}{(x+3)(x-2)}$$

$\because x \neq -3, 2$

$$f(x) = \frac{x-1}{x+3}$$

Range : $R - \left\{1, \frac{1}{5}\right\}$

So 1 is the integer which does not lie in the range of $f(x)$

3. **Ans (14.00)**

$1^2 + 2.2^2 + 3^2 + 2.4^2 + 5^2 + 2.6^2 + \ldots\ldots\ldots$ up to n terms = 1575

Here two case arise

(i) either $n \in$ even (ii) OR $n \in$ odd

(i) if $n \in$ even.

$$\Rightarrow \left(1^2 + 3^2 + 5^2 + \ldots\ldots \frac{n}{2}\text{terms}\right)$$

$$+2\left(2^2 + 4^2 + 6^2 + \ldots\ldots \frac{n}{2}\text{terms}\right)$$

$$\Rightarrow \left(1^2 + 2^2 + 3^2 + 4^2 + \ldots\ldots \text{n terms}\right)$$

$$+4\left(1^2 + 2^2 + 3^2 + \ldots\ldots \frac{n}{2}\text{terms}\right)$$

$$\Rightarrow \frac{n(n+1)(2n+1)}{6} + \frac{4\left(\frac{n}{2}\right)\left(\frac{n}{2}+1\right)(n+1)}{6}$$

$$\Rightarrow \frac{n(n+1)(2n+1)}{6} + \frac{n(n+2)(n+1)}{6}$$

$$\Rightarrow \frac{n(n+1)}{6}[(2n+1) + n + 2] = \frac{n(n+1)^2}{2}$$

Now if $\frac{n(n+1)^2}{2} = 1575$

$\Rightarrow n(n+1)^2 = 2 \times 1575 = 2 \times 5 \times 5 \times 7 \times 3 \times 3 = 14 \times (15)^2 \Rightarrow n = 14$

(ii) so there is no need to check far $n \in$ odd

4. **Ans (469.00)**

Let the three digits be a, ar and ar^2 then number is

$100a + 10ar + ar^2$ ....(i)

Given, $a + ar^2 = 2ar + 1$

or $a(r^2 - 2r + 1) = 1$

or $a(r-1)^2 = 1$ ....(ii)

Also given $a + ar = \frac{2}{3}(ar + ar^2)$

$\Rightarrow \quad 3 + 3r = 2r + 2r^2 \quad \Rightarrow \quad 2r^2 - r - 3 = 0 \quad \Rightarrow$

$(r+1)(2r-3) = 0$

$\therefore \quad r = -1, 3/2$

for $r = -1$, $\quad a = \frac{1}{(r-1)^2} = \frac{1}{4} \notin I \qquad \therefore r \neq -1$

for $r = 3/2$, $\quad a = \frac{1}{\left(\frac{3}{2}-1\right)^2} = 4 \qquad \{\text{from (ii)}\}$

From (i), number is $400 + 10.4.\frac{3}{2} + 4.\frac{9}{4} = 469$

Ans.

5. **Ans (9.00)**

$$T_r = \frac{(1+2+3+\ldots+r)(1+3+\ldots+(2r-1))}{r^2(1^3+2^3+\ldots+r^3)}$$

$$= \frac{\frac{r(r+1)}{2}.r^2}{r^2.\frac{r^2(r+1)^2}{4}} = \frac{2}{r(r+1)} = 2\left[\frac{1}{r} - \frac{1}{r+1}\right]$$

$$\therefore \quad \sum_{r=1}^{10} T_r = 2\left(\frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \ldots + \frac{1}{10} - \frac{1}{11}\right)$$

$$= 2\left(1 - \frac{1}{11}\right) = \frac{20}{11}$$

6. **Ans (5.00)**

$\underbrace{(1+3+5+\ldots+a)}_{\text{let } n_1 \text{ term}} + \underbrace{(1+3+5+\ldots+b)}_{n_2 \text{ term}} = \underbrace{(1+3+5+\ldots+c)}_{n_3 \text{ term}}$

$\Rightarrow n_1^2 + n_2^2 = n_3^2$ where

$a = 2n_1 - 1$

$b = 2n_2 - 1$

$c = 2n_3 - 1$

$a + b + c = 57$

$(2n_1 - 1) + (2n_2 - 1) + (2n_3 - 1) = 57$

$2(n_1 + n_2 + n_3) = 60$

$n_1 + n_2 + n_3 = 30$

$\Rightarrow n_1 = 5, n_2 = 12, n_3 = 13$

$\Rightarrow a = 9, b = 23, c = 25$

MATHS

Section–IV : Matrix Match Type

1. **Ans (A->PS,B->PQRT,C->PS,D->QS)**

(A) $T_q^2 = T_p T_r$

$(aR^{q-1})^2 = aR^{p-1}\,aR^{r-1}$

$2q = p + r$ (A.P.)

(B) If $(a^2 + b^2 + c^2)(b^2 + c^2 + d^2) < (ab+bc+cd)^2$

then a,b,c in G.P.

and if $c(a-b) = a(b-c) \Rightarrow$ a,b,c in H.P.

$\Rightarrow$ then a,b,c,d all are equal.

(C) $\frac{a}{b}\log a^2 b, \frac{b}{c}\log b^2 c, \frac{c}{d}\log c^2 d$

$\frac{1}{r}\log a^3 r, \frac{1}{r}\log a^3 r^4, \frac{1}{r}\log a^3 r^7$ (A.P.)

(D) $2a_2 = a_1 + a_3$ .....(i)

$a_3^2 = a_2 a_4$ .....(ii)

$a_4 = \frac{2a_3 a_5}{a_3 + a_5}$ .....(iii)

from (i), (ii) & (iii) $\Rightarrow a_3^2 = a_1 a_5$.

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ANSWER KEY

(Log,Mod,Tri and TE) GUIDED REVISION # 19

TIME : 60 Min.

M.M. : 78

SOLUTIONS

MATHS

Section–I(i) Single Correct Type

1. Ans (D)

$$\tan(180^\circ + 20^\circ)\left(\frac{\cos^2 10^\circ - \sin^2 10^\circ}{\cos 10^\circ \sin 10^\circ}\right) + 1$$

$$= \tan 20^\circ \left(\frac{\cos 20^\circ}{\sin 20^\circ}.2\right) + 1 = 2 + 1 = 3.$$

2. Ans (D)

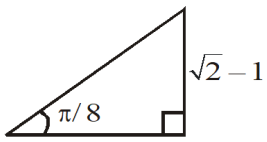
$$\frac{(\cos 1^\circ + \cos 89^\circ)(\cos 2^\circ + \cos 88^\circ)(\cos 3^\circ + \cos 87^\circ)\ldots(\cos 45^\circ + \cos 45^\circ)}{\cos 1^\circ.\cos 2^\circ \cos 3^\circ.\ldots.\cos 45^\circ}$$

$$= \frac{(2.\cos 45^\circ \cos 44^\circ)(2\cos 45^\circ \cos 43^\circ)(2\cos 45^\circ \cos 42^\circ)\ldots.2\cos 45^\circ}{\cos 1^\circ.\cos 2^\circ \cos 3^\circ.\ldots.\cos 45^\circ}$$

$= (\sqrt{2})^{44}.2 = 2^{23} \Rightarrow a + b = 25$

3. Ans (C)

$$\tan\frac{\pi}{8} = \sqrt{2} - 1$$



$$\Rightarrow \sin\frac{\pi}{8} = \frac{\sqrt{2}-1}{\sqrt{4-2\sqrt{2}}}$$

$$= \frac{\sqrt{2}-1}{\sqrt{2\sqrt{2}\left(\sqrt{2}-1\right)}}$$

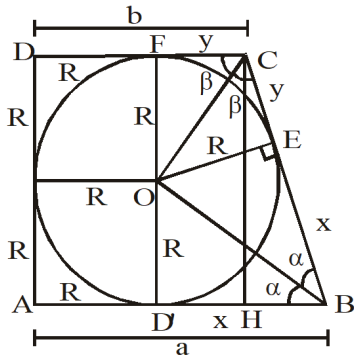
$$= \sqrt{\frac{\sqrt{2}-1}{2\sqrt{2}}} = \sqrt{\frac{1}{2} - \frac{1}{2\sqrt{2}}} = \sqrt{\frac{1}{\sqrt{4}} - \frac{1}{\sqrt{8}}}$$

$\Rightarrow$ m+n = 12

4. Ans (B)

Let ABCD is a right angled trapezium.

Let radius of circle is R



& let BD = BE = x and

CE = CF = y

AB = a

$\because$ R + x = a $\Rightarrow$ x = a - R ..... (1)

& DC = b

R + y = b $\Rightarrow$ y = b - R ..... (2)

In Δ OD'B, $\tan\alpha = \frac{R}{x}$

Now in Δ CHB, $\sin 2\alpha = \frac{CH}{BC}$

$$\Rightarrow \frac{2\tan\alpha}{1 + \tan^2\alpha} = \frac{2R}{x + y} \Rightarrow \frac{\frac{2R}{x}}{1 + \frac{R^2}{x^2}} = \frac{2R}{x + y} \Rightarrow \frac{2Rx}{x^2 + R^2} = \frac{2R}{x + y}$$

$\Rightarrow x^2 + xy = x^2 + R^2 \Rightarrow xy = R^2 \Rightarrow (a - R)(b - R) = R^2$ [From (1) and (2)]

$\Rightarrow x^2 + xy = x^2 + R^2 \Rightarrow xy = R^2 \Rightarrow (a - R)(b - R) = R^2$ [From (1) and (2)]

$\Rightarrow ab - (a + b)R + R^2 = R^2 \Rightarrow ab - (a + b)R = 0$

$\therefore R = \frac{ab}{a + b}$

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5. Ans (D)

$\log_{1/4}\beta = -1 \Rightarrow b = 4$

$\log_{\beta}\alpha = -1 \Rightarrow \alpha = \frac{1}{\beta} = \frac{1}{4}$

$g = \log a 8 = \log 1/48 = -\frac{3}{2}$

Now $\frac{1}{\alpha} + 1 = 4 + 1 = 5$

$\beta^2 + 4\gamma^2 = 16 + 4\left(-\frac{3}{2}\right)^2 = 25$

$\therefore \left(\frac{1}{\alpha} + 1\right)^{\log_{\sqrt{5}}(\beta^2+4\gamma^2)} = 5^{\log_{\sqrt{5}}25} = 625$

6. Ans (A)

For any positive value of x LHS is positive and RHS is negative

$\therefore$ no value of x will satisfy the given equation

Hence there is no solution

7. Ans (B)

$\sin\theta + \cos\theta = \sqrt{2}\sin(\theta + 45^\circ) = \sqrt{2}\sin(145^\circ) = \sqrt{2}\sin 35^\circ$ i.e. $\in (0,\ 1)$

8. Ans (B)

$\frac{\sin^2\theta \cdot \sin\theta}{\cos\theta} + \frac{\cos^3\theta}{\sin\theta} = (1 + \sin 2\theta) + \frac{(\sin^2\theta + \cos^2\theta)}{\sin\theta \cdot \cos\theta}$

$\Rightarrow \frac{\sin^4\theta + \cos^4\theta}{\sin\theta \cdot \cos\theta} = (1 + \sin 2\theta) + \frac{1}{\sin\theta \cdot \cos\theta} \Rightarrow$

$\frac{1 - 2\sin^2\theta\cos^2\theta}{\sin\theta \cdot \cos\theta} = (1 + \sin 2\theta) + \frac{2}{(\sin 2\theta)}$

$\Rightarrow \frac{2\left(1 - \frac{\sin^2 2\theta}{2}\right)}{\sin 2\theta} = (1 + \sin 2\theta) + \frac{2}{\sin 2\theta} \Rightarrow$

$\frac{2\left(1 - \frac{t^2}{2}\right)}{t} = (1 + t) + \frac{2}{t} \Rightarrow$

$\frac{(2 - t^2)}{t} = \frac{t^2 + t + 2}{t}$

$\Rightarrow 2t^2 + t = 0 \Rightarrow t(2t + 1) = 0 \Rightarrow t = 0$

& $t = -\frac{1}{2}$

$\sin 2\theta = 0 \Rightarrow \theta = 0$

or $\sin 2\theta = -\frac{1}{2}$

$\Rightarrow 2\theta = \frac{7\pi}{6}$ or $\frac{11\pi}{6}$

$\Rightarrow \theta = \frac{7\pi}{12}$ or $\frac{11\pi}{12}$

$\therefore$ sum of solution

$= 0 + \frac{7\pi}{12} + \frac{11\pi}{12}$

$= \frac{3\pi}{2}$

9. Ans (C)

$2^{\sec^2 x}\sqrt{\left(\tan y - \frac{1}{2}\right)^2 + \frac{1}{4}} \leq 1$

$\Rightarrow \sec^2 x \geq 1$ & because $\left(\tan y - \frac{1}{2}\right)^2 + \frac{1}{4} \geqslant \frac{1}{4}$

$\therefore \sec^2 x = 1$

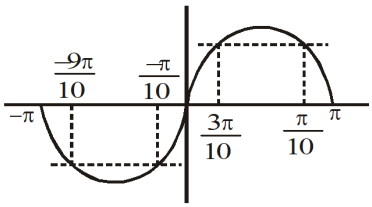
$\Rightarrow x = 0,\ \pi,\ 2\pi$

10. Ans (C)

$-\sqrt{5} < 4\sin x - 1 < \sqrt{5}$

$\Rightarrow \frac{1 - \sqrt{5}}{4} < \sin x < \frac{\sqrt{5} + 1}{4}$

$\therefore -\pi < x < -\frac{9\pi}{10},\ \frac{-\pi}{10} < x < \frac{3\pi}{10}, \frac{7\pi}{10} < x < \pi$



MATHS

Section–I(ii) Multiple Correct Type

11. Ans (A,B,C)

$\sqrt{\cos 2x} + \sqrt{1 + \sin 2x} = 2\sqrt{\sin x + \cos x}$

$\Rightarrow \sqrt{(\cos^2 x - \sin^2 x)} + \sqrt{(\cos x + \sin x)^2}$

$= 2\sqrt{\cos x + \sin x} \Rightarrow \cos x + \sin x = 0$

$\Rightarrow \tan x = -1 \Rightarrow x = n\pi - \frac{\pi}{4}$

Now,

$\sqrt{\cos x - \sin x}\left\{\sqrt{\cos x - \sin x} + \sqrt{\cos x + \sin x} - 2\right\} = 0$

$\Rightarrow \cos x + \sqrt{\cos 2x} = 2$

$\Rightarrow \cos^2 x + 4\cos x - 5 = 0 \Rightarrow \cos x = 1$

$\Rightarrow x = 2n\pi$

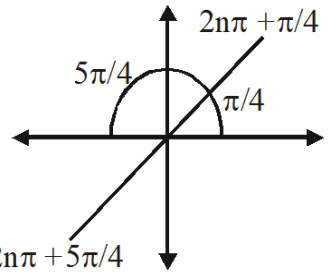
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12. **Ans (A,C)**

(i) when tanx > 0

$\tan^2 x = 1 \Rightarrow \tan x = 1$

$x = n\pi + \frac{\pi}{4}$



(ii) when tanx < 0

$\therefore -\tan x = \cot x$

$\Rightarrow \tan^2(x) = -1$ not possible

13. **Ans (A,C)**

$$\frac{(\sqrt{3}-1)\cos\theta}{2\sqrt{2}} + \frac{(\sqrt{3}+1)\sin\theta}{2\sqrt{2}} = \frac{(\sqrt{3}-1)}{2\sqrt{2}}$$

$\cos\left(\theta - \frac{5\pi}{12}\right) = \cos\frac{5\pi}{12} \Rightarrow \theta - \frac{5\pi}{12} = 2n\pi \pm \frac{5\pi}{12}$

$\Rightarrow 2n\pi + \frac{5\pi}{6}$ or $2n\pi$

14. **Ans (A,D)**

$2x = \cos(\alpha - \beta - \gamma + \delta) - \cos(\alpha - \beta + \gamma - \delta)$

$2y = \cos(\beta - \gamma - \alpha + \delta) - \cos(\beta - \gamma + \alpha - \delta)$ and similarly $2z = \cos(\gamma - \alpha - \beta + \delta) - \cos(\gamma - \alpha + \beta - \delta)$

Adding, $2x + 2y + 2z = 0 \Rightarrow x + y + z = 0$

$\therefore \quad x^3 + y^3 + z^3 - 3xyz = 0$

15. **Ans (B,D)**

$\log_{x+1}(x - 0.5) = \log_{x-0.5}(x+1)$

$\Rightarrow \left[\log_{x+1}\left(x - \frac{1}{2}\right)\right]^2 = 1 \Rightarrow \log_{x+1}\left(x - \frac{1}{2}\right) = 1$ or -1

$\Rightarrow x - \frac{1}{2} = (x+1)$ or $(x+1)^{-1} = x - \frac{1}{2} \Rightarrow x = 1$, $x = -\frac{3}{2}$ (Reject)

16. **Ans (A,B,C)**

$\log_{10}(1.96 - 1) < 0$ Hence (A) is incorrect

$a^2 - a - 2 = 0 \Rightarrow a = 2$ or $\underbrace{-1}_{\text{rej}}$ Hence (B) is incorrect

$m = 7$ & $n = 7^4$ Hence (D) is correct

$\log 0.1(\sqrt{2} - 1) > 0$ Hence (C) is incorrect

Hence (A) (B) (C)

MATHS

Section–II (iii) : Integer Type

1. **Ans (7)**

$\because \angle A = 100° \Rightarrow \angle B = \angle C = 40°$

Now, $(x^2 + y^2) = [(\sin A + \sin B + \sin C)^2 + (\cos A + \cos B + \cos C)^2]$

$= 1 + 1 + 1 + 2\sin A\sin B + 2\sin B\sin C + 2\sin C\sin A + 2\cos A\cos B + 2\cos B\cos C + 2\cos C\cos A$

$= 3 + 2\{(\cos A\cos B + \sin A\sin B) + (\cos B\cos C + \sin B\sin C) + (\cos A\cos C + \sin A\sin C)\}$

$= 3 + 2\{\cos 60° + \cos 0° + \cos(-60°)\}$

$= 3 + 2\left\{\frac{1}{2} + 1 + \frac{1}{2}\right\} = 7$

2. **Ans (0)**

$$\sum_{r=1}^{89} \log_{10}\tan\left(\frac{\pi r}{180}\right) = \left(\log_{10}\tan\frac{\pi}{180} + \log_{10}\tan\frac{89\pi}{180}\right) + \left(\log_{10}\tan\frac{2\pi}{180} + \log_{10}\tan\frac{88\pi}{180}\right) + \ldots\ldots\ldots$$

$$= \left(\log_{10}\tan\frac{\pi}{180} - \log_{10}\tan\frac{\pi}{180}\right) + \left(\log_{10}\tan\frac{2\pi}{180} - \log_{10}\tan\frac{2\pi}{180}\right) + \ldots\ldots\ldots$$

$= 0$

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3. Ans (7)

$$\ell n\left(\frac{a+b}{3}\right) = \left(\frac{\ell na + \ell nb}{2}\right)$$

$$\Rightarrow \ell n\left(\frac{a+b}{3}\right) = \ell n(ab)^{1/2}$$

$$\Rightarrow \left(\frac{a+b}{3}\right)^2 = (ab)$$

$$\Rightarrow a^2 + b^2 = 7ab$$

$$\Rightarrow \frac{a}{b} + \frac{b}{a} = 7$$

4. Ans (3)

$\underbrace{\cos^4 x}_{\leqslant 1} = \underbrace{1 + \sin^{36} x}_{\geqslant 1}$ only equality holds

$\cos^2 x = 1$ and $\sin(x) = 0$

$x = n\pi$

solution are $0, \pi, 2\pi$

5. Ans (4)

$1 + |\sin x| + |\sin^2 x| + \ldots\ldots \infty = 2$

$$\frac{1}{1 - |\sin x|} = 2$$

$$1 - |\sin x| = \frac{1}{2}$$

$$|\sin x| = \frac{1}{2}$$

$$\sin x = \pm\frac{1}{2}$$

6. Ans (6)

$$\frac{1}{\sin\left(\frac{\pi}{4} + x\right)} + \frac{1}{\sin\left(\frac{\pi}{4} - x\right)} = 2\sqrt{2}$$

$$\frac{\sqrt{2}}{\cos x + \sin x} + \frac{\sqrt{2}}{(\cos x - \sin x)} = 2\sqrt{2}$$

$$\frac{\cos x}{\cos^2 x - \sin^2 x} = 1$$

$\cos x = 2\cos^2 x - 1$

$2\cos^2 x - \cos x - 1 = 0$

$2\cos^2 x - 2\cos x + \cos x - 1 = 0$

$\cos x = 1, \cos x = -\frac{1}{2}$

$\lambda_1 = 0, \lambda_2 = \frac{2\pi}{3}$

$\sin^2\lambda_1 + \sin^2\lambda_2 = 0 + \frac{3}{4}$

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GUIDED REVISION

JEE(Main + Advanced) 2025
ENTHUSIAST+LEADER COURSE
SCORE ADVANCED

ANSWER KEY

(Sets, Relation & Statistics) GUIDED REVISION # 20

TIME : 60 Min.

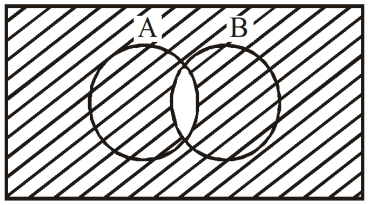
M.M. : 81

SOLUTIONS

MATHS

Section–I(i) Single Correct Type

1. **Ans (A)**



$= \cup - (A \cap B)$

$\Rightarrow \overline{(A \cap B)} = \bar{A} \cup \bar{B}$

2. **Ans (C)**

$n_1 = 100 \quad m = 250$

$\bar{X}_1 = 15 \quad \bar{X} = 15.6$

$V_1(x) = 9 \quad Var(x) = 13.44$

$$\sigma^2 = \frac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2} + \frac{n_1 n_2}{(n_1 + n_2)^2}(\bar{x}_1 - \bar{x}_2)^2$$

$n_2 = 150, \ \bar{x}_2 = 16, V_2(x) = \sigma_2$

$$13.44 = \frac{100 \times 9 + 150 \times \sigma_2^2}{250} + \frac{100 \times 150}{(250)^2} \times 1$$

$\Rightarrow \sigma_2^2 = 16 \Rightarrow \sigma_2 = 4$

3. **Ans (D)**

Given :

$$\text{Mean } (\bar{x}) = \frac{\Sigma x_i}{20} = 10$$

or $\Sigma x_i = 200$ (incorrect)

or $200 - 25 + 35 = 210 = \Sigma x_i$ (Correct)

$$\text{Now correct } \bar{x} = \frac{210}{20} = 10.5$$

again given S.D = 2.5 (σ)

$$\sigma^2 = \frac{\Sigma x_i^2}{20} - (10)^2 = (2.5)^2$$

or $\Sigma x_i^2 = 2125$ (incorrect)

or $\Sigma x_i^2 = 2125 - 25^2 + 35^2$

$= 2725$ (Correct)

$$\therefore \text{ correct } \sigma^2 = \frac{2725}{20} - (10.5)^2$$

$\sigma^2 = 26$

or $\sigma = 26$

$\therefore \alpha = 10.5, \beta = 26$

4. **Ans (B)**

$2^m = 56 + 2^n$

$\Rightarrow \quad 2^m - 2^n = 8 \times 7$

$\Rightarrow \quad 2^{m-3} - 2^{n-3} = 7$

even – even cant be odd

$\therefore 2^{n-3} = \text{odd} = 1$, if $n = 3$

$m = 6$

$m + n = 9.$

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5. Ans (D)

For a, b, c

mean = $\frac{a+b+c}{3}(=\bar{x})$

b = a + c

$\Rightarrow \bar{x} = \frac{2b}{3}$ .....(1)

S.D. (a + 2, b + 2, c + 2) = S.D. (a, b, c) = d

$\Rightarrow d^2 = \frac{a^2+b^2+c^2}{3} - (\bar{x})^2$

$\Rightarrow d^2 = \frac{a^2+b^2+c^2}{3} - \frac{4b^2}{9}$

$\Rightarrow 9d^2 = 3(a^2+b^2+c^2) - 4b^2$

$\Rightarrow b^2 = 3(a^2+c^2) - 9d^2$

6. Ans (A)

Let observations are denoted by xi for $1 \le i < 2n$

$\bar{x} = \frac{\sum x_i}{2n} = \frac{(a+a+\ldots+a)-(a+a+\ldots+a)}{2n}$

$\Rightarrow \bar{x} = 0$

and

$\sigma_x^2 = \frac{\sum x_i^2}{2n} - (\bar{x})^2 = \frac{a^2+a^2+\ldots+a^2}{2n} - 0 = a^2$

$\Rightarrow \sigma_x = a$

Now, adding a constant b then $\bar{y} = \bar{x} + b = 5$

$\Rightarrow b = 5$

and $\sigma_y = \sigma_x$ (No change in S.D.) $\Rightarrow a = 20$

$\Rightarrow a^2 + b^2 = 425$

7. Ans (B)

Reflexive : a + a = 2a need not to be irrational $\Rightarrow$ not reflexive

Symmetric :

If a + b = irrational number , then

b + a = irrational number

$\therefore$ M is symmetric relation

Transistive :

$(1, \sqrt{2}) \in M$

$(\sqrt{2}, 2) \in M$

but $(1,2) \notin M$

$\therefore$ It is not transistive

8. Ans (B)

xRx as sinx = sinx $\therefore$ Reflexive

xRy $\Rightarrow$ yRx $\therefore$ Symmetric

xRy & yRz $\Rightarrow$ xRz $\therefore$ transitive

$\Rightarrow$ Relation is equivalence

9. Ans (C)

$|a| \le b$ and $|b| \le c$

$\Rightarrow |a| \le c$

$\Rightarrow$ relation is transitive

$|-1| \le 5$ but $|5| \not\le -1$

$\Rightarrow$ relation is not symmetric

$|-2| \not\le -2$

$\Rightarrow$ relation is not reflexive

10. Ans (D)

If variance of $\sum x_i$ is v

then variance of $\sum ax_i + b$ will be a^2v

11. Ans (D)

Mean is changed as the observations are changed.

Median will remain same.

12. Ans (B)

$x_1 + x_2 + x_3 + \ldots\ldots + x_n = \bar{x}.n$

$\Rightarrow \frac{x_1}{\alpha} + \frac{x_2}{\alpha} + \frac{x_3}{\alpha} \ldots\ldots\ldots + \frac{x_n}{\alpha} = \frac{\bar{x}.n}{\alpha}$

$\Rightarrow \frac{x_1}{\alpha} + \beta + \frac{x_2}{\alpha} + \beta + \ldots\ldots + \frac{x_n}{\alpha} + \beta = \frac{\bar{x}.n}{\alpha} + n\beta$

$\Rightarrow \frac{\frac{x_1}{\alpha} + \beta + \frac{x_2}{\alpha} + \beta + \ldots\ldots + \frac{x_n}{\alpha} + \beta}{n} = \frac{\bar{x}}{\alpha} + \beta$

13. Ans (B)

Mean square deviation is minimum about mean i.e.

a_{51} and median of the group is a_{51}

$\Rightarrow$ Mode will also be a_{51}

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14. Ans (A)

Median is $(5.5x) = a$

$$\text{Mean deviation} = \frac{\sum |x_i - a|}{10} = 30$$

$$\frac{2(4.5x + 3.5x + 2.5x + 1.5x + .5x)}{10} = 30$$

$$\frac{2(12.5)|x|}{10} = 30$$

$$|x| = 12$$

15. Ans (B)

$$\sigma^2 = \frac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2} + \frac{n_1 n_2}{(n_1 + n_2)^2}(\bar{x}_1 - \bar{x}_2)^2$$

$$\sigma^2 = \frac{6 \times 24 + 3 \times 36}{6 + 3} + \frac{6 \times 3}{(6+3)^2}(11 - 14)^2$$

$$\sigma^2 = \frac{144 + 108}{9} + \frac{18}{81} \times 9 = 28 + 2 = 30$$

16. Ans (B)

$$\bar{x} = \frac{\sum x_i}{100} = 0 \ \&$$

$$\frac{\sum |x_i - \bar{x}|}{100} = 5 \Rightarrow \sum |x_i| = 500$$

$$\Rightarrow \sum x_i^2 + 2 \sum_{1 \le i < j \le 100} |x_i x_j| = (500)^2$$

$$\Rightarrow \frac{\sum x_i^2}{100} = \frac{(500)^2 - 2\sum |x_i x_j|}{100} = 2500 - 1600$$

$$\text{S.D.} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{100}} = \sqrt{900} = 30$$

17. Ans (C)

$$\frac{\Sigma a}{n} = 38 \Rightarrow \Sigma a = (38)(50)$$

$$\frac{\Sigma a - 100}{48} = \frac{(38)(50) - 100}{48} = 37.5$$

18. Ans (D)

$$\frac{\Sigma x_i^2}{n} \geqslant \left(\frac{\Sigma x_i}{n}\right)^2 \Rightarrow \frac{200}{n} \geqslant \frac{(40)^2}{n^2} \Rightarrow n > 8$$

19. Ans (C)

Let x_1, x_2, x_n be n observations, then

$$\bar{X} = \frac{1}{n}\sum x_i$$

Let $y_i = \frac{x_i}{\alpha} + 10$, then

$$\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} y_i = \frac{1}{\alpha}\left(\frac{1}{n}\sum x_i\right) + \frac{1}{n}(10n)$$

$$\Rightarrow \bar{Y} = \frac{1}{\alpha}\bar{X} + 10 = \frac{\bar{X} + 10\alpha}{\alpha}$$

MATHS

Section–II (iii) : Integer Type

1. Ans (1.00)

After adding (C,C) the R is reflexive symmetric and transitive

2. Ans (12.00)

$$5 = \frac{3 + 7 + x + y}{4} \Rightarrow x + y = 10$$

$$\text{Var}(x) = 10 = \frac{3^2 + 7^2 + x^2 + y^2}{4} - 25$$

$$140 = 49 + 9 + x^2 + y^2$$

$$x^2 + y^2 = 82$$

$$x + y = 10$$

$$\Rightarrow (x, y) = (9,1)$$

Four numbers are 21,9,10,8

$$\text{Mean} = \frac{48}{4} = 12$$

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3. Ans (4.00)

$\sum_{i=1}^{18}(X_i-\alpha)=36,\ \sum_{i=1}^{18}(X_i-\beta)^2=90$

$\Rightarrow \quad \sum_{i=1}^{18}x_i=18(\alpha+2),$

$\sum_{i=1}^{18}x_i^2-2\beta\sum_{i=1}^{18}x_i=18\beta^2=90$

Hence $\sum x_i^2=90-18\beta^2+36\beta(\alpha+2)$

Given $\frac{\sum x_i^2}{18}-\left(\frac{\sum x_i}{18}\right)^2=1$

$\Rightarrow 90-18\beta^2+36\beta(\alpha+2)-18(\alpha+2)^2=18$

$\Rightarrow 5-\beta^2+2\alpha\beta+4\beta-\alpha^2-4\alpha-4=1$

$\Rightarrow (\alpha-\beta)^2+4(\alpha-\beta)=0 \Rightarrow |\alpha-\beta|=0 \text{ or } 4$

As α and β are distinct $|\alpha-\beta|=4$

4. Ans (8.00)

Var $(ax_i+b)=a^2$ var (xi)

Variance on doubling each observation

$=2^2\times 16=64$

Std. deviation $=\sqrt{\text{var}}=8$

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5. Ans (24.00)

Mode = 3 median − 2 mean

$= 66-42=24$

6. Ans (1.25)

M.D. $=\frac{\sum|x_i-3|}{4}=1$

$\Rightarrow 2+1+|x-3|=4 \Rightarrow x=4$

Variance $=\frac{\sum(x_i-\bar{x})^2}{4}=\frac{5}{4}$

where $\bar{x}=\frac{1+2+3+4}{4}=\frac{5}{2}$